

Miller 10e + Barash 9e

Complete Chapter Key Points

Cleaned verbatim extraction • 146 chapters • complete continuation sections

Miller 10e chapters 1–87, followed by Barash 9e chapters 1–59

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Chapter 1: The Scope of Modern Anesthetic Practice

- The scope of modern anesthesiology practice includes preoperative evaluation and management of comorbid conditions; intraoperative care; postoperative care, including acute pain management; critical care, resuscitation, and retrieval; and chronic pain management. As a result, anesthesia plays a key role in perioperative medicine and the health care system with a significant impact on global health and the burden of disease.
- Health care continues to evolve, both in scope and practice requirements, with focus set on better health care access and coverage across socioeconomic strata with a mission to provide precision medicine for many endemic disorders. Some of the changes that impact anesthesia care include changing patient populations with demand for more individualized care management, locations in which care is provided, escalating costs of care, quality and safety concerns after the Coronavirus Disease 2019 (COVID-19) pandemic and labor shortages. In addition, expanded research opportunities may arise in part as a result of improved patient data availability combined with in-depth exploration of biobank material. Global conflict, climate change, and pandemics have also had significant impacts on health care. These forces have major implications for the education and training of anesthesia providers, the organization of practice, and the quality and safety of anesthesia care.
- The volume of patients presenting for perioperative and obstetric care continues to expand. Increasingly these patients are older and have significant comorbidities, including obesity and opioid use disorder. These changes have important implications for allocation of health care resources, access, and equity in health care delivery.
- Anesthesia care is expanding beyond the traditional operating room environment into interventional suites, ambulatory surgery centers, physician offices, and other settings. In many countries intensive care is also being led and provided by anesthesia departments and the education of intensivists is based on educational programs initiated by specialist medical colleges and societies with joined missions in anesthesia and intensive care. Anesthesiologists must concentrate on the impact of these changes on the safety and quality of care, including ensuring a process for escalating care to a more appropriate environment if clinically necessary.
- With expansion in surgical and procedural volume, ensuring an adequate workforce has become a global and regional challenge. To expand capability to provide care to a diverse patient population across a number of settings, the supply of both physician and non-physician anesthesia providers has to expand, with some countries needing a larger expansion, while also identifying ways in which to better utilize technology. Increased attention to health promotion and public health measures to reduce the burden of disease is also essential.
- Health care costs continue to escalate internationally, although the increased spending on health care in all environments has not consistently translated into improved quality or health outcomes. In most countries, particularly those that compensate providers based on fee for service models, payment is transitioning to value-based payment that requires providers to document quality, safety, and equity of care. Increasing pharmaceutical costs and costs of advanced technologies, which have a major impact on health care costs and costs of anesthesia care, must be collaboratively addressed.
- Anesthesiology was among the first medical specialties to focus on improving patient safety. Despite the improvement observed, particularly in the perioperative environment, the focus of attention should continue to emphasize safety without neglecting quality improvement, a process designed to improve outcomes and patient experience through documentation of quality of care and systematic change to address shortfalls in care.

- Basic, translational, clinical, and implementation research is vital to advancing anesthesia care and to ensure continuous improvement. Electronic health records have provided access to large databases and novel analytic techniques that can be used to improve care delivery models, costs of care, and clinical outcomes. These landmark achievements need to be combined with systematic build-up of large biobank and biostatistical resources within perioperative care, which can lead to a precision medicine approach to perioperative care. At the same time, identifying resources to support these research initiatives remains challenging.
- The scope of modern anesthetic practice continues to evolve and expand. Many of the disruptions occurring in health care in the 21st century have implications for the role played by clinicians in the practice of anesthesia and overall health care delivery. These disruptions provide exciting opportunities for the next generation of leaders to redefine the scope of practice and ways in which to take leadership roles in advancing care.

Source: Miller 10e, Chapter 1, Key Points, PDF pp. 23–24.

Chapter 2: Anesthesia and Global Health Equity

- More than 5 billion of the world's 7 billion people lack access to safe anesthesia and surgical services, and a similarly significant proportion lack access to fundamentals such as oxygen therapy. Surgical disease accounts for 30% of the global disease burden, yet less than 1% of development assistance for health supports the delivery of anesthesia and surgical care. Lack of access to safe, timely, and affordable anesthesia and surgery kills more than four times as many people as acquired immunodeficiency syndrome (AIDS), tuberculosis (TB), and malaria combined.
- Lack of access to safe anesthesia and surgical services is among the most neglected crises in global health. Anesthesia-related mortality in some low-income countries (LICs) is reported in the 1:100s and is mostly avoidable. The burden of surgical disease is growing and disproportionately affects low- and middle-income countries.
- Pain is one of the top causes of morbidity worldwide, and inadequate access to analgesia is among the most inequitable global public health crises the world faces today. Five and a half billion people have limited or no access to narcotic medications for analgesia. Six high-income countries (HICs) account for 80% of the world's opioid consumption. Key drug control policies, organizations, and politics continue to influence issues of access and abuse, with disproportionately negative impacts on underserved populations.
- Critical shortages and the inequitable distribution of anesthesia providers are significant barriers to increasing access to safe anesthesia and surgical care. The density of surgery, anesthesia, and obstetric providers is 0.7 per 100,000 population for LICs compared to 57 per 100,000 population for HICs.
- Modern anesthesia workforce training and practice models vary widely from country to country. Innovative workforce solutions are needed to increase provider numbers while simultaneously ensuring quality and promoting access for underserved populations.
- Anesthesia, analgesia, and surgical services are feasible in resource-constrained settings and are as cost-effective as many other public health interventions (e.g., vaccinations).
- Issues of access, safety, and resource utilization are relevant to all anesthesia providers. Equipping anesthesia providers with the knowledge and skills needed to address global challenges in anesthesia will become increasingly important to expand access to safe and affordable anesthesia care worldwide.
- Health care providers from multiple disciplines (nursing, surgery, obstetrics, anesthesia, and many more) are needed to provide surgical and perioperative care effectively. Anesthesiologists routinely work across the perioperative disciplines and can play a significant role in improving cohesive advocacy efforts and leading progress in global health equity for surgical disease and analgesia.
- The global anesthesia community is lagging behind other health disciplines in addressing global health challenges. It must rapidly expand investment in initiatives to help characterize (research), address (implementation and policy), and support (financing) global anesthesia challenges.
- Developing infrastructure, expanding the workforce, improving data to drive policy, providing financial risk protection mechanisms for surgical patients, improving referral and prehospital systems, and providing essential medicines are actions the global anesthesia community must prioritize. The need to act promptly is immense and must be balanced but not overshadowed by research agendas.

Source: Miller 10e, Chapter 2, Key Points, PDF p. 32.

Chapter 3: Perioperative Medicine

- Advances in the scientific underpinnings of anesthesia practice, expansion of anesthesia subspecialties, and changes in perioperative care have been instrumental in facilitating advances in surgical care and practice. Many new approaches to surgical and procedural care have been supported by changes in anesthesia practice and scope, allowing surgeons to perform minimally invasive or other interventional procedures not only in the traditional operating room (OR) environment but also in many alternative non-OR settings. As a result of these changes, OR anesthesia and intraoperative care no longer dominate most anesthesia practices. Each practice must understand and respond to the changing needs of the patient population it serves and the expectations of the surgeons and others regarding the need to expand the scope of anesthesia services to support advances in perioperative and periprocedural care.
- The goal for perioperative medicine is to improve patient care, aligning patient goals and expectations with clinical objectives and fostering improved clinical management and decision-making. Perioperative medicine requires coordination and shared decision-making among a diverse group of providers within and beyond the hospital setting.
- Perioperative medicine is not a specialty-specific model of care. Providing optimal perioperative care in the current health care environment requires collaboration with other providers across the health care system. Anesthesiologists are well-positioned to take a more prominent role in perioperative management, integrating their understanding of patients' surgical and medical needs during and after surgical procedures. At the same time, collaboration with hospitalists, medical specialists, and others is critical to improving perioperative care. Anesthesiologists participating in perioperative medical models of care must understand their roles and, in many cases, develop new collaborations with other providers and specialties to coordinate perioperative care and optimize transitions through the continuum of care. The specific models for delivery of a coordinated approach to perioperative care will vary depending on the interest and commitment of the anesthesiologists for assuming an expanded role, established relationships within each health system, as well as the needs of each patient and existing relationship with other providers.
- The subspecialization of anesthesiology creates opportunities for anesthesia practices to participate in perioperative medicine models of care, collaborating with surgeons, other proceduralists, and other medical subspecialists in developing clinical protocols to address patient needs across the continuum of care. When implemented, these models have been demonstrated to improve outcomes, reduce complications, length of inpatient stays, and, in many cases, costs of care. Accomplishing these goals for subspecialties requires support of the anesthesia practice to ensure that the anesthesiologists knowledgeable about the specific needs of the patients are available to fulfill these clinical goals without compromising the commitment to meet clinical demands and support other initiatives vital to the success of the department and health system.
- Various models for perioperative medicine are being successfully adopted in the United States and other countries. The perioperative surgical home (PSH) and Enhanced Recovery After Surgery (ERAS) protocols are examples of new approaches to collaborative multidisciplinary care. While these models vary from one health system or practice to another, each has documented significant benefits for a variety of procedures and clinical situations; these models have also demonstrated the importance and value of aligning perioperative goals for patients, providers, hospitals, and payors.
- While exciting, new approaches to perioperative care are also challenging to implement. Expanding the scope of practice and assuming greater responsibility for managing patient care

throughout the perioperative period requires new knowledge and skills beyond those traditionally required of an anesthesiologist, including not only understanding of the implications of and managing underlying medical conditions and comorbidities but also addressing clinical needs beyond the immediate perioperative period.

- To successfully integrate perioperative medicine into their practice, anesthesiologists must acquire a more comprehensive understanding of the economics of health care and the role perioperative care has in determining costs of care, outcomes, quality, and safety. Advances in perioperative medicine have been demonstrated to reduce costs and improve outcomes. Based on these successes, private and government payors are evaluating alternative payment models to compensate clinicians and health systems based on the documented “value” of care to better align quality, costs, and care goals. Many alternative payment models transfer the financial risk from patients and payors to providers. Anesthesia practices must understand these new payment models, their impact on clinical management, opportunities and threats associated with these expanded roles in perioperative care, and what metrics will be used to determine how the practice will be compensated (or penalized) based on predetermined performance measures.
- The electronic health record and access to “big data” can serve as valuable resources in identifying approaches to optimizing care and outcomes for individual patients based on their unique situations. Effectively taking advantage of the extensive data incorporated in the electronic record will require that anesthesiologists acquire skills in biomedical informatics and data sciences to advance perioperative care. Artificial intelligence (AI) has begun to impact perioperative management with significant potential to improve diagnostic and therapeutic decision-making. At the same time, AI will not and should not replace the critical importance of the relationship between the anesthesiologist, patient, and other providers in determining the best approach to optimizing clinical management. With appropriate integration into clinical practice, however, AI will provide another source of information that should ideally improve clinical performance and decision-making, optimize the care of each patient, and reduce complications. Similarly, advances in precision medicine will improve the perioperative anesthesiologist’s ability to identify ways to optimize the care of each patient based on integration of pharmacogenetics and risk profiling, integrated with AI and machine learning.

Source: Miller 10e, Chapter 3, Key Points, PDF pp. 76–77.

Chapter 4: Informatics in Perioperative Medicine

- Individual computers are connected via networks to share information across many users.
- Information security is about ensuring that the correct information is available only to the correct users at the correct time.
- Health care information storage and exchange is regulated to protect patient privacy.
- Information regarding the provision of anesthesia care is highly structured and organized compared to most health care specialties.
- Anesthesia care documentation systems have evolved in complexity and are widely adopted in the perioperative care of patients in the United States.
- Benefits of electronic documentation of anesthesia care typically emerge from integration with monitoring, scheduling, billing, and enterprise electronic health record (EHR) systems.
- Active and passive decision support tools may suggest typical courses of action or call to attention patterns that are not apparent to the clinician.
- Secondary use of EHR data is valuable for research, operations, and the measurement of quality of care. This may involve the development of statistical models to identify patterns or associations.

Source: Miller 10e, Chapter 4, Key Points, PDF p. 90.

Chapter 5: Quality Improvement in Anesthesia Practice and Patient Safety

- Quality needs to be an integral characteristic of health care delivery systems. Improving the quality of care often requires a reorganization of the way we work. Combining efficiency in perioperative care (especially in the operating room) with safety and the best quality possible remains challenging.
- The growing demand from patients, clinicians, insurers, regulators, accreditors, and purchasers for improved health care quality and safety requires that anesthesiologists and anesthesia team members continually evaluate the quality of the care they provide.
- Improving the quality of care requires measuring performance. Clinicians have an enhanced ability to obtain feedback regarding performance in their daily work, in part because of the increasing use of information systems. Unfortunately, a consensus has not been reached on how to measure quality of care.
- The goal of quality measurement is to learn and improve. The measurement system must fit into an improvement system; clinicians must have the will to work cooperatively to improve, and they must have ideas or hypotheses about changes to the current system of care. Also, the clinical team must have a model for testing and implementing changes that result in improvements.
- A balanced set of structures (how care is organized), processes (what we do), and outcome measures (results of care in terms of patient's health over time) are needed to evaluate the overall quality of care. Outcome measures, including in-hospital mortality rates, have been the basis for evaluating performance and quality. However, using hospital mortality as the sole performance and outcome measure alone provides an incomplete picture of quality, does not include all domains of quality, and does not measure the overall success of the entire cycle of care for a specific medical condition.
- Efforts to improve the quality of care require developing valid, reliable, and practical quality measures. Identifying clinical care that truly achieves excellence would be helpful not only to administering anesthesia but also to health care overall.
- Developing a quality measure requires several steps: prioritizing the clinical area to evaluate; selecting the type of measure; writing definitions and designing specifications; developing data collection tools; pilot-testing data collection tools and evaluating the validity, reliability, and feasibility of measures; developing scoring and analytic specifications; and collecting baseline data.
- The best opportunities to improve quality of care and patient outcomes will come not only from discovering new therapies but also from discovering how to better deliver therapies already known to be effective, i.e., quality improvement.
- Safety is an integral part of quality that is focused on the prevention of error and patient harm. The airline industry is often lauded as an exemplar of safety because it has embraced important safety principles, including the standardization of routine tasks, the reduction of unnecessary complexity, and the creation of redundancies. Anesthesia care teams have adopted these principles, although many opportunities remain to further bolster patient safety.
- Health care providers can organize their quality improvement and patient safety efforts around three key areas: (1) translating evidence into practice, (2) identifying and mitigating hazards, and (3) improving culture and communication. Although each area requires different tools, they all help health care organizations evaluate progress in patient safety and quality.

Source: Miller 10e, Chapter 5, Key Points, PDF p. 109.

Chapter 6: Human Behavior and Simulation in Anesthesia

- Clinical performance is dependent on the knowledge and clinical skills of the providers, while also considering a variety of human factors that impact decision-making and care. The impact of these human factors, including individual, team, and organizational issues within a health care system, has taken on increased significance as a determinant of the quality and safety of anesthesia care.
- Individual performance can be adversely influenced by a number of factors, including distractions, sleep deprivation, fatigue, environmental noise, illness, mental acuity, aging, as well as by social dynamics within and between teams.
- It is necessary to have a clear understanding of known human performance pitfalls such as fixation errors, ineffective team communication, misunderstandings, medication errors, unclear task management, and erroneous assumptions. Addressing these limitations will help to avoid or mitigate the likelihood of an unintended adverse event.
- Personal performance also requires an organizational culture that prioritizes and supports patient quality and safety, identifies and analyzes incidents that impact care, and has systems in place to optimize performance of each individual and team.
- Organizations that support these concepts are referred to as high-reliability organizations (HROs). HROs do not, in and of themselves, prevent all errors, but are able to identify factors that result in potential human error and proactively identify and prevent them.
- Crisis resource management (CRM) provides an approach to understanding and intervening in human performance issues that impact anesthesia care. CRM typically highlights situation awareness, dynamic decision-making, task management, communication, and teamwork.
- Simulation models represent important approaches to assessing both individual and team-based care.
- The specific approach to simulating the clinical environment should support the learning objectives and needs of the target population. For some situations, the use of standardized patients (actors), procedure-specific models, or more sophisticated simulators can provide the most effective training and evaluation of performance.
- The most widely used simulators in health care are computer screen-based simulators; manikin-based simulators are most often used for resuscitation training. More complex simulation team training is accomplished using more sophisticated manikin-based simulators, often referred to as "high-fidelity" or "full scale" patient simulation models.
- The greatest obstacles to providing effective and relevant (high-fidelity) patient simulation training are (a) time requirements for training and evaluation, and (b) providing appropriately trained and skilled simulation instructors to prepare, conduct, debrief, and evaluate the simulation sessions.
- The most important component of high-fidelity patient simulation training is the self-reflective debriefing session after the clinical scenario to trigger self-reflection and insightful analysis, leading to deep learning by the participants.

Source: Miller 10e, Chapter 6, Key Points, PDF p. 133.

Chapter 7: Ethical Aspects of Anesthesia Care

- Medical ethics emphasize principles of respect for patient autonomy, beneficence, nonmaleficence, and justice.
- Medical ethics principles also prioritize justice, equality, diversity, and inclusion (JEDI) in considering how to address specific ethical dilemmas.
- Adult patients with decisional capacity have the right to consent to—and to refuse—medical therapy, even if such therapy would be lifesaving.
- Minor patients have varying degrees of capacity, and have legal rights to make some medical decisions.
- Routine preoperative pregnancy testing has potential for significant patient harm, is not medically indicated, and requires the patient's informed consent.
- The US Supreme Court ruling in *Dobbs v. Jackson* has had major impact on how individual states and the legal system address reproductive health issues and maternal–fetal conflicts, presenting physicians with dilemmas of whether to follow the law, or adhere to ethical practice.
- Advance directives have both ethical and legal weight in allowing patients to express choices; Do Not Attempt Resuscitation (DNAR) orders should not be automatically suspended for anesthesia and/or surgery.
- Medical care at end-of-life requires special medical knowledge and experience on the part of the physician; support and counseling of patients and families; understanding and respect for the patient's autonomy; and respecting religious and cultural practices, and beliefs.
- Research in human subjects balances the needs and rights of the research subject and the hypothetical interests of future patients, and can create physician conflicts of interest that must be acknowledged and addressed.
- Advances in understanding animal cognition have led most biologists to believe that many, if not all, animals are capable of feeling pleasure, pain, anticipation, and fear – placing special obligations on the part of the researcher to reduce suffering.
- US professional physician organizations consistently state that physician participation in executions is unethical, and the American Board of Anesthesiologists states that an anesthesiologist's participation in executions may result in loss of board certification.
- "Conscientious objection" by physicians to providing legally allowable medical care is ethically limited.

Source: Miller 10e, Chapter 7, Key Points, PDF p. 151.

Chapter 8: CONSCIOUSNESS, MEMORY, AND ANESTHESIA

- Mechanisms of consciousness and memory, and their interruption by general anesthetics, are important scientific problems that have clinical relevance for the practice of anesthesiology.
- Consciousness, and its assessment, is a multidimensional phenomenon characterized by wakefulness, awareness, and behavioral expression.
- Anesthetics act at structures in the brainstem, hypothalamus, and basal forebrain that regulate sleep–wake states, which may account for loss of wakefulness.
- Anesthetics disrupt connectivity, communication, and flexible dynamics across cortical and thalamocortical networks, which may account for loss of awareness.
- Memory can be subdivided into explicit (conscious) and implicit (unconscious) recall; an example of explicit episodic recall is remembering a surgical event.
- Suppression of explicit episodic recall is one of the most potent effects of most general anesthetics.
- Effects on the hippocampus, amygdala, and prefrontal cortex—as well as the connectivity of these structures—may account for anesthetic-induced amnesia, even before loss of consciousness.

Source: Miller 10e, Chapter 8, Key Points, PDF p. 172.

Chapter 9: Sleep Medicine

- Sleep is a dynamic neuronal and behavioral state that can be characterized using specific electroencephalographic, electrophysiologic, and behavioral findings.
- Characteristics of sleep can be quantified using questionnaires, actigraphy, or respiratory polygraphy. However, polysomnography, including electroencephalogram, electrooculogram, submental electromyogram, and analysis of breathing, is required to describe the cortical characteristics and immediate physiologic consequences.
- Activation of hypothalamic sleep-promoting pathways including the ventrolateral preoptic nucleus and the median preoptic nucleus induce sleep or sedation.
- Anesthetics activate sleep-promoting neurons and can induce arousable unconsciousness (sedation) at low doses. General anesthesia (unarousable unconsciousness) cannot be achieved via activation of arousal-inhibiting neural circuits alone.
- Control of breathing is altered during sleep and anesthesia, typically leading to a decreased drive to upper airway dilator and respiratory pump muscles.
- Persistent respiratory depressant effects of anesthetics and neuromuscular blocking drugs increase the risk of postoperative respiratory complications, particularly in patients with obstructive sleep apnea.
- Anesthesia and surgery, as well as sedatives and depth of sedation in the intensive care unit, affect sleep architecture, which may lead to adverse outcomes.

Source: Miller 10e, Chapter 9, Key Points, PDF p. 197.

Chapter 10: Cerebral Physiology and the Effects of Anesthetic Drugs

- The brain has a high metabolic rate and receives approximately 15% of cardiac output. Under normal circumstances, cerebral blood flow (CBF) is approximately 50 mL/100 g/min. Gray matter receives 80% and white matter receives 20% of this blood flow.
- Approximately 60% of the brain's energy consumption supports electrophysiologic function. The remainder of the energy consumed by the brain is involved in cellular homeostatic activities.
- CBF is tightly coupled to local cerebral metabolism. When cerebral activity in a particular region of the brain increases, a corresponding increase in blood flow to that region takes place. Conversely, suppression of cerebral metabolism leads to a reduction in blood flow.
- Cerebral autoregulation maintains a relatively stable CBF with varying mean arterial pressure (MAP). There is a substantial amount of inter-individual variability in this autoregulation, not only in the lower and upper limits but also in the width and slope of the plateau. With acknowledgment of this variability, the lower limit in most humans is not less than 70 mm Hg.
- CBF is also under chemical regulation. CBF varies directly with arterial carbon dioxide tension in the arterial partial pressure of carbon dioxide (PaCO₂) range of 25 to 70 mm Hg. When arterial partial pressure of oxygen (PaO₂) decreases to less than 60 mm Hg, CBF dramatically increases. Reductions in body temperature influence CBF primarily by suppression of cerebral metabolism.
- Systemic vasodilators (e.g., nitroglycerin, nitroprusside, hydralazine, calcium channel blockers) vasodilate the cerebral circulation and can, depending on the MAP, increase CBF. Vasopressors such as phenylephrine, norepinephrine, ephedrine, and dopamine do not directly affect global CBF; their effect on CBF is via their effect on arterial blood pressure. When the MAP is less than the lower limit of autoregulation, vasopressors increase the MAP and thereby increase CBF. If the MAP is within the limits of autoregulation, then vasopressor-induced increases in systemic pressure either have no effect or a modest effect on CBF depending upon the slope of the plateau.
- All volatile anesthetics suppress the cerebral metabolic rate (CMR) and, except for halothane, can produce burst suppression of the electroencephalogram (EEG). At the level of EEG burst suppression, the CMR is reduced by approximately 60%. Volatile anesthetics have dose-dependent effects on CBF. In doses less than the minimal alveolar concentration (MAC), CBF is modestly decreased. In doses larger than 1 MAC, direct cerebral vasodilation results in an increase in CBF and cerebral blood volume.
- Barbiturates, etomidate, and propofol decrease the CMR and can produce burst suppression of the EEG; at that level, the CMR is reduced by approximately 60% and there is a corresponding reduction in CBF because blood flow and metabolism coupling are preserved. Opiates effect minor decreases in CBF and CMR. Sedative doses of benzodiazepines have only a minor effect on CMR and CBF; greater doses that can induce general anesthesia result in a significant reduction in CBF and CMR. In contrast, ketamine can increase the CMR with a corresponding modest increase in blood flow.
- Brain stores of oxygen and substrates are limited, and the brain is extremely sensitive to decreases in CBF. Symptoms of cerebral ischemia are apparent when CBF is reduced by ~ 40% from resting, basal flow in awake individuals. The difference between the basal flow and the level at which symptoms of cerebral ischemia occur is referred to as cerebrovascular flow reserve. Transient reductions in CBF that are of brief duration do not injure the brain given this reserve;

sustained reduction in CBF, even within the reserve, can lead to cerebral injury. Severe decreases in CBF lead to rapid neuronal death.

- Barbiturates, propofol, ketamine, volatile anesthetics, and xenon have neuroprotective efficacy and can reduce ischemic cerebral injury in experimental models. This anesthetic neuroprotection is sustained only when the severity of the ischemic insult is mild; with moderate to severe injury, long-term neuroprotection is not achieved. The neuroprotective efficacy of anesthetics in humans is limited.

Source: Miller 10e, Chapter 10, Key Points, PDF p. 233.

Chapter 11: Neuromuscular Physiology and Pharmacology

- The neuromuscular junction (NMJ) contains the distal motor neuron nerve terminal, Schwann cell, and muscle acetylcholine receptors (AChRs), which together provide an array of receptors and substrates for drug action. Neuromuscular transmission is predominantly dependent on acetylcholine (ACh) as the natural transmitter. When ACh is released from the presynaptic nerve terminal, it binds to the AChRs that are present either presynaptically or postsynaptically. The AChRs, depending on their structural composition, are classified as muscle subtype AChRs or neuronal subtype AChRs.
- There are two types of muscle relaxants, the depolarizing and nondepolarizing muscle relaxants (NDMRs), and they act as agonists and antagonists (blockers) of the AChRs, respectively. Neuromuscular transmission is impeded when NDMRs prevent access of ACh to its binding site on the postsynaptic nicotinic AChRs.
- If the concentration of NDMR is increased, another, noncompetitive action—block of the ion channel—is superimposed. The postsynaptic paralytic effects of muscle relaxants can be enhanced by the actions of the relaxant on presynaptic AChRs, which can be detected as muscle weakness or fade that occurs with repetitive nerve stimulation. Fade can also be seen when the postsynaptic AChRs alone are functionally blocked (e.g., by bungarotoxin) or when the number of AChRs is decreased (e.g., myasthenia gravis). Hence, the NMJ is a complex and dynamic system in which the phenomena produced by drugs are composites of actions that vary with the drug, dose, activity at the nerve terminal and muscle, the AChR subtype, time after administration, presence of anesthetics or other drugs, and age and condition of the patient.
- Inhibition of acetylcholinesterase enzyme by anticholinesterases (e.g., neostigmine) increases the concentration of ACh in the synaptic cleft, which can compete with and displace the NDMRs and thus reverse the paralysis. These anticholinesterase drugs also have other effects, including the nerve terminal and the receptor, by an allosteric mechanism. Acute bolus or prolonged administration of anticholinesterases can have deleterious effects on neuromuscular function in otherwise healthy patients. The modified cyclodextrin, sugammadex, is a novel compound that reverses paralysis of only steroidal muscle relaxants by encapsulation of the drug.
- Depolarizing compounds (e.g., succinylcholine) initially react with the AChRs and, like ACh, open AChR ion channels during depolarization of the end-plate membrane. However, in contrast to ACh, succinylcholine is not subject to hydrolysis by acetylcholinesterase and therefore remains in the junction. Soon after the administration of succinylcholine, some receptors are desensitized, and, although occupied by the agonist, they do not open to allow current to flow to depolarize the muscle membrane area.
- If depolarizing relaxant is administered at higher-than-usual concentrations, or is allowed to remain at the junction, then other effects occur. Higher concentrations of relaxant have effects on presynaptic structures, and the combination of presynaptic and postsynaptic effects plus secondary ones on muscle and nerve homeostasis results in the complicated phenomenon known as phase II block. Other drugs used clinically (e.g., Botox), effect paralysis by acting on the motor nerve and therefore indirectly on muscle. Infection with clostridial toxins (*Clostridium botulinum*, gas gangrene) can lead to systemic paralysis because of decreased release of ACh from the nerve terminal. NDMRs administered for 24 hours or longer can have effects on the postsynaptic AChRs and simulate a denervation state (chemical denervation) manifested by upregulated postsynaptic AChRs. Magnesium given to preeclamptic pregnant patients decreases the release of ACh with a potential for muscle weakness in the mother or newborn. In recognizing these sites and mechanisms, we begin to bring our theoretical knowledge closer to explaining the phenomena

observed when these drugs are exposed to living humans.

- Altered AChR expression on the postjunctional membrane occurs in normal and diseased states and complicates responses to muscle relaxants. In certain pathologic states (e.g., denervation, stroke, sepsis, burns, immobilization, chronic use of NDMRs), AChRs are upregulated, with increased expression of the immature isoform and de novo expression of the nicotinic $\alpha 7$ acetylcholine receptors ($\alpha 7$ AChRs). Altered functional and pharmacologic characteristics of the immature (fetal or γ -subunit-containing receptors) and $\alpha 7$ AChRs expressed in pathologic conditions result in increased sensitivity to succinylcholine with hyperkalemia and resistance (insensitivity) to NDMRs.
- Genetic mutations in the AChRs, that result in prolonged or fast open-channel time, can lead to a myasthenia-like state, even in the presence of normal receptor numbers. The weakness is usually related to ineffective depolarization or to the altered open-channel time, or to both.

Source: Miller 10e, Chapter 11, Key Points, PDF pp. 288–289.

Chapter 12: Respiratory Physiology and Pathophysiology

- Removal of carbon dioxide (CO₂) is determined by alveolar ventilation, not by total (minute) ventilation.
- Dead space ventilation can be dramatically increased in patients with chronic obstructive pulmonary disease and pulmonary embolism to more than 80% of minute ventilation.
- Respiratory drive is regulated by central and peripheral chemoreceptors and by stretching/irritation receptors in the lungs and rib cage.
- Hypoxemia can be caused by alveolar hypoventilation, diffusion impairment, ventilation–perfusion mismatch, and right-to-left shunt.
- Almost all anesthetics reduce skeletal muscle tone, which decreases functional residual capacity (FRC) to levels close to the awake residual volume.
- Atelectasis during anesthesia is caused by decreased FRC and the use of high fraction of inspired oxygen (FiO₂), including breathing oxygen before induction of anesthesia.
- General anesthesia causes ventilation–perfusion mismatch (airway closure) and shunts (atelectasis).
- Venous admixture is due to $\dot{V}_A/\dot{Q}$ mismatch (response to increased FiO₂) and shunts (unresponsive to increased FiO₂).
- Hypoxic pulmonary vasoconstriction is blunted by most anesthetics, and this results in increased ventilation–perfusion mismatching.
- Respiratory work is increased during anesthesia as a consequence of reduced respiratory compliance and increased airway resistance.

Source: Miller 10e, Chapter 12, Key Points, PDF p. 310.

Chapter 13: Cardiac Physiology

- The cardiac cycle is the sequence of electrical and mechanical events during a single heartbeat.
- Cardiac output is determined by the heart rate, myocardial contractility, and preload and afterload.
- Most cardiomyocytes consist of myofibrils, which are rod-like bundles that form the contractile elements within the cardiomyocyte.
- The basic working unit of contraction is the sarcomere.
- Gap junctions are responsible for the electrical coupling of small molecules between cells.
- Action potentials have four phases in the heart.
- The key player in cardiac excitation–contraction coupling is the ubiquitous second messenger calcium.
- Calcium-induced sparks are spatially and temporally patterned activations of localized calcium release that are important for excitation–contraction coupling and regulation of automaticity and contractility.
- Hormones that have cardiac action can be synthesized and secreted by cardiomyocytes or produced by other tissues and delivered to the heart.
- Cardiac reflexes are fast-acting reflex loops between the heart and central nervous system that contribute to the regulation of cardiac function and the maintenance of physiologic homeostasis.

Source: Miller 10e, Chapter 13, Key Points, PDF p. 342.

Chapter 14: Gastrointestinal and Hepatic Physiology, Pathophysiology, and Anesthetic Considerations

- The gastrointestinal (GI) tract forms a long tube from mouth to anus and its main functions are motility, digestion, absorption, excretion, and circulation. Each component of the GI tract has specific functions.
- The acinus is the basic functional unit of the liver. It is organized around the flow of blood from the portal vein to the central vein through the sinusoids. Hepatocytes in the acinus are organized into zones based on their proximity to the portal vein or central vein. Zone 1 or periportal hepatocytes are closer to the portal vein and receive oxygen and nutrient-rich blood. Zone 3 or perivenous hepatocytes are closer to the central veins and receive oxygen-poor blood. Hepatocytes in different zones serve different anatomic functions.
- The gallbladder primarily functions to store bile, while the pancreas has a variety of specialized cells to carry out both endocrine and exocrine functions.
- The GI tract is innervated by the autonomic and enteric systems, while the liver, gallbladder, and pancreas are innervated by the autonomic and sensory systems.
- The GI tract contains a microbiome of approximately 100 trillion microorganisms that perform various functions to aid digestion.
- The liver plays an integral role in carbohydrate, protein, lipid, and bile metabolism. It is also responsible for protein synthesis. Albumin is the most abundant plasma protein produced by the liver.
- Drug and toxin excretion is carried out by the hepatocytes by first polarizing the molecules, then conjugating them to make them more hydrophilic. Drugs excreted in the bile may be reabsorbed through enterohepatic circulation leading to prolonged effects.
- Standard laboratory panels used to evaluate the hepatobiliary system help define broad categories of hepatobiliary pathology: hepatitis, hepatobiliary dysfunction, or insufficient protein synthesis. Specific diagnoses often require clinical context and radiologic studies.
- The effect of general anesthesia on the GI tract is multifaceted. Medications, as well as their hemodynamic side effects, can alter GI tract function. Opioids have an adverse effect on bowel function. Many efforts are underway to decrease the use of opioids in GI surgery.
- Hemodynamic changes, bowel manipulation, and open abdominal surgeries can produce major effects on the anatomy and function of the GI tract including postoperative ileus, inflammatory states, mesenteric ischemia, and partial or total disruption of myogenic continuity.
- Volatile anesthetics reduce mean arterial pressure and cardiac output, leading to a reduction in portal blood flow in a dose-dependent manner. The hepatic arterial buffer response is preserved with isoflurane, sevoflurane, and desflurane leading to the preservation of total hepatic blood flow.
- Advanced liver disease impairs the elimination of many drugs including vecuronium, rocuronium, morphine, meperidine, and benzodiazepines. Dosing should be adjusted in the setting of liver failure.
- Elective surgery is contraindicated in patients with acute hepatitis or liver failure. Patients with chronic hepatitis can safely undergo elective surgery. Hepatotoxic drugs should be avoided and hepatic perfusion maintained. Child–Turcotte–Pugh class and Model for End-Stage Liver Disease (MELD) score can be used to predict risk of perioperative mortality.

- Cirrhosis from chronic liver disease can lead to liver failure, which can lead to significant systemic dysfunction: coagulopathy, hyperdynamic circulation, encephalopathy, hepatopulmonary dysfunction, hepatorenal dysfunction, and portopulmonary hypertension. The definitive treatment for hepatic failure is liver transplantation.
- Enhanced Recovery After GI Surgery (ERAS) protocols and evidence-based practices help to preserve natural GI physiology and are associated with shorter hospitalizations. These protocols focus on: optimal perioperative pain control; nutrition; avoidance of unnecessary tubes, lines, and drains; temperature; fluid management; and early ambulation.

Source: Miller 10e, Chapter 14, Key Points, PDF p. 364.

Chapter 15: Renal Anatomy, Physiology, Pharmacology, and Evaluation of Function

- A molecule in the plasma must pass in succession through the endothelial fenestration, glomerular basement membrane (GBM), and epithelial slit diaphragm to cross the filtration barrier and enter the tubular fluid. The capillary endothelium restricts the passage of cells, while the GBM restricts passage of albumin and larger molecules. The negatively charged glycoproteins of the GBM limit the passage of other negatively charged proteins. Thus, the filtration barrier is size selective and charge selective. The epithelial cells maintain the integrity of this capillary bed by producing several key signaling molecules and endocytosis of membrane receptors.
- A primary determinant of glomerular filtration rate (GFR) is glomerular filtration pressure, which depends on renal artery perfusion pressure and the balance between afferent and efferent arteriolar tone. With decreases in afferent arteriolar pressure or blood flow, mediators such as catecholamines, angiotensin II, and arginine vasopressin (AVP) constrict efferent arterioles to maintain glomerular filtration pressure. This change is reflected by an increase in GFR.
- Tubuloglomerular feedback is a critical mechanism for renal autoregulation. When GFR is increased, distal tubular NaCl delivery is enhanced and the increase in chloride concentration is sensed by the macula densa to trigger the renin–angiotensin cascade. The result is afferent arteriolar vasoconstriction, which in turn reduces GFR. Autoregulation enables the kidney to maintain solute and water regulation independently of wide fluctuations in arterial blood pressure. Tubular water reabsorption determines urinary flow rate and is closely related to the hydrostatic pressure in the peritubular capillaries. Hypotension of any cause results in decreased urinary flow rate that may be correctable only when arterial blood pressure is restored toward normal.
- The tubule has an enormous capacity for reabsorption of water and NaCl. Each day, 180 L of protein-free glomerular ultrafiltrate is formed, of which almost 99% of the water and 99% of the sodium is reabsorbed. The ability of the kidney to concentrate urine is dependent on the interaction of at least three processes: (1) generation of a hypertonic medullary interstitium by the countercurrent mechanism and urea recycling, (2) concentration and then dilution of tubular fluid in the loop of Henle, and (3) the action of AVP, also known as antidiuretic hormone, in increasing water permeability in the last part of the distal tubule and collecting ducts. The juxtaglomerular apparatus provides an important regulating system for arterial blood pressure, salt, and water homeostasis.
- The interactions of the sympathoadrenal axis, renin–angiotensin–aldosterone system, and AVP will respond to hypotension and hypovolemia by promoting vasoconstriction and the retention of salt and water. Prostaglandins and natriuretic peptides facilitate vasodilation and the excretion of salt and water excretion when there is hypertension and hypervolemia.
- Plasma osmolality is tightly controlled. The threshold for AVP secretion by the neurohypophysis is between 280 and 290 mOsm/kg. Even mild dehydration results in rapid antidiuresis, and urine osmolality can increase from 300 to 1200 mOsm/kg. Decreases in intravascular volume stimulate AVP secretion.
- Serum creatinine reflects the balance between creatinine production from muscle and creatinine excretion by the kidney, which is dependent on GFR. Creatinine generation rate varies with muscle mass, physical activity, protein intake, and catabolism. However, when these processes are in equilibrium and renal function is stable, serum creatinine is a useful marker of GFR. The relationship between serum creatinine and GFR is inverse and exponential. A doubling of the serum creatinine implies a halving of GFR.

Source: Miller 10e, Chapter 15, Key Points, PDF p. 408.

Chapter 16: Basic Principles of Pharmacology

- Pharmacokinetics describes the relationship between drug dose and drug concentration in plasma or at the site of drug effect over time. For anesthetic drugs, the processes of distribution and elimination (metabolism and excretion) govern this relationship.
- The time course of intravenously administered drugs is a function of distribution volume and clearance. Estimates of distribution volumes and clearances, pharmacokinetic parameters, are derived from mathematical formulas fit to measured blood or plasma concentrations over time following a known drug dose.
- Front-end kinetics refer to alterations in cardiac output that substantially influence the pharmacokinetic behavior of anesthetic drugs in terms of onset and duration of effect. Context-sensitive decrement time, which is defined as the time required to reach a certain plasma concentration after a termination of long infusion, characterizes the back-end kinetics.
- Hysteresis refers to the time delay between changes in plasma concentration and drug effect. Hysteresis accounts for the time required for drug to diffuse from the plasma to the site of action plus the time required, once drug is at the site of action, to elicit a drug effect.
- Pharmacodynamics describes what the drug does to the body, particularly the relationship between drug concentration and pharmacologic effect.
- The effect-site concentration describes a mathematically derived virtual location where an anesthetic drug exerts its effect. This approach cannot describe a mechanism of drug action (e.g., drug–receptor interaction).
- A single anesthetic drug has multiple effects (i.e., analgesia, ventilatory depression, loss of response to laryngoscopy, and changes in the electroencephalogram) that typically occur at various effect-site concentrations.
- The concentration range where changes in drug effect occur is known as the dynamic range. Concentrations outside the dynamic range do not yield much change in drug effect. Levels below the dynamic range are ineffective and those above the dynamic range do not provide additional effect.
- Anesthesia is the practice of applied drug interactions. Anesthetic management rarely includes administration of a single drug, but rather a combination of drugs to achieve desired levels of hypnosis, analgesia, and muscle relaxation. Hypnotics, analgesics, and muscle relaxants all interact with one another such that rarely does one drug, when administered in the presence of other drugs, behave as if it were administered alone.
- Pharmacokinetic and pharmacodynamic principles characterize the magnitude and time course of drug effect, but because of complex mathematics, they have limited clinical utility. Advances in computer simulation have brought this capability to the point of real-time patient care in the form of drug displays.
- Many additional issues including, but not limited to, patient's demographics and medical history must be considered in determining the correct drug(s) and correct doses for an individual patient. Some of these include age; body habitus; gender; chronic exposure to opioids, benzodiazepines, or alcohol; presence of heart, lung, kidney, or liver disease; and the extent of blood loss or dehydration.
- Some patient characteristics (e.g., obesity and age) influence anesthetic drug behavior, while the impact of other patient characteristics (chronic opioid use, hepatic and renal failure) remain

poorly understood.

Source: Miller 10e, Chapter 16, Key Points, PDF p. 428.

Chapter 17: Inhaled Anesthetics: Mechanisms of Action

- Anesthesia consists of separable and independent components, each of which involves distinct, but possibly overlapping, mechanisms at different sites in the central nervous system.
- The potencies of general anesthetics correlate with their solubility in oil, indicating the importance of interactions with predominantly hydrophobic targets.
- General anesthetics act by binding directly to amphiphilic cavities in proteins. Binding sites have been identified by a combination of site-directed mutagenesis and high-resolution structural analysis of anesthetic binding.
- Mutations made to render putative protein targets insensitive to inhaled anesthetics have been expressed in mice but have not generated breakthroughs analogous to the success of this strategy with intravenous anesthetics.
- The effects of inhaled anesthetics cannot be explained by a single molecular mechanism. Rather, several targets contribute to the component actions comprising the anesthetic effects of each anesthetic. However, these effects do converge on a limited number of states underlying the behavioral effects.
- The immobilizing effect of inhaled anesthetics involves actions in the spinal cord, whereas sedation/hypnosis and amnesia involve supraspinal mechanisms that interact with endogenous memory, sleep, and consciousness pathways and networks.
- Volatile inhaled anesthetics enhance inhibitory synaptic transmission postsynaptically by potentiating ligand-gated ion channels activated by γ -aminobutyric acid (GABA) and glycine, extrasynaptically by enhancing GABA receptors, and presynaptically by enhancing basal GABA release.
- Inhaled anesthetics suppress excitatory synaptic transmission presynaptically by reducing glutamate release (volatile anesthetics) and postsynaptically by inhibiting excitatory ionotropic receptors activated by glutamate (gaseous, and to some extent volatile, anesthetics).
- Inhaled anesthetics directly activate certain two-pore domain potassium channels, whose effects are likely to impact on both pre- and postsynaptic mechanisms.
- No comprehensive theory of anesthesia yet describes the sequence of events leading from the interaction between an anesthetic molecule and its targets to the behavioral effects.

Source: Miller 10e, Chapter 17, Key Points, PDF p. 455.

Chapter 18: Inhaled Anesthetic Uptake, Distribution, Metabolism, and Toxicity

- The alveolar anesthetic concentration (FA) or partial pressure (Palv) is important because it is the driving force determining anesthetic uptake into blood and target tissues in the central nervous system, and it can be monitored as a readout of anesthetic dosage. Palv is influenced by both delivery and uptake of anesthetic gas.
- Inhaled anesthetic delivery to patients can be increased with larger fresh carrier gas flows, vaporizer output settings, and minute ventilation.
- Initial anesthetic uptake into blood increases with greater pulmonary blood flow (cardiac output) and high blood solubility of anesthetic gas. Increased uptake (as with a highly blood-soluble drug or high cardiac output) slows anesthetic induction because it slows the rate of rise of Palv. Conversely, low anesthetic solubility in blood is associated with rapid onset and offset of anesthesia.
- Uptake of anesthetic into blood slows as blood and tissue partial pressures increase, resulting in higher anesthetic partial pressure in mixed venous blood.
- The higher the inspired anesthetic concentration, the less it diminishes because of uptake (the concentration effect). At 100% inspired concentration of a gas, uptake reduces the volume of gas in alveoli, but does not reduce Palv. During inhalation of anesthetic mixtures containing high nitrous oxide (N₂O) concentrations, the reduction of alveolar volume produced by rapid N₂O uptake sustains or increases the concentrations of other alveolar gases (the second gas effect).
- Factors that affect anesthetic uptake similarly affect pulmonary clearance of anesthetics. The rate of clearance is also context sensitive – that is, equivalent drops in alveolar and brain anesthetic concentrations are slower after a long exposure to inhaled anesthetic compared to a short exposure at equal depth.
- Toxicities of inhaled anesthetics that last beyond the exposure period are primarily associated with their biotransformation (metabolism). These toxic effects are usually produced in the tissues, such as liver and kidney, in which metabolism occurs. Modern inhaled anesthetics undergo less metabolism than older drugs, resulting in less hepatic and renal toxicity.
- Halothane hepatitis is a potentially fatal syndrome of fulminant liver damage after exposure to reactive metabolites produced by oxidation of volatile anesthetics. These metabolites covalently modify liver proteins, creating neohaptens that elicit an immune response against hepatocytes. The incidence of the syndrome varies with different anesthetics, paralleling the extent of drug metabolism: halothane >> enflurane > isoflurane > desflurane.
- Defluorination of some inhaled anesthetics occurs in both the liver and kidney, raising fluoride concentrations in blood. High fluoride is associated with toxicity characterized by high-output renal failure and is almost exclusively seen after prolonged exposure to methoxyflurane. Sevoflurane metabolism also results in high fluoride levels in blood but does not damage kidneys. Factors that enhance the toxicity of methoxyflurane relative to sevoflurane include its higher tissue solubility, slower clearance, and higher degree of renal metabolism, resulting in high intrarenal fluoride levels for an extended time.
- In laboratory mammals, including nonhuman primates, all general anesthetics alter synapse and neural circuit formation during critical periods of brain development, resulting in abnormal memory and behavior. Clinical studies in young children suggest that single brief (<3 hours) anesthetic exposures are not associated with significant neurocognitive deficits relative to

unexposed controls (see also Chapter 72). Postoperative delirium and cognitive decline/dysfunction (POCD) in elderly patients is another concern (see Chapter 78). Based on animal and clinical studies, both postsurgical neuroinflammation and exposure to general anesthesia may contribute to POCD.

- Volatile anesthetics react with strong bases, particularly potassium hydroxide (KOH), in carbon dioxide (CO₂) absorbents, resulting in production of several potentially toxic substances. Sevoflurane degrades to form compound A, which is associated with renal damage in rodents, but not in humans. This differential toxicity is associated with differential renal metabolism in rodents versus humans. Dry CO₂ absorbents in the breathing circuit react with inhaled anesthetics, releasing carbon monoxide and heat. New CO₂ absorbent materials that lack strong base chemicals prevent these reactions and subsequent potential harm to patients.
- N₂O is unique among anesthetics in its ability to inhibit methionine synthase by oxidizing a cofactor, vitamin B₁₂. In certain susceptible patients or with multiple frequent exposures to N₂O, methionine synthase inhibition can lead to hematologic and neurologic dysfunction. After prolonged N₂O exposure, methionine synthase inhibition also increases blood homocysteine, which is hypothesized to increase vascular inflammation and the risk of thrombosis. Large clinical trials show that N₂O exposure does not increase the risk of cardiovascular morbidity in most patients, but N₂O should be avoided in patients with deficiencies in dietary vitamin B₁₂ intake or absorption or vitamin B₁₂-dependent metabolism.
- Inhaled anesthetics, when scavenged and ejected into the atmosphere, contribute to both global warming and ozone depletion. Minimizing the environmental impact of inhaled anesthetics requires reducing waste, through the routine use of low fresh gas flows and/or by use of new technologies that trap scavenged anesthetic gases. Reprocessing and reusing trapped anesthetic gases further reduce the environmental impact of drug production.

Source: Miller 10e, Chapter 18, Key Points, PDF pp. 484–485.

Chapter 19: Pulmonary Pharmacology of Inhaled Anesthetics

- Inhaled anesthetics affect every part of lung physiology, and their pulmonary pharmacology is complex.
- Volatile anesthetics produce bronchodilation by reducing cytoplasmic ionized calcium concentration and/or calcium sensitivity of airway smooth muscle. Volatile anesthetics also attenuate an increase in airway resistance induced by chemical and mechanical stimulation.
- Inhaled anesthetics reduce the rate of mucus clearance and affect function of type II alveolar cells potentially contributing to postoperative pulmonary complications.
- Inhaled anesthetics produce a biphasic response in pulmonary vascular smooth muscle. Although inhibition of hypoxic pulmonary vasoconstriction by volatile anesthetics is an overall small effect, it can contribute to worsening of hypoxemia in patients with underlying pulmonary disease.
- Inhaled anesthetics depress respiratory function through a diminished respiratory drive and an increase in upper airway collapsibility. Following extubation of the trachea, volatile anesthetics can severely impair peripheral chemoreceptor inputs and hypoxic arousal reflexes even at residual concentrations.
- Inhaled anesthetics produce dose-dependent reductions in tidal volume and minute alveolar ventilation, cause tachypnea, and blunt ventilatory response to hypercapnia and hypoxia.
- During anesthesia with volatile agents, diaphragmatic function remains relatively well preserved, whereas activity of the inspiratory rib cage muscles is significantly depressed, resulting in insufficiency of breathing or paradoxical breathing.
- Volatile anesthetics can compromise upper airway patency. Even at low concentrations, upper airway obstruction can occur in susceptible patients, including elderly, obese, and critically ill patients.
- Inhaled anesthetics vary in their ability to irritate airways and augment defensive airway reflexes. Sevoflurane causes the least amount of subjective airway irritation and is the anesthetic of choice for inhaled induction of anesthesia in infants and children.
- Experimental and clinical evidence suggest therapeutic potential of isoflurane and sevoflurane in acute lung injury.
- Despite recent concerns regarding the use of nitrous oxide, a lack of strong evidence does not justify abandoning it from clinical practice, especially considering its favorable cost-effectiveness.
- Xenon produces a fast on and off action and is a promising alternative for sedation in the critical care setting.

Source: Miller 10e, Chapter 19, Key Points, PDF p. 521.

Chapter 20: Inhaled Anesthetic Delivery Systems

- The modern anesthesia workstation has evolved into a complex device with several safety features. However, if there is any possibility that the workstation or the breathing circuit is malfunctioning, ventilating the patient using an oxygen cylinder and a manual ventilation bag is the top priority. When in doubt, ventilate and oxygenate the patient first via another method and troubleshoot later.
- The most important part of the pre-anesthesia workstation checkout procedure is to verify the presence of a self-inflating resuscitation bag and that an alternative oxygen source (E-cylinder) is available.
- The Diameter Index Safety System (DISS) is designed to prevent the misconnection of hospital gas supply lines to the anesthesia workstation. The Pin Index Safety System (PISS) is designed to prevent incorrect gas cylinder connections to the anesthesia workstation. Despite these safety features, no system is immune to misconnections.
- In the event of hospital pipeline crossover or contamination, two actions must be taken: the backup oxygen cylinder valve must be opened, and the wall supply sources must be disconnected. If the wall source remains connected, the suspect hospital pipeline gas will continue to flow to the patient.
- The oxygen flush valve provides a high flow of 100% oxygen directly to the patient's breathing circuit, allowing the anesthesia provider to overcome circuit leaks or to rapidly increase the inspired oxygen concentration. Improper use of the oxygen flush valve can be associated with barotrauma or patient awareness.
- When using nitrous oxide, there is a risk of delivering a hypoxic mixture to the patient. "Fail-safe" valves and nitrous oxide/oxygen proportioning systems help minimize this risk, but they are not 100% fail-safe.
- The low-pressure section (LPS) of the gas supply system includes the flow control valves, flowmeters or flow sensors, and the anesthetic vaporizers. This section of the anesthesia workstation is most vulnerable to leaks, which can cause delivery of a hypoxic gas mixture or an inadequate concentration of anesthetic agent to the patient. The workstation must be checked for leaks before delivery of any anesthetic.
- The oxygen analyzer is the only protection against a hypoxic mixture within the low-pressure section of the pneumatic system.
- Anesthesia workstations with a one-way check valve in the LPS require a manual negative-pressure leak test. On machines without a check valve in this location, manual positive-pressure testing or automated testing is used to test the LPS for leaks.
- On machines with manually controlled anesthetic vaporizers, internal vaporizer leaks can be detected only when the vaporizer is turned on. This is true even during automated machine self-tests. Machines with electronically controlled vaporizers can check the installed vaporizer during self-test.
- Variable bypass vaporizers route a portion of the fresh gas flow into a vaporizing chamber to create the desired anesthetic concentration. Injection-type vaporizers utilize microprocessor control to inject small amounts of anesthetic liquid into an evaporating chamber.
- Desflurane's low boiling point and high vapor pressure make it unsuitable for a variable bypass vaporizer. A variable bypass vaporizer misfilled with desflurane could theoretically cause delivery

of a hypoxic mixture and a massive overdose of inhaled desflurane.

- The major advantage of the circle breathing system is the capability to rebreathe exhaled gas, including volatile anesthetic. The major disadvantage is its complex design requiring multiple valves.
- Before an anesthetic agent is administered, the circle system must be checked both to rule out leaks and to verify flow.
- Increasing the fresh gas flow rate into the circle breathing system causes less rebreathing of volatile anesthetic gas and more waste gas. To avoid rebreathing of carbon dioxide, a carbon dioxide absorbent is essential to the circle system's function.
- Older absorbent formulations with larger quantities of strong base catalyst had the potential to interact with inhaled anesthetic agents to produce potentially harmful degradation products. Carbon dioxide absorbents without strong bases such as potassium hydroxide or sodium hydroxide decrease this risk, allowing for low-flow anesthesia.
- The Mapleson breathing circuits are simple, lightweight breathing systems that support both spontaneous and manual ventilation. None are economical for volatile anesthetic use, as they do not readily support carbon dioxide absorbent use.
- Contemporary anesthesia ventilators support a wide variety of ventilation modes similar to intensive care unit ventilators. Anesthesia ventilators differ from intensive care unit ventilators in that they must support the rebreathing of exhaled gases. Types of anesthesia ventilators include bellows, piston, volume reflector, and blower. Each design has its own benefits and limitations.
- For bellows-type anesthesia ventilators, ascending bellows (bellows that ascend during the expiratory phase) are safer than descending bellows (bellows that descend during the expiratory phase) because disconnections are readily manifested by failure of ascending bellows to refill.
- Piston ventilators can potentially draw room air into the breathing circuit if a leak is present. The volume reflector ventilator compensates for leaks with 100% oxygen. Both are susceptible to lower-than-expected levels of inhaled anesthetic.
- Anesthesia providers should know how their machines react to changes in fresh gas flow. On older anesthesia machines, the portion of fresh gas flow that occurs during inspiration is added to the tidal volume. Therefore, increased fresh gas flow leads to increased tidal volume and increased airway pressure during positive-pressure ventilation. Newer-generation anesthesia workstations either decouple the fresh gas flow from the inspired tidal volume or compensate for the fresh gas flow in calculating the amount of gas to deliver as tidal volume.
- The anesthesia gas scavenging system protects the operating room from waste anesthesia gases. Active systems, which apply vacuum suction to the scavenge system, are most common in contemporary operating rooms. Obstruction of, or inadequate vacuum to, the scavenging system transfer tubing can result in increased breathing circuit pressure or discharge of waste anesthesia gases to the operating room depending on the design.
- Volatile anesthetic agents are greenhouse gases. Desflurane and nitrous oxide have higher greenhouse potential than other agents in use. Strategies for reducing the environmental footprint of inhaled anesthesia include eschewing desflurane and nitrous oxide, keeping fresh gas flows low, and considering alternatives to inhalational anesthesia.
- The American Society of Anesthesiologists (ASA) Recommendations for Pre-Anesthesia Checkout Procedures (2008) serves as an excellent template for the creation of machine-specific checkout procedures. However, it is not a one-size-fits-all checklist for all the commercially available

anesthesia machines.

Source: Miller 10e, Chapter 20, Key Points, PDF pp. 559–560.

Chapter 21: Intravenous Anesthetics

- The introduction of thiopental into clinical practice in 1934 marked the beginning of modern intravenous (IV) anesthesia. Today, a wide variety of IV anesthetics, analgesics, and hypnotic agents are used for induction and maintenance of anesthesia and sedation in a wide variety of circumstances.
- The most commonly used IV anesthetic is propofol, an alkylphenol presently formulated in a lipid emulsion. Propofol provides rapid onset and offset with context-sensitive decrement times of approximately 10 minutes when infused for less than 3 hours and less than 40 minutes when infused for up to 8 hours. Its mechanism of action is likely the enhancement of gamma-aminobutyric acid (GABA)-induced chloride currents. Propofol causes a dose-dependent decrease in arterial blood pressure predominantly through a decrease in systemic vascular resistance and causes moderate respiratory depression. A unique action of propofol is its antiemetic effect, even at concentrations less than those producing sedation.
- Barbiturates were the most commonly used IV drugs administered to induce anesthesia prior to the introduction of propofol. With the introduction of propofol, barbiturate use has essentially been eliminated as a routine anesthetic agent. Thiopental provides rapid onset and offset when used as a single dose, but it accumulates rapidly with repeated or prolonged administration, thus postponing recovery from anesthesia. Methohexital has a rapid onset and offset similar to propofol for procedures lasting less than 2 hours. The barbiturates are administered as sodium salts diluted in a water base at an alkaline pH. Similar to propofol, the barbiturates provide their hypnotic effects largely through action on the GABA receptor. Barbiturates provide cerebral protection and are, apart from induction of anesthesia, used primarily for this purpose. They cause a moderate dose-dependent decrease in arterial blood pressure (primarily because of peripheral vasodilation) and respiratory drive. The barbiturates are contraindicated in patients with porphyria.
- Benzodiazepines are used primarily for anxiolysis and amnesia or for conscious sedation. The water-soluble benzodiazepine midazolam is most frequently used intravenously as part of anesthetic management because of its rapid onset and offset compared with other benzodiazepines (e.g., diazepam). The onset time of midazolam is slower than that of propofol and barbiturates, and its offset, especially with larger doses or a prolonged infusion, is considerably longer than that of propofol or methohexital and may be prolonged in hepatic and renal failure. The benzodiazepines act through the GABA receptor. Flumazenil is a specific benzodiazepine antagonist. It can be used to reverse the effects of benzodiazepines but should be used with caution because its antagonizing effect often lasts shorter than the benzodiazepine effect that it is supposed to antagonize. The benzodiazepines generally produce only a mild decrease in arterial blood pressure and mild to moderate respiratory depression. Remimazolam is the most recently introduced benzodiazepine with an ultra-short duration of action due to its rapid clearance through plasma esterases.
- Ketamine is a phencyclidine derivative that acts primarily, but not entirely, as antagonist of the N-methyl-D-aspartate (NMDA) receptor. It produces a dissociative state of hypnosis and analgesia. It has been used for induction and maintenance of anesthesia. Ketamine is associated with significant adverse psychological effects from larger doses and has several other side effects. It is used now primarily for its analgesic properties. It has rapid onset and relatively rapid offset, even after an infusion of several hours. It has sympathomimetic effects that preserve cardiac function. Ketamine has minimal effect on respiration and tends to preserve autonomic reflexes. In addition, recent studies have demonstrated that ketamine may play a role as an antidepressant.

- Etomidate is an imidazole derivative used primarily for induction of anesthesia, especially in elderly patients and patients who have cardiovascular instability. It has a rapid onset of effect and a rapid offset even after a continuous infusion. One disadvantage of etomidate is that a single dose used to induce anesthesia results in inhibition of adrenocortical synthesis and possible increased mortality when administered to intensive care unit (ICU) patients. The major advantage of etomidate is its minimal effect on the cardiovascular and respiratory systems.
- Dexmedetomidine has become a commonly used IV anesthetic. It is a highly selective α_2 -adrenergic agonist that produces sedation, sympatholysis, hypnosis, and analgesia. Dexmedetomidine is also approved for ICU sedation of initially intubated and mechanically ventilated patients, up to 24 hours. It can be administered for procedural sedation in non-intubated patients, including when undergoing invasive or radiologic procedures. In addition, it might have benefit in preventing delirium. Its primary action is as an agonist on α_2 -receptors in the locus coeruleus. It has minimal effect on respiration. Heart rate and cardiac output show a concentration-dependent decrease.
- Droperidol, a butyrophenone and major tranquilizer, has been used to produce a state of neuroleptanesthesia previously, but is now rarely administered due to potential side effects, including prolongation of the QT interval. As a result, droperidol has been withdrawn in several countries. In the US, it is accompanied by a black box warning; its use is limited to the treatment of postoperative nausea and vomiting (PONV); low-dose droperidol continues to be used for antiemetic therapy in many European countries (also see Chapter 76).

Source: Miller 10e, Chapter 21, Key Points, PDF pp. 619–620.

Chapter 22: Opioids

- Opioids are a vital part of the analgesic component of anesthesia and often form the foundation for postoperative pain management.
- Opioids suppress pain by targeting multiple sites throughout the nervous system including action in brain, spinal cord, and peripheral nervous systems.
- An increased understanding of the molecular pharmacology of opioid receptors and opioid-induced cellular responses allows utilization of innovative techniques for analgesia and development of novel opioid analgesics with fewer side effects.
- Opioids also affect multiple organ systems, including the respiratory and cardiovascular systems, and can cause a variety of adverse effects. Proper dosing and monitoring may allow these adverse effects to be minimized. Understanding of the mechanism of the adverse effects will contribute to better management of patients.
- Pharmacokinetic and pharmacodynamic properties of opioids are affected by a variety of factors, such as age, body weight, organ failure, shock, and drug interactions. To properly administer opioids, these factors should be taken into consideration.
- Although new opioid delivery systems such as transdermal patches provide certain clinical advantages, they can also pose additional risks such as respiratory depression.
- Opioid analgesics serve a critical role in acute pain management but their role in the long-term treatment of chronic non-cancer pain has been called into question due to increased risks of overdose, addiction, and opioid use disorders.

Source: Miller 10e, Chapter 22, Key Points, PDF p. 672.

Chapter 23: Intravenous Drug Delivery Systems

- The pharmacokinetics of anesthetic drugs are described by multicompartment models. Accurate intravenous drug delivery requires adjusting the maintenance infusion rates to consider the accumulation of the drug in the peripheral tissues.
- Biophase is the site of action of a drug. Initiation, maintenance, and titration of intravenous anesthetics must account for the delay in equilibration between plasma and the site of drug effect.
- Some drug effects directly reflect the concentration of the drug in the biophase (direct-effect models). Other drug effects reflect the alteration of feedback systems by anesthetics (indirect-effect models). The influence of opioids on ventilation reflects the dynamic influence of opioids on the feedback between ventilation and carbon dioxide and is thus an example of an indirect drug effect.
- The target concentration in the effect site is the same as the target concentration in plasma at steady state. Effect-site requirements are influenced by patient physiologic characteristics, surgical stimulation, and concurrent drug administration. Ideally, target concentrations should be set for the hypnotic (volatile anesthetic or propofol) and the analgesic (opioid) that properly accounts for the synergy between them.
- To achieve an effective target concentration, the conventional teaching of administering an initial dose as calculated by the product of target concentration and volume of distribution, followed by a maintenance rate as calculated as the product of target concentration and clearance, is inaccurate. The initial dose may be calculated as the product of target concentration and volume of distribution at peak effect. Maintenance rates must initially account for the distribution of drug in peripheral tissues and should only be reduced to the product of target concentration and clearance after equilibration of plasma and peripheral tissue concentrations.
- The terminal half-life does not reflect the clinical time course of drug plasma concentration. The context-sensitive decrement time is the time for a given decrement in drug concentration, as a function of the duration of infusion that maintains a steady plasma concentration. Context-sensitive decrement times properly incorporate the multicompartment behavior of intravenous anesthetics. The context-sensitive half-time is the time for a 50% decrement in concentration.
- Alfentanil, fentanyl, sufentanil, remifentanyl, propofol, thiopental, methohexital, etomidate, ketamine, midazolam, dexmedetomidine, and remimazolam can all be administered as a continuous intravenous infusion. Specific caveats, infusion rates, and titration guidelines are presented in this text.
- Target-controlled infusions (TCIs) use pharmacokinetic models to determine intravenous anesthetic administration rates required to achieve specified plasma or effect-site drug concentrations. Various plasma and effect-site targeting TCI systems are commercially available worldwide (except within the United States) to administer hypnotics and opioids.
- Closed-loop drug delivery systems have used the median electroencephalographic frequency, bispectral index (BIS), or auditory-evoked potentials to control intravenous anesthetic delivery. Although these systems have generally performed well clinically, they remain under investigation.

Source: Miller 10e, Chapter 23, Key Points, PDF p. 754.

Chapter 24: Pharmacology of Neuromuscular Blocking Drugs and Antagonists (Reversal Agents)

Neuromuscular Blocking Agents

- Two different populations of nicotinic acetylcholine receptors exist at the mammalian neuromuscular junction. In the adult, the nicotinic acetylcholine receptor at the postsynaptic (muscular) membrane is composed of $\alpha 2\beta\delta\epsilon$ subunits, while the fetal (immature) receptor is composed of $\alpha 2\beta\gamma\delta$. The presynaptic (neuronal) nicotinic receptor is a pentameric complex composed of $\alpha 3\beta 2$ subunits. Each of the two α subunits of the postsynaptic receptors has a ligand (acetylcholine) binding site.
- Nondepolarizing neuromuscular blocking drugs produce neuromuscular block by competing with acetylcholine for the postsynaptic α subunits. In contrast, succinylcholine acts directly with the recognition sites and produces prolonged depolarization that results in decreased sensitivity of the postsynaptic nicotinic acetylcholine receptor and inactivation of sodium channels so that propagation of the action potential across the muscle membrane is inhibited.
- Different patterns of stimulation examine neuromuscular block at different areas of the motor end plate. Depression of the response to single-twitch stimulation is likely caused by block of postsynaptic nicotinic acetylcholine receptors, whereas fade in response to tetanic and train-of-four stimuli results mainly from block of presynaptic nicotinic receptors.
- Succinylcholine is the only available depolarizing neuromuscular blocking drug for clinical use. It is characterized by rapid onset of effect and ultra-short duration of action because of its rapid hydrolysis by butyrylcholinesterase.
- Available nondepolarizing neuromuscular blocking drugs can be classified according to chemical class (aminosteroid, benzylisoquinolinium, or other compounds) or by duration of action (long-, intermediate-, and short-acting drugs) of equipotent doses.
- The speed of onset is inversely proportional to the potency of nondepolarizing neuromuscular blocking drugs. Except for atracurium, molar potency is highly predictive of a drug's rate of onset of effect. Rocuronium has a molar potency that is approximately 13% that of vecuronium and 9% that of cisatracurium. Its onset of effect is more rapid than either of these neuromuscular blocking drugs.
- Neuromuscular block develops faster, lasts a shorter time, and recovers faster in the more centrally located neuromuscular units (e.g., laryngeal adductors, diaphragm, and masseter muscle) than in the more peripherally located adductor pollicis muscle.
- Many long-acting neuromuscular blocking drugs undergo minimal or no metabolism, and they are primarily eliminated, largely unchanged, by renal excretion. Neuromuscular blocking drugs of intermediate duration of action have faster distribution and more rapid clearances than the long-acting drugs because of multiple pathways of degradation, metabolism, and elimination. Mivacurium, a short-acting neuromuscular blocking drug, is cleared rapidly and almost exclusively by metabolism by butyrylcholinesterase.
- After the administration of nondepolarizing neuromuscular blocking drugs, it is essential to ensure adequate return of normal neuromuscular function using objective (quantitative) means of monitoring. Residual neuromuscular paralysis decreases upper esophageal tone, coordination of the esophageal musculature during swallowing, and hypoxic ventilatory drive. Residual paralysis can increase health care costs and the patient hospital length of stay, morbidity, and mortality.

Antagonism

- Appropriate antagonism of a nondepolarizing neuromuscular block is essential to avoid adverse patient outcomes. Adequate recovery of muscle strength should be present, and the residual effects of neuromuscular blocking drugs should be antagonized pharmacologically or undergo spontaneous recovery.
- Sufficient recovery from neuromuscular block for tracheal extubation can be confirmed by an adductor pollicis train-of-four ratio of at least 0.90 (or 1.0 if acceleromyography is used). Quantitative neuromuscular monitoring is the only method of assessing whether a safe level of recovery of muscular function has occurred.
- Residual neuromuscular block is not a rare event in the postanesthesia care unit, particularly after neostigmine antagonism: approximately 30% to 50% of patients can have train-of-four ratios less than 0.90 following tracheal extubation in the absence of quantitative monitoring. Occurrence of residual block after sugammadex is less common, but still present.
- Patients with train-of-four ratios less than 0.90 in the postanesthesia care unit are at increased risk of critical respiratory events: hypoxemic events, impaired control of breathing during hypoxia, airway obstruction, postoperative pulmonary complications, symptoms of muscle weakness, and prolonged postanesthesia care unit and hospital stay. Appropriate management of neuromuscular block based on quantitative monitoring will eliminate residual block and reduce the risk of postoperative complications.
- Neostigmine, pyridostigmine, and edrophonium inhibit the breakdown of acetylcholine, resulting in an increase in acetylcholine in the neuromuscular junction. However, there is a "ceiling" effect to the maximal concentration of acetylcholine that can be achieved with these drugs. Neostigmine in the dose range of 30 to 50 mcg/kg body weight antagonizes shallow to minimal levels of neuromuscular block.
- Sugammadex is a modified γ -cyclodextrin that shows a high affinity for the steroidal neuromuscular blocking drugs rocuronium and vecuronium (and pipecuronium). Sugammadex forms a tight inclusion complex with these steroidal neuromuscular blocking drugs, thereby inactivating their effects and facilitating rapid reversal of neuromuscular block.
- Sugammadex reverses both moderate and deep neuromuscular block with a dose of 2.0 mg/kg and 4.0 mg/kg, respectively. An immediate reversal of neuromuscular block induced by rocuronium (given in doses up to 1.2 mg/kg) is possible when using a dose of sugammadex of 16 mg/kg. Reversal of neuromuscular block by sugammadex is frequently, but not always, rapid.

Source: Miller 10e, Chapter 24, Key Points, PDF pp. 814–815.

Chapter 25: Local Anesthetics

- Local anesthetics block voltage-gated sodium channels and thereby interrupt initiation and propagation of impulses in axons, but they have a wide variety of other biologic actions, desirable and undesirable.
- Currently available local anesthetics are of two chemical classes: aminoesters and aminoamides.
- The low potency and lack of specificity of available local anesthetics are due in part to the very weak structural constraints at their binding site on the sodium channel. Most of their features derive from the requirement for high solubility, rapidly reversible protonation, and rapid diffusion in both aqueous environments and in the lipid phases of biologic membranes.
- Reversible protonation of the tertiary amine group tends to make local anesthetics less charged at more basic pH and more charged at neutral or acidic pH; the neutral base forms are more soluble in lipid environments, whereas the charged acid forms, the more potent species, are more soluble in aqueous environments.
- Aminoesters are metabolized primarily by plasma esterases, and aminoamides are metabolized primarily by hepatic cytochrome P450-linked enzymes.
- The principal systemic toxicities of local anesthetics involve the heart (including atrioventricular conduction block, arrhythmias, myocardial depression, and cardiac arrest) and the brain (including agitation, lethargy, seizures, and generalized central nervous system depression). Hypoxemia and acidosis exacerbate these toxicities. Resuscitation after bupivacaine overdose is particularly difficult. Therefore, prevention of intravascular injection or overdose is crucial, and major nerve blockade should involve incremental, fractionated dosing. Ultrasound can reduce the incidence of systemic toxicity and Intralipid can help in the resuscitation.
- Local anesthetics are directly toxic to nerve at the concentrations supplied in commercial solutions. Intra-neural concentrations during regional anesthesia are generally (but not always) below a threshold for toxicity because of spread of solutions through tissues and diffusion gradients from extraneural injection sites into nerve. Injection into a constrained tissue space increases the risk for local toxicity.
- Optimal use of local anesthetics in regional anesthesia requires an understanding of: (1) the individual patient's clinical situation; (2) the location, intensity, and duration of regional anesthesia and analgesia required; (3) anatomic factors affecting deposition of drug near nerves; (4) proper drug selection and dosing; and (5) ongoing assessment of clinical effects after administration of a local anesthetic.
- Single-stereoisomer formulations (as opposed to a racemic mixtures) have been developed in an effort to reduce systemic toxicity and improve sensory selectivity but true sensory selectivity has not been achieved with drugs currently available.
- The most important research avenues to improve local anesthetic action are slow-release formulations, targeting of specific sodium channel subtypes, and targeting of nociceptive fibers.

Source: Miller 10e, Chapter 25, Key Points, PDF p. 895.

Chapter 26: Immune Implications of Anesthesia Care and Practice

- Anesthetic agents and opiates have been reported to have multiple effects on immune cell functions. The specific effects vary widely between studies and between cells and individual agents.
- The available data suggest that, in some cases, the anesthetic approach may affect immune function and outcomes in the operative and perioperative settings.
- There is abundant evidence that surgery and tissue damage have substantially more impact than anesthetic agents on the functional immune outcomes of patients.
- Evidence is lacking in integrated systems (e.g., animal models, human studies, in silico), to recommend one anesthetic over another to protect or bolster immune function.
- Further advances in artificial intelligence and machine learning are anticipated to deepen the understanding of the relationship between various practices and patient immune outcomes. This may eventually support tailoring anesthetics and using immune-targeted precision medicine to improve outcomes of surgical and critical care patients.

Source: Miller 10e, Chapter 26, Key Points, PDF p. 925.

Chapter 27: Risk of Anesthesia

- Perioperative risk is multifactorial and may occur as a result of anesthesia-, surgery-, and/or patient-specific factors.
- Anesthesia-related (and surgery-related) risk is typically defined as morbidity and mortality occurring within 30 days of surgery, although events that occur at later points may still be related to anesthesia and/or surgery.
- The overall risk of anesthesia relates to both specific, organ-based complications and the rapidity with which they are managed (i.e., rescued).
- In the literature on anesthesia-related risk, the rates of morbidity and mortality reported across studies show a substantial variability in part attributable to the wide variety of definitions used in these studies.
- Historical studies of anesthesia-related risk identified anesthesia-related respiratory depression as the major cause of death and coma totally attributable to anesthesia. This finding prompted the creation of postanesthesia care units (PACUs).
- Research into anesthesia-related cardiac arrest has found it to be attributable to medication administration, airway management, and technical problems of central venous access.
- Multivariate modeling can be used to determine specific factors associated with an increased likelihood of adverse postoperative events, and it has been used to define a range of clinical risk indices to predict postoperative outcomes.
- Surveys of maternal mortality suggest that although the absolute rate of complications attributable to anesthesia has not decreased over time, the increased use of regional anesthesia may have led to improvements in outcome.
- Medication-related and cardiovascular events were the most common causes of cardiac arrest in the Pediatric Perioperative Cardiac Arrest (POCA) Registry.
- Growth in the number and variety of surgical procedures performed in hospital outpatient departments, ambulatory surgery centers, and physician offices creates novel challenges for assessing and managing perioperative risk.
- Initiatives established over time by the Anesthesia Patient Safety Foundation, the American Society of Anesthesiologists (ASA), and others have sought to decrease the potential risks of anesthesia through systems-level improvements, standardization of care processes, human-factors engineering, and simulation-based training.
- Emerging evidence suggests that the choice of anesthetic drugs, ventilator strategies, or technique may impact patient outcomes.

Source: Miller 10e, Chapter 27, Key Points, PDF p. 939.

Chapter 28: Preoperative Evaluation

- The anesthesia preoperative evaluation, which is the clinical foundation for guiding perioperative patient management, reduces perioperative morbidity and enhances patient outcome.
- The fundamental purpose of preoperative evaluation is to obtain pertinent information regarding the patient's medical history, formulate an assessment of the patient's perioperative risk, and develop a plan for any clinical optimization.
- The pre-anesthesia evaluation should include a focused clinical examination, documentation of comorbid illness, reduction of patients' anxiety through education, assurance that preexisting medical conditions are optimally managed, selective referrals to medical specialists, ordering of preoperative investigations, initiation of interventions intended to decrease risk, discussion of aspects of perioperative care, arrangements for appropriate postoperative care, and recommendations to delay or cancel the surgical procedure, if deemed appropriate.
- Comorbid medical diseases can influence anesthesia and perioperative management, thus requiring the anesthesiologist to be knowledgeable in many aspects of internal medicine.
- Patients require appropriate preoperative diagnostic and laboratory studies consistent with their medical history and the proposed surgical procedure. Arbitrary routine preoperative testing cannot be justified because it is costly and often clinically inappropriate.
- The anesthesiologist-led preoperative evaluation outpatient clinic can enhance operating room efficiency, decrease day-of-surgery cancellations or delays, reduce hospital costs, and enhance the quality of patient care.
- New and updated preoperative evaluation consensus and evidence-based guidelines published by multiple medical specialties have important influences on the preparation of patients for anesthesia and surgery.
- Anesthesiologists must be aware of, and be compliant with, increasing regulatory and reporting requirements involving preoperative issues by health care agencies.
- As the perioperative medical specialist, the anesthesiologist is uniquely positioned to evaluate the risks associated with anesthesia or surgery, discuss these risks with the patient, and manage them perioperatively in collaboration with the surgical team, referring physician, and possibly other medical specialists.

Source: Miller 10e, Chapter 28, Key Points, PDF p. 970.

Chapter 29: Anesthetic Implications of Concurrent Diseases

- Comorbidities increase the risks of anesthesia and the need for alterations in monitoring and management.
- Diseases need to be optimized preoperatively.
- Adverse outcomes increase with major surgeries.
- Evaluation of patients preoperatively assists planning of anesthesia and postoperative care.
- Cardiac complications are common in surgical patients. Heart failure and ischemic disease pose some of the highest risks.
- End-organ dysfunction and the degree of glucose control in diabetic patients are important.
- In patients with pulmonary disease, the following are assessed: dyspnea, coughing and the production of sputum, recent respiratory infection, hemoptysis, wheezing, previous pulmonary complications, smoking history, and physical findings.
- Risk factors for perioperative renal dysfunction include advanced age, chronic kidney disease, heart failure, ischemic heart disease, and diabetes
- Anemia is associated with increased risk and should be evaluated and treated in patients anticipating surgery.
- Adrenergic receptor blocking drugs are generally recommended in patients undergoing resection of pheochromocytomas.

Source: Miller 10e, Chapter 29, Key Points, PDF p. 1048.

Chapter 30: Patient Positioning and Associated Risks

- Patient positioning is a major responsibility that requires the cooperation of the entire surgical team.
- Many patient positions that are used for surgery result in undesirable physiologic consequences including significant cardiovascular and respiratory compromise. Anesthetic agents blunt natural compensatory mechanisms, exaggerating physiologic consequences to positional changes.
- Peripheral nerve injuries, although rare, represent 22% of cases in the 1990 to 2007 American Society of Anesthesiologists' Closed Claims Project. The mechanisms of injury are stretching, compression, and ischemia. Patient positioning is often a suspected contributory factor, although precautions have usually been taken and no specific cause for the injury is known.
- The American Society of Anesthesiologists first issued a Practice Advisory in 2000 for the prevention of perioperative peripheral neuropathies that was updated in 2019. However, very few of the studies reviewed met the standard for a scientifically proven relationship between intervention and outcome.
- Anesthesia administered outside the operating room presents special challenges with regard to patient positioning because of monitoring and equipment limitations and differences in the work environment and culture.
- Perioperative visual loss (POVL) is a rare but serious injury that appears more frequently after cardiac, spine, and orthopedic joint surgery.
- Causes of POVL include central or branch retinal artery occlusion, anterior and posterior ischemic optic neuropathy, cortical blindness, acute glaucoma, and acute expansion of gas bubbles placed in the eye in retinal surgery.
- Signs and symptoms of visual loss in the postoperative period may be subtle and can be incorrectly attributed to the residual effects of anesthetic drugs. Any patient reporting eye pain, an inability to perceive light or motion, complete or partial loss of visual fields, decreased visual acuity, or loss of pupil reactivity must be evaluated immediately by an ophthalmologist.
- The most common cause of perioperative central and branch retinal artery occlusion is compression of the eye. During cardiac surgery, emboli may occlude the retinal arteries.
- Patients who undergo prolonged operative procedures in the prone position with large blood loss are at increased risk for development of ischemic optic neuropathy. Other factors conferring a risk during spine surgery include male sex, obesity, the use of a Wilson frame, and intravascular fluids administered perioperatively.
- Patients should be informed of the risk for visual loss accompanying lengthy surgical procedures with the patient positioned prone and with anticipated large blood loss. Both anesthesia and surgery personnel, together, should develop a plan by which informed consent for this complication may be facilitated.

Source: Miller 10e, Chapter 30, Key Points, PDF p. 1112.

Chapter 31: Neuromuscular Disorders and Other Genetic Disorders

- Malignant hyperthermia (MH) is a pharmacogenetic disorder inherited primarily in an autosomal dominant pattern.
- MH susceptibility (MHS) is linked to 230 mutations in the skeletal muscle ryanodine receptor (RyR1) and four mutations in the calcium voltage-gated channel subunit alpha1 S (CACNA1S) genes that encode two Ca^{2+} channels necessary for skeletal muscle excitation–contraction coupling.
- Physical interactions between L-type Ca^{2+} channel (Cav1.1) and RyR1 tightly regulate initiation and termination of skeletal muscle excitation–contraction coupling.
- Skeletal muscle accounts for approximately 40% of body weight and inherent changes in its metabolism have profound impacts on whole-body metabolism and physiology.
- Carriers of MH mutations can exhibit mild to moderate muscle impairments in the absence of triggering agents but are rarely diagnosed.
- Carriers of MH mutations are susceptible to anesthetic-triggered runaway skeletal muscle metabolism, which if not promptly treated is lethal.
- Signs of MH, including increased end-tidal CO_2 , increased core temperature, muscle rigidity, tachycardia, and more, are consequences of the fulminant hypermetabolic crisis.
- Exposure to triggering agents or heat stress leads to acute loss of RyR1/Cav1.1 channel regulation, rapid accumulation of Ca^{2+} within the sarcoplasm, and a hypermetabolic crisis that stimulates adenosine triphosphate (ATP) utilization by pumps attempting to restore resting Ca^{2+} balance among sarcoplasmic reticulum, mitochondrial, and extracellular compartments.
- Dantrolene markedly attenuates myoplasmic calcium (Ca^{2+}) concentrations and thereby restores resting Ca^{2+} balance and metabolism, with reversal of clinical signs.
- Evaluation of persons susceptible to MH includes an in vitro contracture test (IVCT) and caffeine/halothane contracture test (CHCT), and evaluation of DNA to identify mutations.
- Currently DNA testing alone can be used to evaluate 42 human mutations and all swine, equine, and canine MH.
- Future MH goals include advancement of genetic evaluations in North American and European medical programs and stronger finances to support genetic studies, the identification of the mode of action of dantrolene, a determination of the immediate cause of triggering MH, and the development of effective, noninvasive tests for MH susceptibility.
- The absence of mutations in dystrophin, along with dystrophin-associated glycoproteins, is involved in sarcolemmal stability. Its defects are responsible for Duchenne muscular dystrophy (DMD) and Becker muscular dystrophy (BMD).
- Whereas the risk for an MH mutation in DMD and BMD patients is like that in the general population, the incidence of MH-like anesthetic events has been reported to be 0.002 with DMD and 0.00036 with BMD.
- Succinylcholine is contraindicated in DMD and BMD patients because of the risk of rhabdomyolysis and hyperkalemia due to their unstable sarcolemmal membrane.

- Amyotrophic lateral sclerosis patients have exaggerated respiratory depression and sensitivity to sedatives and hypnotics and have higher risk for aspiration and pulmonary complications.
- For patients with fibrodysplasia ossificans progressive, awake nasal fiberoptic intubation is the first choice for airway management under non-emergent conditions, and intramuscular injection and regional anesthesia should be avoided as heterotopic ossification may exacerbate after muscle injury.
- Patients with limb-girdle muscular dystrophy are generally sensitive to opioids and muscle relaxants; short-acting anesthetic agents are preferred.
- Patients with glycogen storage disease type I should be monitored frequently to avoid hypoglycemia when they are fasted, and lactate-containing solutions should be avoided in these patients.
- Cardiomyopathy and cardiac failure are present in many patients with glycogen storage disease type II, and deterioration of diaphragmatic function can lead to postoperative acute respiratory failure in these patients.
- Anesthesia has a significant impact on mitochondrial function. For patients with mitochondrial diseases and mitochondrial myopathies, the implications of the patients' comorbidities and metabolic status are more important than the specific choice of anesthetic agents.
- Guillain-Barré syndrome is an immune-mediated acute polyradiculopathy. Autonomic dysfunction may lead to hemodynamic instability with an exaggerated response to anesthesia induction agents or stimulating interventions such as laryngoscopy.
- Prolonged immobility in critically ill patients is associated with a relative increase in immature AChRs that can lead to an insensitivity to nondepolarizing neuromuscular blocking agent (NMBA), while the sensitivity to depolarizing neuromuscular blockers is increased with risk for life-threatening succinylcholine-induced hyperkalemia. Succinylcholine is therefore contraindicated in this condition.
- Myasthenia gravis is a disorder of neuromuscular transmission caused by autoantibodies against the muscle-type AChRs. Because of the decreased number of functional AChRs, these patients are very sensitive to nondepolarizing NMBAs but resistant to succinylcholine.
- For myasthenia gravis patients, a preoperative vital capacity <2 L is thought to be associated with a higher risk of intensive care unit admission. Other risk factors include pyridostigmine dose >750 mg, preoperative bulbar symptoms, high anti-AChR antibody titers, prior history of myasthenic crisis, and an expected blood loss >1 L.
- Reversal of neuromuscular blockade with sugammadex is a practical alternative to the management of many of these disorders, if rocuronium or vecuronium is used. The combination of rocuronium and sugammadex has improved the anesthetic management for some of these challenging disorders.

Source: Miller 10e, Chapter 31, Key Points, PDF pp. 1145–1146.

Chapter 32: Cardiovascular Monitoring

- Monitoring of the electrocardiogram (ECG) provides continuous monitoring of heart rate, identification of arrhythmias and conduction abnormalities, and detection of myocardial ischemia.
- Bedside ECG monitors often apply a filter to the ECG signal that reduces low-frequency artifact, such as baseline wander, and high-frequency artifact, such as 60-Hz electromagnetic interference. However, such filters may distort or exaggerate ST-segment deviations or obscure pacemaker spikes.
- An anterolateral precordial lead (V3, V4, or V5) should be selected for the most sensitive detection of myocardial ischemia.
- Demand-mediated subendocardial ischemia resulting in ST-segment depression is the most commonly observed form of perioperative ischemia. ST-segment depression is most commonly observed in an anterolateral precordial lead regardless of the coronary territory responsible.
- Supply-mediated transmural ischemia resulting in ST-segment elevation is uncommonly observed intraoperatively except during cardiac operations. In contrast to ST-segment depression, ST-segment elevation is indicative of the myocardial territory and coronary artery involved.
- Most automated noninvasive arterial blood pressure measuring devices use an oscillometric measurement technique and rarely cause complications. Caution should be exercised in patients who cannot complain of arm pain, those with irregular rhythms that force repeated cuff inflation, and those receiving anticoagulant therapy.
- The Allen test for palmar arch collateral arterial flow is not a reliable method to predict complications from radial artery cannulation. Despite the absence of anatomic collateral flow at the elbow, brachial artery catheterization for perioperative blood pressure monitoring is a safe alternative to radial or femoral arterial catheterization.
- The accuracy of a directly recorded arterial pressure waveform is determined by the natural frequency and damping coefficient of the pressure monitoring system. Optimal dynamic response of the system will be achieved when the natural frequency is high, thereby allowing accurate pressure recording across a wide range of damping coefficients.
- The preferred position for alignment (or "leveling") of external pressure transducers for measuring arterial or central venous pressure (CVP), which eliminates confounding hydrostatic pressure artifacts, lies approximately 5 cm posterior to the sternomanubrial junction. The more conventional location for the reference level used for hemodynamic monitoring including central venous and pulmonary artery pressures, is the mid-thoracic level, which corresponds most closely to the mid-left atrial position and is located halfway between the anterior sternum and the bed surface in the supine patient.
- Because of wave reflection and other physical phenomena, the arterial blood pressure recorded from peripheral sites has a wider pulse pressure than when measured more centrally.
- Dynamic measures of cardiac preload, such as stroke volume and pulse pressure variation, are better predictors of intravascular volume responsiveness than static indicators, such as CVP and pulmonary artery wedge pressure.
- Selecting the best site, catheter, and method for safe and effective central venous cannulation requires that the physician consider the purpose of catheterization, the patient's underlying medical condition, the intended operation, and the skill and experience of the physician performing the procedure. Right internal jugular vein cannulation is preferred due to its

consistent, predictable anatomic location and its relative ease of access intraoperatively.

- Mechanical complications from central venous catheters can be decreased by the use of ultrasound vessel localization, venous pressure measurement before large-catheter insertion, and radiographic confirmation that the catheter tip lies outside the pericardium and parallel to the walls of the superior vena cava.
- CVP is the result of a complex and diverse interplay among many different physiologic variables, the main ones being venous return and cardiac function. No simple relationship exists between CVP and circulating blood volume. Despite this, important pathophysiologic information can be obtained by careful assessment of the CVP waveform morphology.
- Catheter misuse and data misinterpretation are among the most common complications of central venous and pulmonary artery catheters.
- Pulmonary artery wedge pressure is a delayed and damped reflection of left atrial pressure. The wedge pressure provides a close estimate for pulmonary capillary pressure in many cases, but it may underestimate capillary pressure when postcapillary pulmonary vascular resistance is increased, as in patients with sepsis.
- Use of central venous, pulmonary artery diastolic, or pulmonary artery wedge pressures as estimates of left ventricular preload is subject to many confounding factors, including changes in diastolic ventricular compliance and juxtacardiac pressure.
- Pulmonary artery catheter monitoring has not been shown to improve patient outcomes. Reasons cited for these results include misinterpretation of catheter-derived data and failure of hemodynamic therapies that are guided by specific hemodynamic indices.
- Thermodilution cardiac output monitoring, the most widely used clinical technique, is subject to measurement errors introduced by rapid intravenous fluid administration, intracardiac shunts, and tricuspid valve regurgitation.
- Mixed venous oxygen saturation is a measure of the adequacy of cardiac output relative to body oxygen requirements. This measurement is also dependent on the arterial oxygen saturation and hemoglobin concentration.
- All minimally invasive and noninvasive cardiac output monitors have their own sources of potential error. Awareness of potential sources of error and clinical limitations is key.

Source: Miller 10e, Chapter 32, Key Points, PDF pp. 1190–1191.

Chapter 33: Perioperative Echocardiography and Point-of-Care Ultrasound (POCUS)

- Perioperative applications of transesophageal echocardiography (TEE) include monitoring, diagnosis, and procedural guidance. The role of TEE in cardiac surgery, noncardiac surgery, interventional procedures, and critical care continues to evolve.
- In noncardiac surgery, intraoperative TEE serves as a routine monitor or as a rescue tool in life-threatening emergencies.
- TEE aids in decision-making during cardiac surgery, and three-dimensional (3D) TEE adds incremental value in intraoperative assessments.
- TEE is integral to structural heart interventions and is a key component of multimodality imaging.
- Echocardiography plays a vital role in the diagnosis and management of shock.
- Applications of transthoracic echocardiography (TTE) in the perioperative setting are expanding.
- Focused ultrasound is performed and interpreted at the point of care and addresses specific questions relevant to the clinical context. The application includes the perioperative period and ICU. It includes assessment of organ systems for diagnosis, treatment, and procedural guidance.
- Critical care societies have proposed distinct competencies and guidelines for critical care echocardiography beyond those included in perioperative TEE guidelines. The skill set of the intensivist performing echocardiography overlaps with that of the intraoperative echocardiographer and includes other unique components.
- Basic knowledge of echocardiography is an expectation of anesthesiology training.
- Simulation is an effective training tool when integrated within a multimodal, curriculum-based approach to echocardiography education.

Source: Miller 10e, Chapter 33, Key Points, PDF p. 1249.

Chapter 34: Implantable Cardiac Pulse Generators: Pacemakers and Cardioverter-Defibrillators

- Identify the type of cardiovascular implantable electronic device (CIED) (pacemaker, transvenous defibrillator, subcutaneous defibrillator), as well as the generator manufacturer and model of the CIED.
- Contact the physician or clinic managing the patient's CIED in the preoperative period to obtain appropriate records and a perioperative recommendation (Heart Rhythm Society [HRS]).
- Obtain a copy of this interrogation and the perioperative recommendation from the CIED physician (HRS). Ensure that the implantable cardioverter-defibrillator (ICD) treatment settings are appropriate and that the CIED will pace the heart.
- Consider replacing any CIED near its elective replacement period in a patient scheduled to undergo either a major surgical procedure or a surgical procedure where monopolar electrosurgery will be used within 15 cm of the generator (or leads).
- Determine the patient's underlying rate and rhythm to determine the need for backup (external) pacing support.
- Ensure that all magnet behavior (pacing, suspension of shock therapy) is appropriate if magnet use is planned.
- Program minute ventilation rate responsiveness "off," if present.
- Consider increasing the lower rate limit to provide optimal oxygen delivery for major procedures.
- If electromagnetic interference is likely, (1) disable antitachycardia therapy if a defibrillator is present and (2) consider asynchronous pacing for some pacing-dependent patients. Magnet application may be acceptable for some ICDs (disable antitachycardia therapy) or pacemakers (provide asynchronous pacing). Asynchronous pacing from an ICD always requires reprogramming.
- Monitor cardiac rhythm with the pulse oximeter (plethysmography) or arterial waveform analysis.
- Use a bipolar electrosurgical unit (ESU), if possible; if not possible, then pure "cut" electrosurgery is better than "blend" or "coag," and ESU should be applied in short bursts (<4 seconds) separated by at least 2 seconds.
- Place the ESU dispersive electrode in such a way to prevent electricity from crossing the generator–heart circuit.
- If the ESU causes ventricular oversensing with pacing quiescence or atrial oversensing with inappropriate ventricular pacing, limit the effect by using short ESU bursts, reducing the output of the ESU, relocating the dispersive electrode, or placing a magnet over the pacemaker (not indicated for ICDs).
- Some patients require postoperative interrogation, especially if preoperative reprogramming took place. For "low-risk cases," HRS states that this interrogation can take place in an ambulatory setting up to 1 month postoperatively. Some rate enhancements can be reinitiated, and a determination of optimum heart rate and pacing parameters should be made. Any patient with disabled antitachycardia therapy must be monitored until the antitachycardia therapy is restored.

Chapter 35: Neurophysiologic Monitoring

- The pathway at risk during the surgical procedure must be amenable to monitoring.
- The monitor must provide reliable and reproducible data.
- If evidence of injury to the pathway is detected, there must be some intervention possible.
- If changes in the neurophysiologic monitor are detected, and no intervention is possible, the monitor may only be of prognostic value to the current patient. Nonetheless, the monitor may identify surgical steps that may be altered in future surgeries for the benefit of those patients.
- For simple techniques, such as monitoring of the facial or recurrent laryngeal nerve, there is good evidence supporting their clinical use. For more complex monitoring approaches, such as multimodality monitoring involving dedicated technologists and neurophysiologists, evidence for the benefit of monitoring is sparser, reflecting the fact that effectiveness depends on the specific context and that tools for studying such complex interventions are still under development.
- Anesthesiologists can improve the efficacy of monitoring by maintaining good physiologic homeostasis and stable levels of anesthesia during parts of the surgical procedure that place the nervous system at greatest risk.
- Good communication between surgeon, anesthesiologist, and neurophysiologist is essential to optimizing the utility of monitoring.
- Based on clinical experience and nonrandomized studies, four practice patterns for use of neurologic monitoring have emerged:
 - There are procedures for which monitoring is recommended and used by most centers.
 - There are procedures for which monitoring is used frequently in some centers, but not in others.
 - There are procedures for which there is no clear clinical experience or evidence indicating that monitoring is useful.
 - There are procedures in which monitoring is used selectively for patients believed to be at higher-than-usual risk for intraoperative neurologic injury.

Source: Miller 10e, Chapter 35, Key Points, PDF p. 1320.

Chapter 36: Monitoring the Brain's Response to Anesthesia and Surgery

- The reasons for monitoring the brain's response to anesthesia are to aid and confirm optimal personalized precision drug delivery, and to prevent or detect adverse events such as unintended consciousness.
- In response to surgical nociceptive input, cardiovascular responses, somatic movement, and consciousness are controlled by different systems and are not closely correlated with each other.
- The spectrogram presents the totality of the frontal electroencephalogram (EEG) in one glance.
- An alpha-delta EEG pattern is the primary target for most anesthesia drug delivery because it indicates adequate drug-induced thalamocortical hyperpolarization in the face of surgical nociceptive stimulation.
- For the same EEG pattern, a somatic response to noxious stimulation is common with propofol anesthesia, less common with sevoflurane, and least common with the addition of an opioid.
- Use of EEG monitoring decreases intraoperative awareness, with recall by more than a half as compared to using clinical monitoring, but the advantage is much less when compared to end-tidal agent monitoring.
- EEG is very sensitive to all sorts of noise. Muscle (EMG) activity has a huge effect on all EEG indices, and is a signal that the patient is grimacing.
- Clinicians should be aware that EEG indices are less reliable in the elderly, who may have an acceptable level of anesthesia even at relatively high index values.
- To use EEG monitoring effectively, anesthesia providers need to fully understand its failings.

Source: Miller 10e, Chapter 36, Key Points, PDF p. 1359.

Chapter 37: Respiratory Monitoring

- Intraoperative respiratory monitoring is a fundamental component of the American Society of Anesthesiologists' standards for basic anesthetic monitoring. Monitoring of oxygenation and ventilation is essential for the safe conduct of an anesthetic.
- A thorough understanding of the physiologic and technologic principles underlying respiratory monitoring is essential for its appropriate clinical application.
- The majority of respiratory monitors in clinical use provide information at the systemic and whole-lung level from which inferences are made regarding the regional lung and tissue-level conditions.
- The degree of invasiveness of utilized monitors should be determined by clinical requirements.
- Pulse oximetry is a noninvasive, reliable, and simple method for continuously monitoring the fractional arterial oxygen saturation.
- Ventilation–perfusion mismatch, shunt, and hypoventilation are the most common causes of hypoxemia in the perioperative period. Monitoring of gas exchange, and its response to various interventions, may differentiate etiologies of hypoxemia.
- Mixed venous oxygen saturation (SvO_2) allows for monitoring of the global balance between oxygen delivery and oxygen consumption. Its measurement provides information on gas exchange, cardiac output, and global oxygen consumption.
- Systems utilizing near-infrared spectroscopy are used clinically to monitor regional tissue oxygenation, particularly in the brain. The value of regional tissue oxygenation monitoring for clinical management is currently being established.
- Capnography is the primary quantitative method to assess ventilation in the perioperative period. Besides providing physiologic information on ventilation, pulmonary blood flow, and aerobic metabolism, capnography is important for verifying the endotracheal tube positioning, and determining the integrity of the breathing circuit.
- End-tidal carbon dioxide (CO_2) is not always a reliable approximation of arterial CO_2 tension, especially in the presence of significant heterogeneity in the distribution of ventilation and perfusion.
- Measurement of the pressures, flows, and volumes associated with ventilation is necessary to optimize mechanical ventilation, as well as to detect pathophysiologic mechanical derangements of the respiratory system (i.e., increased airway resistance or reduced lung compliance).
- Imaging techniques have emerged as important tools for respiratory monitoring. Lung ultrasonography is increasingly utilized in emergency and perioperative settings, allowing for prompt bedside assessments of pulmonary abnormalities such as pneumothorax, lung edema, consolidation, and pleural effusions. Electrical impedance tomography is another noninvasive imaging technique that provides information on lung aeration and recruitment.
- Current approaches to respiratory monitoring primarily assess pulmonary mechanical and global gas exchange processes. The monitoring of tissue and subcellular respiration remains a desirable goal for future innovation.

Source: Miller 10e, Chapter 37, Key Points, PDF p. 1383.

Chapter 38: Renal Pathophysiology and Treatment for Perioperative Ischemia and Nephrotoxic Injury

- The incidence of perioperative acute kidney injury (AKI) (previously referred to as acute renal failure) varies, depending on the definition used. Although uncommon, AKI requiring dialysis is associated with extremely high morbidity and mortality rates.
- The mechanism for perioperative AKI is complex and commonly involves multiple factors including ischemia–reperfusion injury, inflammation, and toxins.
- Repeated direct perioperative assessments of renal hemodynamics or tubular function are impractical; therefore, indirect assessments, such as trends of serum creatinine concentrations, are the best practical perioperative tools to assess renal function.
- Intraoperative urine formation depends on many factors and is not validated as a measure of the risk of postoperative renal dysfunction. Yet postoperatively, patients with low intraoperative urine output may develop renal dysfunction. Therefore, urine output should be carefully monitored in the intraoperative setting.
- Early biochemical markers for kidney injury remain in development; the role of tests such as cystatin C and IGFBP7*TIMP-2 in the clinical management of perioperative patients is an area of active interest and investigation.
- As part of preoperative risk assessment, both serum creatinine and proteinuria/albuminuria can provide important and useful information.
- Intraoperative hypotension and hypovolemia are significant risk factors for AKI.
- With regard to fluid management, the use of balanced salt solutions avoids hyperchloremic metabolic acidosis and is likely preferable in most circumstances, including in patients with end-stage kidney disease. Balanced salt solutions may be associated with lower rates of AKI, though this association is at best modest.
- Volume overload is a risk factor for adverse outcomes in patients with AKI, but caution needs to be applied to the use of goal-directed therapy algorithms, as restrictive fluid protocols have recently been associated with an increased risk of AKI.
- Renal replacement therapy may be indicated for severe AKI: at present, data do not support the use of one modality over another, nor the use of early dialysis.

Source: Miller 10e, Chapter 38, Key Points, PDF p. 1436.

Chapter 39: Neuromuscular Monitoring

- Good evidence-based practice dictates that clinicians always quantitate the extent of neuromuscular block by objective monitoring.
- The neuromuscular block should be adjusted to ensure optimal surgical conditions. In most procedures, one or two responses to train-of-four (TOF) stimulation will suffice. To avoid involuntary diaphragmatic movements, a deeper level of neuromuscular block is required (i.e., one to five responses to posttetanic count [PTC]).
- Adequate recovery of postoperative neuromuscular function cannot be guaranteed without objective neuromuscular monitoring.
- Objective neuromuscular monitoring is essential for management of neuromuscular blockade intraoperatively and its reversal for postoperative care. Neuromuscular blocking agents should not be given in the intensive care unit without proper monitoring.
- It is impossible to exclude with certainty clinically significant residual neuromuscular block by clinical evaluation of recovery of neuromuscular function.
- Residual postoperative neuromuscular block causes decreased chemoreceptor sensitivity to hypoxia, functional impairment of the pharyngeal and upper esophageal muscles, impaired ability to maintain an open upper airway, and an increased risk of hypoxemic events, as well as the development of postoperative pulmonary complications.
- Absence of tactile fade in the response to TOF stimulation, tetanic stimulation, and double-burst stimulation does not exclude significant residual block.
- To exclude clinically significant residual neuromuscular block, the TOF ratio must exceed 0.9 when measured mechanically or electromyographically and 1.0 when measured acceleromyographically.
- If adequate recovery (TOF ≥ 0.9 –1.0) has not been documented objectively at the end of the surgical procedure, the neuromuscular block should be antagonized.
- Antagonism of the neuromuscular block with a cholinesterase inhibitor should not be initiated before at least all four responses to TOF stimulation are observed.
- Antagonism of the neuromuscular block achieved by rocuronium and vecuronium can be initiated at all levels of block with the selective relaxant binding agent sugammadex.

Source: Miller 10e, Chapter 39, Key Points, PDF p. 1454.

Chapter 40: Airway Management in the Adult

- One of the fundamental responsibilities of the anesthesiologist is to mitigate the adverse effects of anesthesia on the respiratory system by maintaining airway patency and ensuring adequate ventilation and oxygenation. The term airway management refers to this practice and is a cornerstone of anesthesia.
- Successful airway management requires a range of knowledge and skill sets – specifically, the ability to predict difficulty with airway management and to formulate an airway management strategy, as well as the skills to execute that strategy using the wide array of airway devices available.
- The American Society of Anesthesiologists' (ASA) Practice Guidelines for Management of the Difficult Airway and the accompanying difficult airway algorithm provide guidelines for evaluating the airway and preparing for difficult airway management and can guide clinical decision-making when an anesthesiologist is faced with a known or potentially difficult airway. Cognitive aids, such as the Vortex Approach and the new ASA Difficult Airway infographics that accompany the ASA adult and pediatric difficult airway algorithms, are useful to help implement airway algorithms in an emergency.
- A detailed understanding of airway anatomy is essential for successful airway management.
- A complete evaluation of the airway and knowledge of difficult airway predictors can alert the anesthesiologist to the potential for difficulty with airway management and allow for appropriate planning.
- Apneic oxygenation can be used to prolong the duration of apnea without desaturation and is increasingly being adopted during the management of both difficult and routine airways.
- Application of local anesthesia to the airway or induction of general anesthesia is usually required to facilitate airway management, to provide comfort for the patient, and to blunt airway reflexes and the hemodynamic response to airway instrumentation.
- Over the past 40 years, supraglottic airways (SGAs) have been one of the most important developments in airway devices.
- Tracheal intubation establishes a definitive airway, provides maximal protection against aspiration of gastric contents, and allows for positive-pressure ventilation with higher airway pressures than via a face mask or SGA.
- Flexible scope intubation of the trachea in an awake, spontaneously ventilating, and cooperative patient is the gold standard for the management of the anticipated difficult airway.
- Invasive airway interventions are indicated when attempts at establishing a noninvasive airway fail. The anesthesiologist should become proficient with techniques for cricothyrotomy.
- Extubation is a critical component of airway management with the potential for significant complications. Extubation of the trachea must be planned in advance and includes a strategy for reintubation should the patient be unable to maintain an adequate airway after extubation.

Source: Miller 10e, Chapter 40, Key Points, PDF p. 1476.

Chapter 41: Spinal, Epidural, and Caudal Anesthesia

- The distal termination of the spinal cord varies from L3 in infants to the lower border of L1 in adults.
- The speed of neural blockade depends on the size, surface area, and degree of myelination of the nerve fibers exposed to the local anesthetic.
- Blockade of the peripheral (T1–L2) and cardiac (T1–T4) sympathetic fibers is responsible for the decrease in arterial blood pressure (systemic vascular resistance and cardiac output) associated with neuraxial techniques.
- Cerebrospinal fluid volume and local anesthetic baricity are the most important determinants for the spread (i.e., block height) of spinal anesthesia.
- The use of smaller-gauge needles decreases the incidence of post-dural puncture headache.
- Serious neurologic complications associated with neuraxial blockade are rare, but the risk increases in the elderly and those with preexisting spinal pathology.
- The use of low-molecular-weight heparin and potent platelet inhibitors increases the risk of epidural hematoma from neuraxial blockade.
- Epidural blood patches are greater than 90% effective in relieving post-dural puncture headache.
- Local anesthetic systemic toxicity can result from unintentional administration of the drug into an epidural vein.
- Use of neuraxial blockade, particularly when used as the sole anesthetic, can reduce perioperative morbidity and may reduce mortality.

Source: Miller 10e, Chapter 41, Key Points, PDF p. 1523.

Chapter 42: Peripheral Nerve Blocks and Ultrasound Guidance for Regional Anesthesia

- Regional anesthesia is only successful when local anesthetic is administered or injected close to the targeted nerve/s. From the inception of regional anesthesia several techniques have been used and are available to facilitate the correct placement of local anesthetic, including the paresthesia-seeking approach, peripheral nerve stimulator, and ultrasound guidance.
- There are no data to support the superiority of one nerve localization technique—paresthesia, nerve stimulation, ultrasound—over another regarding reducing the risk of nerve injury.
- Ultrasound imaging can elucidate the structure of peripheral nerves and adjacent anatomic structures for regional block. Peripheral nerves have a characteristic honeycomb echotexture, formed by the internal pattern of connective tissue and nerve fibers.
- Ultrasound provides real-time imaging for needle tip placement and drug injection. Successful local anesthetic injections clarify the border of the nerve and track along the nerve path and its branches. Ultrasound guidance results in more consistent procedure times for peripheral nerve blocks and can be applied to many regional anesthesia procedures. Anatomic variation in nerve position and course, which is a potential source of block failure, can be directly visualized.
- Ultrasound-guided fascial plane blocks target the space between two fasciae rather than discrete peripheral nerves and are becoming increasingly popular alternatives to traditional central and peripheral nerve techniques.
- Lipid emulsion administration improves the success of resuscitation from cardiac arrest if given immediately after recognition of local anesthetic systemic toxicity.
- Prior to performing a regional block, checklists are recommended to make rare adverse events even less common and improve standardized management when they do, thereby improving patient safety.

Source: Miller 10e, Chapter 42, Key Points, PDF p. 1571.

Chapter 43: Perioperative Fluid and Electrolyte Therapy

- Intravenous fluid therapy is a core part of perioperative practice and has the potential to influence patient outcomes.
- Water makes up approximately 60% of total body weight, the proportion varying widely with age and body composition. The ratio between the water volume within the intracellular and functional extracellular compartments is approximately 2:1.
- The endothelial glycocalyx forms a protein-poor intravascular fluid layer at the vessel perimeter; it has been integrated into a revised version of the Starling equation and updated model of capillary fluid movement.
- Sodium is the dominant extracellular cation and is responsible for much of extracellular fluid osmolality. Dysnatremia is typically associated with disturbed extracellular fluid volume.
- Potassium is the dominant intracellular cation with a key role in the maintenance of transmembrane potentials. Dyskalemia may be accompanied by impaired function of excitable tissues.
- Calcium is a key intracellular second messenger with roles in neuromuscular function, cell division, and oxidative pathways.
- Magnesium has a diverse range of physiologic effects and provides many therapeutic applications in the perioperative setting.
- Acid–base balance is relevant to fluid therapy because fluids containing supraphysiologic levels of chloride may cause an iatrogenic acidosis; the clinical relevance of this acidosis is debated.
- Intravenous fluids have a range of physiologic effects and should be considered drugs with indications, contraindications, dose ranges, cautions, and side effects.
- The physiologic challenge of the perioperative period may lead to a wide variety of disturbances in fluid and electrolyte balance.
- Clinical evidence to guide perioperative fluid therapy is lacking in many areas and cannot be directly extrapolated from general critical care trials.
- A balance must be found between inadequate fluid administration—allowing tissue hypoperfusion—and the adverse effects of excess intravenous fluids and toxicities related to fluid constituents.
- Goal-directed fluid therapy (GDT) may help determine fluid needs for an individual patient in the perioperative setting, and some evidence supports its ability to reduce postoperative morbidity.
- No clear consensus exists on which intravenous fluid is associated with the best clinical outcomes in the perioperative setting. Comparisons of “balanced” with “unbalanced” and “crystalloid” with “colloid” fluids are being studied in many clinical settings.
- The approach to fluid and electrolyte management should be adapted based on patient and surgical factors.

Source: Miller 10e, Chapter 43, Key Points, PDF p. 1600.

Chapter 44: Perioperative Acid–Base Balance

- The presence of a significant acid–base abnormality often signals a sinister underlying problem.
- All acid–base abnormalities result from alterations in the dissociation of water.
- Only three factors independently affect acid–base balance: the PaCO_2 , the strong ion difference (SID), and the total concentration of weak acids (ATOT).
- Respiratory acidosis and alkalosis are caused by hypercarbia and hypocarbia, respectively. Metabolic acidosis is caused by decreased SID or increased ATOT. Decreased SID results from accumulation of metabolic anions (shock, ketoacidosis, and renal failure), hyperchloremia, and free water excess. Increased ATOT results from hyperphosphatemia.
- Metabolic alkalosis is caused by increased SID or decreased ATOT. SID increases due to sodium gain, chloride loss, or free water deficit. ATOT decreases in hypoalbuminemia and hypophosphatemia. This is particularly common in critical illness.
- Most acid–base disorders are treated by reversal of the cause.

Source: Miller 10e, Chapter 44, Key Points, PDF p. 1649.

Chapter 45: Patient Blood Management: Transfusion Therapy

- Blood transfusion is safer now than at any other time in history. Advances in donor screening, improved testing, automated data systems, and changes in transfusion medicine practices account for these increases in safety.
- Although the patient's overall condition is of prime importance, hemoglobin (Hb) values remain a primary component for transfusion decisions with either a restrictive or liberal strategy. In general, a transfusion trigger of 6 to 8 g/dL Hb (restrictive strategy) can be tolerated by patients.
- Preoperative anemia is an independent and potentially modifiable risk factor for postoperative morbidity and mortality. Preoperative treatment with iron (oral or parenteral) is of increasing interest and importance.
- The term patient blood management has become synonymous with appropriate transfusion strategy.
- The addition of plasma and sometimes platelets to packed red blood cells (PRBCs) is described by the term transfusion ratios. For example, 1 unit of PRBCs with 2 units of plasma with 1 unit of platelets would be 1:2:1.
- Acute lung injury is the leading cause of transfusion-related morbidity and mortality.
- Transmission of infection is a small and declining cause of transfusion-related morbidity and mortality.
- Fresh whole blood has gained renewed interest in patients with major blood loss and related coagulopathy.
- Although storage lesions of red blood cells increase over time, there is no evidence that blood stored for short periods compared with moderately long periods contributes to worse clinical outcomes. However, as newer solutions extend the shelf life of blood, this may need continued evaluation, particularly in high-risk groups.
- Synthetic hemoglobin carriers may be a therapeutic option for patients who cannot receive standard allogenic blood transfusions, but these products can carry significant additional cardiovascular and renal risks.
- Cell salvage therapy may reduce the risk of transfusion.

Source: Miller 10e, Chapter 45, Key Points, PDF p. 1676.

Chapter 46: Patient Blood Management: Coagulation

- Normal hemostasis is a balance between generation of a localized hemostatic clot and uncontrolled thrombus formation.
- The extrinsic pathway of coagulation begins with exposure of blood plasma to tissue factor and represents the initiation phase of plasma-mediated hemostasis.
- The intrinsic pathway amplifies and propagates the hemostatic response to maximize thrombin generation.
- The common pathway generates thrombin, forms fibrin, and cross-links fibrin strands to produce an insoluble fibrin clot.
- Routine preoperative coagulation testing of all surgical patients is costly and lacks predictive value for detection of hemostatic abnormalities. Testing should be based on the preoperative history and physical examination and the planned surgery.
- Antiplatelet agents and anticoagulants are used to reduce the formation of blood clots in the setting of coronary or cerebral atherosclerosis or after vascular thrombosis.
- Thrombolytic therapy is used to break up or dissolve blood clots.
- Procoagulant drugs (antifibrinolytics, factor replacements, prothrombin complex concentrate) help control blood loss during surgery.
- Perioperative management of patients who require chronic anticoagulation or antiplatelet therapy involves balancing the risk of surgical bleeding against the risk of developing postoperative thromboembolism.

Source: Miller 10e, Chapter 46, Key Points, PDF p. 1716.

Chapter 47: Management of the Patient with Chronic Pain

- The normal physiology of neuronal function, receptors, and ion channels is altered during persistent pain.
- Chronic pain should be classified into cancer- and non-cancer-related pain. Both include components of neuropathic, inflammatory, and musculoskeletal pain.
- Interdisciplinary management of chronic pain should include specialists in psychology, physical therapy, and anesthesiology. Structured programs based on the biopsychosocial concept with simultaneous engagement of such specialists in a team are most successful.
- Drugs used for chronic pain include opioids, nonsteroidal anti-inflammatory drugs, antiepileptics, antidepressants, topical analgesics (e.g., capsaicin, local anesthetics, opioids), and adjuvants (e.g., local anesthetics, α_2 -agonists, baclofen, botulinum toxin, antiemetics, laxatives).
- Interventional management of chronic pain includes continuous catheter techniques (peripheral, epidural, intrathecal), and stimulation techniques (e.g., transcutaneous electrical nerve stimulation, spinal cord stimulation).
- Perioperative management of patients with chronic pain involves the use of opioid and nonopioid analgesics, and evaluation for dependence and addiction. The anesthesiologist is co-responsible with the surgeons regarding the discharge management with the special focus on opioid therapy (opioid stewardship).

Source: Miller 10e, Chapter 47, Key Points, PDF p. 1747.

Chapter 48: Palliative Medicine

- Palliative care is an interprofessional approach to symptom management and decision-making for patients with a wide range of serious illnesses and is not limited to patients who are imminently expected to die.
- Palliative care teams reduce costs and decrease the burden of symptoms for patients with serious illnesses.
- Physicians receive limited training in discussing difficult topics and tend to focus on details and use medical jargon in conversations with seriously ill patients.
- Patients and families want physicians to be truthful and empathetic and to participate in shared decision-making.
- While patients and families desire a prognosis, they understand and accept that physicians have difficulty prognosticating the future for an individual patient.
- Opioids administered in small doses are effective in treating dyspnea and do not hasten death in patients receiving palliative care.
- The presence of treatment limitations should prompt a thorough discussion of the perioperative plan between the patient and the surgical team.

Source: Miller 10e, Chapter 48, Key Points, PDF p. 1769.

Chapter 49: Anesthesia for Thoracic Surgery

- Patients undergoing pulmonary resection should have a preoperative assessment of their respiratory function in three areas: lung mechanical function, pulmonary parenchymal function, and cardiopulmonary reserve (the “three-legged stool” of respiratory assessment).
- Patients with underlying lung disease have a decreased risk of respiratory complications when pulmonary resections are performed with video-assisted thoracoscopic surgery (VATS).
- After lung resection surgery, it is usually possible to wean and extubate patients with adequate predicted postoperative respiratory function in the operating room provided they are “AWaC” (alert, warm, and comfortable).
- Interventions that have been shown to decrease the incidence of respiratory complications in high-risk patients undergoing thoracic surgery include cessation of smoking, physiotherapy, and thoracic epidural analgesia.
- Geriatric patients are at high risk for cardiac complications, particularly cardiac arrhythmias, after large pulmonary resections. Preoperative exercise capacity is the best predictor of postthoracotomy outcome in the older patient.
- The ability to perform fiberoptic bronchoscopy and a detailed knowledge of bronchial anatomy are necessary for anesthesiologists to provide reliable lung isolation.
- Use of double-lumen endobronchial tubes (DLTs) is the standard method of providing lung isolation in adults. Bronchial blockers are a reasonable alternative for lung isolation in patients with abnormal upper or lower airways.
- With the use of intravenous anesthetic techniques or low-dose volatile anesthetics, hypoxemia during one-lung ventilation (OLV) occurs infrequently.
- Tidal volumes during OLV of 4–6 mL/kg ideal body weight are generally used in combination with titrated positive airway pressure (PEEP) and recruitment maneuvers to decrease potential injury to the ventilated lung.
- The underlying principle of management of a patient with a bronchopleural fistula is to secure lung isolation before positive-pressure ventilation or repositioning the patient for surgery.
- Anesthetic management of a patient with an anterior or superior mediastinal mass should be guided by the patient’s symptoms, the preoperative computed tomography (CT) scan, and the echocardiography findings. The fundamental principle of anesthetic management for these patients is *noli ponti ignii consumere* (“don’t burn your bridges”).
- Ultrasound-guided nerve block techniques combined with multimodal analgesia are a reasonable alternative to epidural analgesia for postoperative pain management in low- to moderate-risk patients.

Source: Miller 10e, Chapter 49, Key Points, PDF p. 1807.

Chapter 50: Anesthesia for Cardiac Surgical Procedures

- Cardiac surgical procedures include coronary revascularization with or without the use of cardiopulmonary bypass (CPB). These procedures can be performed in combination with valvular repair or replacement, depending upon the patient's disease process. Additional cardiac procedures can range from valve repairs and replacements, surgical management of heart failure, e.g. placement of ventricular assist devices, transcatheter aortic valve replacement (TAVR), intervention for structural heart disease, pericardiocentesis, occlusion of left atrial appendage, ablation procedures for arrhythmias, surgical correction of congenital heart defects and heart transplant. In addition, traumatic cardiac and major vascular injuries present another aspect of cardiac surgical patients that cardiac anesthesiologists routinely take care of.
- Choice of anesthetic induction agents and of the maintenance agents will depend upon the patient's disease process and the comorbidities with which the patient presents. Additionally, choice of anesthetic agents will also be determined by the fact if the surgical procedure is elective or emergent. Management of anesthesia includes close control of hemodynamics in the prebypass period.
- Effects of CPB on coagulation pathway, especially platelet function, are mediated through hemodilution, hypothermia and contact activation by bypass circuit materials.
- Issues in the postoperative period can range from relatively minor to major issues. Arrhythmias, low output cardiac failure, hypovolemia and electrolyte abnormalities are a few of the problems of a long list of problems that need to be addressed when present or anticipated.
- Guidelines for management of specific issues are periodically published by various associations, including, The Society of Thoracic Surgeons (STS), Society of Cardiovascular Anesthesiologists, American Society of ExtraCorporeal Technology, European Association of Cardio-Thoracic Surgery and European Association of Cardiothoracic Anesthesiology.
- In recent years there has been a decrease in the incidence of stroke after cardiac surgery. Major factors responsible for postoperative stroke include particulate or micro gaseous emboli, cerebral hypoperfusion and inflammatory response.
- STS guidelines recommend blood glucose control of less than 180 mg/dl. Although many institutions practice an even tighter control of glucose. Hypoglycemia is a risk if a tighter control is pursued, especially as the CPB run is winding down.
- Enhanced sympathetic tone secondary to poor pain control can result in myocardial stress and inability of the patient to effectively clear pulmonary secretions. Therefore, for both cardiac and pulmonary recovery of a patient good postoperative pain control is necessary.

Source: Miller 10e, Chapter 50, Key Points, PDF p. 1887.

Chapter 51: Anesthesia for Correction of Cardiac Arrhythmias

- Cardiac arrhythmias are caused by disorders of impulse formation, disorders of impulse conduction, or both. Cardiac arrhythmias may be life-threatening because of a reduction in cardiac output and/or myocardial blood flow or precipitation of a more serious arrhythmia.
- Radiofrequency ablation is the therapy of choice for many types of cardiac arrhythmias.
- Electrophysiologic studies are used to map out normal and abnormal intracardiac structures. In this process, the mechanism of arrhythmia is delineated and ablation can be performed at the same time.
- Pacing technologies have been developed to treat heart failure resulting in increases in pulse pressure, left ventricular stroke volume, cardiac index, and wedge pressure.
- Implantable pacemakers are placed for treatment of symptomatic bradycardia with the ability to respond to changing hemodynamic demands.
- The development of implantable cardioverter-defibrillators (ICDs) to terminate ventricular tachyarrhythmias by delivering high-voltage shocks to the ventricle has revolutionized therapy for cardiac arrhythmias.
- The main purpose of ICD placement is to prevent sudden cardiac death resulting from hemodynamically unstable ventricular arrhythmias.
- An ICD can be placed for cardiac resynchronization. Cardiac resynchronization therapy improves heart failure symptoms, quality of life, exercise capacity, and electrocardiographic variables.
- Anesthetic management of patients for correction of cardiac arrhythmias depends on associated comorbid illness and the procedure that is planned.

Source: Miller 10e, Chapter 51, Key Points, PDF p. 2007.

Chapter 52: Anesthesia for Vascular Surgery

- Perioperative management of patients undergoing vascular surgery requires an understanding of the underlying pathophysiology of the specific vascular lesion.
- Major vascular surgery is particularly challenging to the anesthesiologist because these are high-risk operations in a patient population with a high prevalence of either overt or occult coronary artery disease, which is the leading cause of perioperative and long-term mortality after vascular surgery.
- Accurate clinical assessment of the pretest probability of significant coronary artery disease is necessary for prudent use and rational interpretation of preoperative cardiac testing.
- Guidelines on perioperative cardiovascular evaluation and care suggest that coronary intervention is rarely necessary to simply decrease the risk for surgery unless such intervention is indicated, irrespective of the preoperative context. Prophylactic coronary revascularization has not been shown to reduce perioperative or long-term morbidity after major vascular surgery. Medical therapy is the cornerstone for the management of coronary artery disease.
- Patients should take their usual cardiovascular medications throughout the perioperative period. Antiplatelet therapy requires special consideration and must be individualized to each patient.
- Prevention and treatment of perioperative myocardial ischemia require careful control of the determinants of myocardial oxygen supply and demand. ST-segment monitoring, particularly with computerized ST-segment analysis, should be used to detect myocardial ischemia during the perioperative period.
- Initiation of perioperative β -adrenergic blocker therapy has potential benefits and risks.
- All patients who require statin therapy on an ongoing basis should also receive statins in the perioperative period.
- The clinical usefulness of any intraoperative monitoring technique ultimately depends on patient selection, accurate interpretation of data, and appropriate therapeutic intervention.
- Maintenance of vital organ perfusion and function by the provision of stable perioperative hemodynamics is more important to overall outcome after aortic surgery than is the choice of anesthetic drug or technique.
- The pathophysiology of aortic cross-clamping and unclamping is complex and depends on many factors, including the level of the cross-clamp, the extent of coronary artery disease and myocardial dysfunction, intravascular blood volume and distribution, activation of the sympathetic nervous system, and the anesthetic drugs and techniques.
- The degree of preoperative renal insufficiency is the strongest predictor of postoperative renal dysfunction.
- Endovascular aortic surgery has become an established, less invasive alternative to conventional open aortic repair. Endoleak, or the inability to obtain or maintain complete exclusion of the aneurysm sac from arterial blood flow, is a complication specific to endovascular aortic repair.
- The primary clinical utility of cerebral monitoring during carotid endarterectomy is to identify patients in need of carotid artery shunting; second, such monitoring is used to identify patients who may benefit from an increase in arterial blood pressure or change in surgical technique.
- Postoperative hypothermia is associated with many undesirable physiologic effects and may

contribute to adverse cardiac outcome.

Source: Miller 10e, Chapter 52, Key Points, PDF p. 2022.

Chapter 53: Anesthesia for Neurologic Surgery and Neurointerventions

- For the purposes of planning a strategy for controlling intracranial pressure (ICP), the four subcompartments of the intracranial space should be considered: cells, interstitial and intracellular fluid, cerebrospinal fluid (CSF), and blood.
- The clinician should make a preoperative assessment of the probable intracranial compliance reserve as the basis for selection of appropriate anesthetic drugs and techniques.
- The venous side of the cerebral circulation is a largely passive compartment that is often the cause of increased ICP, or “tightness,” of the surgical field.
- Cerebral perfusion pressure (CPP) should be supported at or near normal waking levels in patients with recent cerebral injuries (e.g., traumatic brain injury [TBI], subarachnoid hemorrhage [SAH]), and spinal cord injury because of low resting cerebral blood flow and impaired autoregulation.
- When neurosurgical procedures are performed in the sitting position, blood pressure should be corrected to the level of the external auditory meatus and mean arterial pressure (MAP) should be maintained at 60 mm Hg in normotensive adults.
- Monitoring for venous air embolism (VAE) in at-risk situations includes the precordial Doppler and end-tidal carbon dioxide analysis.
- Despite encouraging preclinical data, therapeutic mild hypothermia cannot be advocated in the care of head-injured patients in the intensive care unit (ICU) or during the operative management of patients with intracranial aneurysms because of negative human trial results for those patient groups.
- The most important consideration in the anesthetic management of patients undergoing clipping or coiling after acute SAH is the prevention of paroxysmal hypertension with its attendant risk of aneurysm rerupture. Nonetheless, adequate perfusion pressure is needed if temporary clips are used during management of cerebral aneurysm.
- Although induced hypotension is rarely used electively in aneurysm surgery, the clinician must be ready to reduce blood pressure immediately and accurately in the event of aneurysm rupture.
- Tracheal intubation of a head-injured patient with an undefined cervical spine injury can be accomplished using rapid sequence induction with manual in-line stabilization (the occiput held rigidly to the backboard), with only a very small risk of injury to the spinal cord.
- CPP ($CPP = MAP - ICP$) should be supported to a target range of 60 to 70 mm Hg in the first 48 to 72 hours after TBI in adults.
- Hypocapnia has the potential to cause cerebral ischemia, particularly in a recently injured brain and in a brain beneath retractors; it should be used only when absolutely necessary for the control of critically increased or uncertain ICP.

Source: Miller 10e, Chapter 53, Key Points, PDF p. 2071.

Chapter 54: Anesthesia for Bariatric Surgery

- In the United States more than two out of three people are overweight or obese. Worldwide, more people are obese than are malnourished. Obesity is becoming the largest single preventable cause of death and represents major morbidity and mortality.
- Metabolic syndrome includes abdominal obesity, decreased high-density lipoprotein, insulin resistance, glucose intolerance, and hypertension, and is present in approximately 34% of the adult population in the United States alone.
- The single greatest risk factor for sleep apnea is obesity, with the majority of obese patients having increased oral and pharyngeal tissue. This makes ventilation, intubation, and extubation more challenging.
- Choices for medical management of obesity are limited and efficacious achievement with just medical management is uncommon. Changes in behaviors are important for success.
- Surgery for obesity is recommended at a body mass index (BMI) of 40 kg/m² or BMI over 30 kg/m² with comorbidities expected to respond to weight loss, secondary to surgical therapy, like hypertension, diabetes, and hypercholesterolemia. In clinical trials, long-term survival is better in the surgically treated group over the medically managed.
- Preoperative evaluation should focus on cardiopulmonary issues and securing an airway, along with other concerns such as diabetes, hypertension, and obstructive sleep apnea.
- Anesthetic drugs should be tailored based on lipid solubility and awareness of lingering respiratory depression effects.
- Patient preparation and positioning are keys to successful airway management. Preoperative pressure support ventilation should be used adjunctively if possible.
- Intraoperative ventilation is assisted by complete paralysis, moderate positive end-expiratory pressure, tidal volumes based on ideal body weight, and recruitment maneuvers as needed.
- Common serious postoperative complications are deep vein thrombosis and staple line issues.
- Obese patients presenting for non-weight loss surgery benefit from anesthetic approaches similar to that used for bariatric surgery and endoscopic procedures.

Source: Miller 10e, Chapter 54, Key Points, PDF p. 2124.

Chapter 55: Anesthesia and the Renal and Genitourinary Systems

- Innervation of the intra-abdominal components of the genitourinary system—the kidney and the ureter—is primarily thoracolumbar (T8–L2). The nerve supply of the pelvic organs—the bladder, prostate, seminal vesicles, and urethra—is primarily lumbosacral with some lower thoracic input.
- The spinal level of pain conduction for the external genitourinary organs is S2–4, except for the testes (T10–L1).
- The kidneys receive 15% to 25% of the total cardiac output, with most of this blood directed to the renal cortex. Renal medullary papillae are more vulnerable to ischemic insults. Kidneys successfully autoregulate their blood flow between 60 and 160 mm Hg mean arterial pressures.
- The glomerular filtration rate (GFR) is the best measure of glomerular function. Creatinine clearance is a good measure of the GFR; urine output is not.
- Hypervolemia, acidemia, hyperkalemia, cardiorespiratory dysfunction, anemia, and bleeding disturbances are manifestations of chronic renal failure.
- Serum creatinine, most commonly used as a marker of renal function, has several limitations. Newer biomarkers, such as serum cystatin C, are better and earlier measures of acute kidney injury and the risk of end-stage renal disease, as well as related mortality.
- Although renal transplantation reverses most of the abnormalities associated with end-stage renal disease, dialysis improves only some and introduces additional complications.
- Extracorporeal shock wave lithotripsy (ESWL) historically caused significant physiologic changes related to immersion in a water bath, but newer generations have eliminated the water bath and hence those risks. Shock waves can cause clinically insignificant dysrhythmias. Pregnancy and untreated bleeding disorders are contraindications to ESWL.
- For patients presenting with renal tumors, 5% to 10% of the tumors extend into the renal vein, inferior vena cava, and right atrium. Complications ranging from circulatory failure to embolization of tumor may occur during surgical resection. Cardiopulmonary bypass may be necessary for tumors invading major vessels.
- Several approaches to prostate resection have had significant impact on anesthesia and perioperative management. Newer techniques such as laser prostatectomy have replaced transurethral resection of the prostate (TURP), making TURP syndrome a rare event. TURP syndrome is a constellation of symptoms caused by the absorption of hypotonic bladder irrigants. Cardiovascular and neurologic changes result due to hypoosmolality, hyponatremia, hyperglycinemia, hyperammonemia, and hypervolemia.
- Radical prostatectomy may cause significant blood loss, and intraoperative venous air emboli can occur. Regional anesthesia with spontaneous ventilation is associated with less blood loss than general anesthesia and intermittent positive pressure ventilation. Other advantages of epidural anesthesia include a decreased incidence of deep vein thrombosis and the initiation of preemptive analgesia. Whether outcomes are dependent on the choice of anesthesia is not clear.
- Robotic radical prostatectomy is associated with reduced blood loss and postoperative pain compared with open radical prostatectomy. Anesthetic concerns are related to steep head-down tilt and pneumoperitoneum and include hypercarbia, hypoxemia, increased intraocular and intracranial pressures, decreased perfusion pressure to lower extremities, and positional injuries.

- Laparoscopic surgery in urology frequently requires insufflation of carbon dioxide into the retroperitoneal space. In lengthy procedures, pneumomediastinum and subcutaneous emphysema of the head and neck may occur.
- Anesthetic concerns of robotic-assisted surgery include the length of surgical time, intravenous fluid management, pneumoperitoneum, and positioning. The most frequent reported complications are peripheral neuropathies, corneal abrasions, vascular complications (including compartment syndrome, rhabdomyolysis, and thromboembolic disease), and the effects of edema.
- Postoperative renal dysfunction after genitourinary procedures can occur due to a variety of causes including significant hemorrhage, large intraoperative fluid shifts, nephrotoxic medications or post-renal urinary retention or obstruction.

Source: Miller 10e, Chapter 55, Key Points, PDF p. 2148.

Chapter 56: Anesthesia for Abdominal Organ Transplantation

- Outcomes in abdominal organ transplant patients continue to improve due to improved organ preservation and surgical and anesthetic management.
- Kidney and liver transplantation has expanded globally; active transplant programs exist in all regions of the world.
- Organ demand remains greater than organ supply secondary to evolving end organ disease patterns in solid organ transplant candidates.
- Novel methods have emerged to expand donor pools including use of living organ donation, extended criteria donors, and machine perfusion preservation techniques.
- Organ allocation policies are often regulated on a national or multinational level; many liver transplant allocation systems are based on the Model for End-Stage Liver Disease (MELD) score.
- Solid organ transplant candidates are now older and have a higher prevalence of chronic diseases than previously documented.
- Cardiovascular disease has a significant impact on solid organ transplant candidates; preoperative detection and treatment of cardiovascular disease is a crucial pretransplant step.
- Knowledge of the multiorgan pathophysiology of end-stage renal, liver, and intestinal failure and of advanced diabetes is necessary for the anesthetic management of solid organ transplant patients.
- Intraoperative management strategies for kidney transplant include maintenance of renal perfusion pressure and prevention of early postoperative complications.
- Intraoperative management strategies for liver transplant include the use of dedicated anesthetic teams, massive transfusion protocols, coagulation assessment, and advanced hemodynamic monitoring.
- Pancreas transplantation provides definitive treatment for patients with diabetes mellitus and may include combined kidney transplantation in diabetic patients with renal failure.
- The perioperative management of pancreas transplant requires close monitoring of blood glucose levels.
- Intestinal and multivisceral transplants are definitive treatment options for patients with bowel motility disorders, desmoid tumors, or parenteral nutrition (PN)- related liver disease.
- Transplant recipients are at risk for immunosuppression-related sequelae including infection, malignancy, and cardiovascular disease.
- The COVID-19 pandemic had a significant impact on solid organ transplantation on a global level; transplant candidates and recipients are more adversely affected by COVID-19 disease compared to the general population.

Source: Miller 10e, Chapter 56, Key Points, PDF p. 2183.

Chapter 57: Anesthesia for Organ Procurement

- The shortage of organs available for transplantation is a worldwide problem.
- The discrepancy between the number of patients waiting for organ transplantation and the available organs remains significant.
- Most organs in the United States are donated after brain death, with smaller portions from living organ donors and after circulatory death.
- Brain-death donors have physiologic alterations that must be actively managed to ensure that the organs are suitable for transplantation.
- Determining brain death and circulatory death should follow national guidelines and local institutional protocols.
- The anesthesiologist must have an awareness of the ethical and legal issues related to the declaration of death that precedes organ donation.
- Expansion of the donor pool through the inclusion of extended criteria, such as high-risk donors, addresses the organ shortage and improves time to transplant, often for the recipients most in need of transplantation.
- The use of extended criteria high-risk organs significantly impacts recipient outcomes and presents challenges to perioperative management.
- Ischemia–reperfusion injury in organ transplantation is unavoidable; however, management strategies can lessen the likelihood of postoperative graft failure.
- Goal-directed donor management can improve the number of organs transplanted per donor.
- Living organ donor kidney transplantation remains an important donor source in the United States, whereas the use of living donors for liver transplantation varies by country.
- New technologies, including machine perfusion after procurement, are promising to mitigate the effects of prolonged preservation time, to increase the donor pool, and to improve transplant recipient outcomes.

Source: Miller 10e, Chapter 57, Key Points, PDF p. 2229.

Chapter 58: Anesthesia for Obstetrics

- The normal physiologic changes of pregnancy begin in the first trimester, affect all organ systems, and alter pharmacokinetic and pharmacodynamic responses to many drugs commonly used in anesthesia.
- Maternal–fetal exchange of most drugs and other substances occurs primarily by diffusion. The rate of diffusion and peak levels in the fetus depend on maternal-to-fetal concentration gradients, maternal protein binding, molecular weight of the substance, lipid solubility, and the degree of ionization of that substance.
- All patients in labor are considered to have a full stomach and an increased risk for pulmonary aspiration of gastric contents during induction and emergence from anesthesia and aspiration prophylaxis should be considered before all surgical procedures during pregnancy.
- Uterine blood flow increases progressively during pregnancy from approximately 100 mL/min in the nonpregnant state to between 700 and 900 mL/min (~10% of cardiac output) at term gestation. Consequently, hemorrhage can be rapid during pregnancy and carries significant morbidity. Early recognition with timely intervention, optimal team performance, and appropriate blood product transfusions are essential to good patient outcomes.
- Uterine and placental blood flow depend on maternal cardiac output and are directly related to uterine perfusion pressure and inversely related to uterine vascular resistance. Decreased perfusion pressure can result from maternal hypotension secondary to hypovolemia, aortocaval compression, sympathetic blockade, and decreased systemic resistance from either general or neuraxial anesthesia. Prophylactic or therapeutic phenylephrine in boluses or as an infusion reduces the incidence and severity of hypotension from spinal anesthesia for cesarean delivery. In comparison to ephedrine, phenylephrine results in less fetal acidosis.
- During pregnancy, the maternal oxyhemoglobin dissociation curve shifts to the right with pregnancy while the fetal oxyhemoglobin dissociation curve lies to the left. This facilitates oxygen transfer from maternal to fetal hemoglobin. Fetal O₂ saturation does not exceed 60% even with 100% O₂ delivery to the pregnant patient. Due to increased minute ventilation, the maternal arterial partial pressure of carbon dioxide (PaCO₂) decreases from 40 mm Hg to approximately 30 mm Hg during the first trimester. This reduction facilitates carbon dioxide transfer across the placenta, which is primarily limited by blood flow and not diffusion.
- Neuraxial analgesia is the most reliable and effective method of reducing pain during labor. Adequate analgesia is achieved with blockade of T10 to L1 during the first stage of labor and requires extension to include S2 to S4 during the second stage of labor.
- Neuraxial analgesia may prolong the second stage of labor in comparison to unmedicated birth or intravenous opioid analgesia but does not increase the risk for cesarean delivery. Epidural analgesia utilized early compared to late in labor does not increase the risk for cesarean delivery or prolong the first stage of labor.
- Remifentanyl patient-controlled analgesia (PCA) offers superior pain relief to nitrous oxide and less fetal effects than other intravenous opioid analgesics, but its analgesic effects are inferior to epidural analgesia, and it requires careful maternal oxygenation and ventilation monitoring.
- Hypertensive disorders of pregnancy complicate 5% to 10% of worldwide pregnancies and can cause maternal and fetal mortality. Patients with preeclampsia are at increased risk for cerebral hemorrhage, pulmonary edema, and coagulopathy. Systolic and diastolic blood pressure higher than 160/110 mm Hg should be treated to limit complications.

- Sepsis is a leading cause of maternal morbidity and mortality in the UK and United States. Early identification and treatment have been shown to improve outcomes.

Source: Miller 10e, Chapter 58, Key Points, PDF p. 2244.

Chapter 59: Anesthesia for Fetal Surgery and Other Fetal Therapies

- Most fetal anomalies are not appropriate for in utero treatment. A condition appropriate for fetal therapy should cause ongoing irreversible harm to the fetus and be mitigated by early treatment that occurs at a gestational age before the fetus can tolerate ex utero neonatal intervention.
- A multidisciplinary approach with open communication is essential to the success of each fetal intervention.
- Safety of the pregnant patient and the principle of “do no harm” should be the primary consideration in determining the most appropriate therapeutic option. A thorough maternal and fetal evaluation and frank discussion of risks and benefits by all team members with the birthing parent is required to determine an appropriate care plan.
- Although open fetal surgery typically requires general anesthesia, most minimally invasive percutaneous techniques can be effectively performed using infiltration of local anesthesia or neuraxial anesthesia techniques.
- Randomized controlled clinical trials show improved outcomes with fetoscopic laser photocoagulation of placental vessels to treat twin-to-twin transfusion syndrome, intrauterine open fetal surgery to treat myelomeningocele and fetal tracheal balloon occlusion in severe congenital diaphragmatic hernia.
- In addition to anesthetic considerations associated with nonobstetric surgery during pregnancy, fetal surgery requires additional planning for fetal anesthesia and analgesia, various modalities of fetal monitoring, maintenance of uterine relaxation, preparation for emergent events (e.g., fetal bradycardia, maternal hemorrhage), and postoperative tocolysis.
- Membrane separation, preterm premature rupture of membranes, and preterm labor and delivery remain the most common complications and significant causes of morbidity and suboptimal outcomes with fetal surgical interventions.
- Further research into optimal anesthetic techniques for various fetal interventions is essential to improve patient outcomes and advance the field of fetal surgery.

Source: Miller 10e, Chapter 59, Key Points, PDF p. 2289.

Chapter 60: Anesthesia for Orthopedic Surgery

- Degenerative joint disease is the foremost medical condition leading to surgery in the United States. With continued population growth and an increasing percentage of older people, a 400% increase in the demand for hip and knee arthroplasty and spine procedures is projected by 2040.
- Many orthopedic operations including total joint arthroplasties and spine surgeries are classified as intermediate surgical risk, with 30-day cardiac death or myocardial infarction occurring with an incidence of 1% to 5%. Since preoperative cardiac risk evaluation is often inconclusive in high-risk patients with limited mobility, a higher level of monitoring and perioperative troponin measurements should be considered to assess for perioperative cardiac events.
- Comprehensive preoperative geriatric assessment is increasingly used in the perioperative care of older patients who may have multiple comorbid conditions and who have suffered fragility fractures. Prehabilitation programs can be useful in these patients to reduce frailty and improve surgical outcome.
- Fractures of the proximal femur following falls are common in older patients and are associated with high morbidity and mortality. Early surgery (<24 hours) has been associated with reduced pain and length of hospital stay. Patients with significant medical comorbidities that delay surgery for more than 4 days have a higher mortality.
- As compared to the lateral position, beach chair position offers superior surgical exposure and access for most shoulder surgeries, less distortion of muscle anatomy, and less tension on the brachial plexus. Cerebral perfusion pressure can decrease by 15% in sitting patients under general anesthesia, and in patients with cerebrovascular disease detrimental decrease in cerebral blood flow can occur if the systemic pressure is not carefully monitored and maintained. The role of the sitting position on postoperative neurologic outcome remains, however, controversial.
- The most common complications after total hip arthroplasty and total knee arthroplasty are cardiac events, pulmonary embolism, pneumonia and respiratory failure, and infections. Older patients with major comorbidities including cardiac disease, pulmonary disease, and diabetes should have a complete preoperative medical evaluation.
- Cemented fixation of the femoral prosthesis can be complicated by the bone cement implantation syndrome, resulting in intraoperative hypotension, hypoxia, or even cardiac arrest. Invasive hemodynamic monitoring with an arterial catheter and possibly also a central venous catheter should be considered. Treatment with a potent inotropic agent such as epinephrine may be required. Pulsatile lavage of the femoral canal and drilling a vent hole in the femur before prosthesis insertion can also ameliorate the hemodynamic consequences of this devastating complication.
- Correction of spinal deformities can be associated with large intraoperative blood loss, and measures to minimize blood transfusion should be considered. Deliberate controlled hypotension has been employed but must be used with caution in older adults, those with cardiovascular disease, or those at risk for ischemic complications and postoperative vision loss. Antifibrinolytic agents may be considered to limit blood loss but should be avoided in patients with a history of thromboembolic events, coronary stents, or renal impairment. Pain can be treated using a multimodal analgesic approach.
- Intraoperative neurophysiologic monitoring is increasingly employed for spine surgeries and is currently recommended for procedures with increased risk for spinal cord injury including correction of spine deformities, resection of intramural tumors, unstable spine trauma, Chiari malformation, spinal cord vascular malformations, as well as those with risk for root damage and

in patients with significant risk for compression neuropathies.

- Perioperative visual loss after spine surgery can be caused by anterior or posterior ischemic optic neuropathy, retinal ischemia, cortical blindness, or posterior reversible encephalopathy. Direct pressure on the eye should be avoided and patients should be positioned so that the head is level with or higher than the heart. The patient's head should be maintained in a neutral forward position without significant neck flexion, extension, lateral flexion, or rotation. Staged spine surgery procedures can reduce the risk of perioperative visual loss and should be considered in high-risk patients.

Source: Miller 10e, Chapter 60, Key Points, PDF p. 2329.

Chapter 61: Geriatric Anesthesia

- Normal aging is associated with changes in physiology and an increase in many pathologic conditions.
- The number and impact of normal and pathologic conditions varies significantly across older individuals.
- Preoperative screening recommendations and guidelines for older patients can provide a useful starting point to evaluate and optimize care.
- Best intraoperative practices follow from an understanding of geriatric physiology and awareness of medications, which are contraindicated in the older population.
- Postoperative care tailored to the needs of high-risk adults may benefit the highest-risk patients such as palliative care consultation and delirium prevention measures.

Source: Miller 10e, Chapter 61, Key Points, PDF p. 2368.

Chapter 62: Anesthesia for Trauma

- Successful perioperative anesthesia care for patients with severe injuries requires understanding of trauma system design and surgical priorities.
- Emergency airway management for trauma patients is based on having a clear plan, such as the American Society of Anesthesiologists' algorithm for difficult airways adapted for trauma. The safest and most effective approach is the rapid sequence induction of anesthesia and intubation with in-line cervical stabilization, followed by direct laryngoscopy or video laryngoscopy. The use of cricoid pressure remains controversial.
- Recognition and treatment of hemorrhagic shock is the cornerstone of advanced trauma life support. Hemorrhagic shock indicates the need for rapid operative treatment, with the possibility of a damage control resuscitative approach. Although establishing an adequate airway remains the initial priority, apparent hemorrhage should be concurrently addressed through the immediate application of tourniquets or direct pressure and transfusion of blood products.
- Recognizing the impact of early coagulopathy in trauma, damage control resuscitation should be employed, with an emphasis on maintenance of blood composition by early transfusion of red blood cells, plasma, and platelets and viscoelastic monitoring (see also Chapter 46) when available. Hypotensive resuscitation is an appropriate tactic for trauma patients with hemorrhagic shock, although vasopressors may be indicated in some patients.
- Management of patients with severe traumatic brain injury (see also Chapter 53) requires monitoring and maintenance of cerebral perfusion and oxygenation for successful operative and intensive care management.
- Operative timing for the surgical management of traumatic injuries, including orthopedic trauma, must be balanced between early definitive repairs and the potential for worsening overall physiologic stress.
- Trauma anesthesiology includes a substantial critical care practice component (Chapters 79 and 80).

Source: Miller 10e, Chapter 62, Key Points, PDF p. 2389.

Chapter 63: Prehospital Care for Medical Emergencies and Trauma

- Following World War II, anesthesiologists (among others) led the development of prehospital and retrieval medicine (PHRM).
- Emergency medical service (EMS) systems differ both between and within countries. Broadly, the most common models of prehospital care in high-income nations are paramedic-led, physician-led, physician-paramedic crews, or a combination of these models depending on the clinical situation. In the United States, paramedics provide prehospital care, and it is rare for physicians to work in the prehospital environment. In much of Europe, on the other hand, physicians are commonplace. While in Australia and New Zealand, the crew mix varies from region to region. In many European countries, EMS physicians lead the prehospital care for patients requiring advanced life support.
- The core of successful PHRM practice is founded in high standards of basic and advanced life support, and standardized, team-based decision-making.
- Rapid, simultaneous assessment, triage, and treatment form the cornerstone of prehospital care – the use of a primary survey and limited diagnostic adjuncts can ensure transport to the most appropriate care setting.
- In major trauma, prehospital care must limit the time spent on the scene, control hemorrhage, and expedite transport to a trauma center, via the most timely means.
- Prehospital intravenous fluid resuscitation for major trauma varies in approach. Patients with penetrating torso injuries and hemorrhagic shock may benefit from limited intravenous fluid resuscitation and permissive hypotension, particularly in urban settings. Prevention of the lethal triad of hypothermia, acidosis, and coagulopathy is of paramount importance. Although the evidence for the type of prehospital resuscitation fluid is equivocal, crystalloid (especially >1500 mL) is associated with worse outcomes in major trauma. If crystalloid is used, it should be used sparingly. In acute coronary syndrome and stroke, achieving rapid reperfusion of the ischemic tissue is the priority. Because only specialized centers provide 24-hour cardiac catheter service or stroke teams, rapid transport to acute myocardial infarction or stroke centers is critical. In response to a myocardial infarction, morphine, oxygen, nitrates, and aspirin are the main components of prehospital therapy. Fibrinolysis for myocardial infarction has been used with success in the prehospital setting but requires clinical governance by appropriate physicians to ensure suitable patient selection and adequate risk management.
- The future of EMS will likely see an increased use of telehealth, which can narrow the time gap between the hospital and the field. This will facilitate improved field and hospital diagnostics and treatments and ensure more efficient handoffs when arriving at the receiving hospital.
- Autonomous and semi-autonomous vehicles may also play an important role in EMS and PHRM delivery in the future (e.g., drone delivery of blood products, equipment, or personnel to the scene – or even of patients away from it!)

Source: Miller 10e, Chapter 63, Key Points, PDF p. 2442.

Chapter 64: Biologic, Natural, and Human-Induced Disasters: The Role of the Anesthesiologist

- Anesthesiologists possess a combination of knowledge and procedural skills that make them critical participants during the immediate and ongoing response to large-scale disasters. To do so, they must be willing and able to modify their traditional workflow to best meet the needs of victims.
- Natural disasters such as earthquakes, hurricanes, floods, tsunamis, and tornadoes require significant resources to address the ramifications of these events; as a result, they can cause massive interruptions in the normal provision of health care.
- Anesthesiologists who travel from resource-rich to resource-limited environments to provide rescue medical care must cope with a lack of familiarity with infectious diseases that are uncommon in resource-rich nations.
- In addition to addressing the acute challenges experienced during these disasters, anesthesiologists and other providers who travel to disaster-ravaged areas are exposed to a variety of physical and mental stressors that impact their immediate response and can have lasting effects after the event.
- Because of the very high early mortality and personal impact on local health care workers following many disasters, anesthesiologists who travel to countries experiencing them may provide the greatest value by temporarily replacing local health care workers lost in the disaster.
- While hurricanes tend to have fewer immediate fatalities than earthquakes or tsunamis, the ability of hurricanes and subsequent floods to cripple the provision of medical and surgical care has numerous downstream consequences, including long-term effects on medical education and clinical care because of loss of faculty and flood damage to educational and clinical facilities.
- When disasters induce shortages of medications and fluids, as Hurricane Maria did in 2017 and the COVID-19 pandemic did in 2020, distribution of limited resources becomes a challenge, requiring a coordinated approach to global conservation of existing supplies and avoidance of waste.
- Since many disasters cannot be anticipated, such as the September 11, 2001 attack on New York City, developing and updating existing prospective plans to address triage of victims and coordination of services among diverse hospital systems and trauma centers become critical to appropriate resource use and early access to life-saving care.
- In the eight years between 2015 and 2022, over 19,000 people were shot and killed or wounded in the United States in a mass shooting.¹ Accordingly, all anesthesiologists must be prepared to provide care to often multiple victims in the aftermath of these heinous events.
- As acts of terrorism have accelerated in recent decades, anesthesiologists need to have knowledge and skills required to care for victims of chemical, biologic, radiologic, or nuclear (CBRN) disasters. The impact of these events will not only affect those in the immediate vicinity of the act, but also the entire health system. Anesthesiologists must be prepared to alter their traditional roles and develop appropriate triage methods to best meet the needs of all patients and the community following such an event.
- One of the highest priorities for health care workers after a CBRN event is to take precautions against becoming the next (second) victim. Appropriate personal protective equipment (PPE) is extremely important after these events and must be readily available to protect health care

workers.

- During an outbreak of a highly dangerous infectious disease, anesthesiologists must understand the mechanisms of transmitting the disease and implement appropriate isolation and management strategies. For many such situations, they also serve as consultants in collaboration with infectious disease specialties in the development of optimal care pathways to minimize risk to other patients and health care workers.
- Pandemics place incredible strain on health care systems around the world. To address these challenging situations and ensure sufficient staff capable of managing the clinical issues, anesthesiologists can often utilize their broad knowledge and clinical and procedural skills to provide patient care outside of their traditional roles and environments.

Source: Miller 10e, Chapter 64, Key Points, PDF pp. 2456–2457.

Chapter 65: Anesthesia for Ophthalmic Surgery

- Ophthalmic procedures, which vary from 15-minute cataract surgery to more lengthy cornea and retina procedures, are considered to be “low risk.” However, the patient population is at higher risk because of the extremes of age and commonly associated comorbidities.
- Although many ophthalmic surgeries (cataract, glaucoma, simple vitrectomy) are short, the daily volume of procedures is high, necessitating an efficient and well-trained group of providers knowledgeable about the diverse procedures and patient-specific implications to ensure patient safety.
- Most ophthalmic procedures can be performed under topical anesthesia or orbital block in combination with monitored anesthesia care (MAC).
- Orbital blocks, particularly retrobulbar blocks, can have severe complications, including retrobulbar hemorrhage, and retrograde spread of local anesthetic into the subarachnoid space causing brainstem anesthesia, loss of consciousness, and respiratory arrest.
- General anesthesia is indicated for 20%–30% of ophthalmic surgeries. It is required for pediatric ophthalmic surgery; complex procedures, some of which require muscle relaxation; and adults who cannot undergo MAC for a variety of reasons.
- Strabismus surgery is an independent risk factor for postoperative vomiting in pediatric patients.
- A working knowledge of the anatomy and physiology of the eye is essential to providing safe anesthesia care. Anesthesiologists should understand the effects of anesthetic drugs and interventions on intraocular pressure and systemic effects of ophthalmologic medications.
- Preoperative laboratory testing is not necessary for patients undergoing cataract surgery and has not been shown to reduce postoperative complications.
- Succinylcholine can be safely administered to unfasted patients who have an open-globe injury that requires emergent vision-saving surgery under general anesthesia.
- Most ophthalmic procedures can be safely performed in the outpatient setting. To prevent a prolonged postoperative anesthesia care unit stay or unanticipated hospital admission requires careful patient evaluation, optimization of underlying medical conditions, adequate pain control, prevention of postoperative nausea and vomiting, and maintenance of hemodynamic stability during the perioperative period.

Source: Miller 10e, Chapter 65, Key Points, PDF p. 2486.

Chapter 66: Anesthesia for Otolaryngologic and Head–Neck Surgery

- Difficult airways are more frequently encountered in patients undergoing ear, nose, and throat (ENT) surgery, especially for cancer. Reviewing the results of a preoperative computed tomography (CT) scan or an endoscopic airway examination may help identify pathologic features likely to produce airflow obstruction or complicate tracheal intubation.
- Although the tracheas of ENT patients are often intubated using ordinary polyvinyl chloride endotracheal tubes (ETTs), microlaryngeal, laser-safe, and wire-reinforced tubes are frequently employed to improve access to the operative site, ensure laser safety, and minimize risk of tube compression during the surgical procedure.
- Endotracheal intubation in an awake patient (awake tracheal intubation [ATI]) with a flexible bronchoscope is commonly used when intubation following the induction of general anesthesia is predicted to be difficult or impossible.
- Fiberoptic endotracheal intubation performed in an awake patient (ATI) is usually well-tolerated, is gentle on the airway, and does not require force to obtain glottic exposure.
- When an airway disorder is so extensive that awake endotracheal intubation is too risky or impractical, tracheostomy performed using local anesthesia (with or without judicious intravenous sedation) is usually the best alternative. In urgent or emergent airway management situations, a cricothyrotomy may be preferable to a tracheostomy because cricothyrotomy generally takes much less time to complete for experienced physicians.
- For some head and neck surgical procedures, such as parotid resections or to provide facial nerve monitoring during ear surgery, the need for electrical testing of the facial nerve precludes the extended use of neuromuscular blocking drugs.
- In many patients with head and neck disorders, gentle emergence from anesthesia, free of coughing and straining, is vitally important to prevent emergence rebleeding as a result of venous engorgement and hypertensive episodes associated with agitation or discomfort.
- Bleeding following tonsillectomy within the first 24 hours is classified as a primary bleed and usually occurs within the first 6 postoperative hours. A secondary bleed (after 24 hours) can also occur as long as several days later.
- Upper airway and facial trauma create challenging situations for anesthesiologists. Facial trauma can produce unremitting bleeding as well as the risk of aspiration of teeth, blood, bone, and tissue fragments, and can be associated with cervical spine injury. Airway trauma can result from blunt or penetrating injuries, burns, inhalational injury, and iatrogenic causes. In all of these situations, initial management is dictated by the degree of airway compromise, presence of respiratory distress, as well as the clinical experience of the clinician and what equipment is available.
- Intubating the trachea of a patient with laryngeal trauma may result in further injury to the airway or even complete airway loss. If intubation is attempted, it is advised to use a fiberoptic bronchoscope with a small ETT. If there is airway disruption, initiation of positive-pressure ventilation may initiate or worsen subcutaneous emphysema. In some cases, a tracheostomy may be the most prudent approach to securing the airway.
- Causes of stridor include inhaled foreign bodies, bilateral vocal cord palsy, airway edema, angioedema, epiglottitis, traumatic injury, subglottic stenosis, and other pathologic entities. Regardless of the etiology, the initial assessment should focus on whether endotracheal

intubation or a surgical approach to the airway is immediately necessary to rescue the patient from death or injury. For patients with stridor but satisfactory gas exchange, a helium:oxygen (heliox) blend of inspired gas administered with a non-rebreathing face mask or high-flow nasal oxygen can be helpful to improve airflow temporarily while assessing and managing the cause for stridor.

- The chosen anesthetic technique for endoscopic procedures varies with patient and lesion particulars (site, size, and nature), clinical preferences, and the selected surgical tools (laser, rigid bronchoscope). Total intravenous anesthesia is a popular choice for many of these cases.
- Lasers are often used to vaporize otolaryngologic lesions; when used, special precautions are needed to prevent accidental thermal injury or an airway fire, a potentially deadly complication that may also occur during tracheotomy surgery. Oxygen concentration must be kept to a minimum when a significant potential for an airway fire exists. Moreover, nitrous oxide should not be used during airway surgery because it supports combustion like oxygen.

Source: Miller 10e, Chapter 66, Key Points, PDF pp. 2508–2509.

Chapter 67: Anesthesia for Robotic Surgery

- Robotic surgery has witnessed exponential growth: since 2000, more than 8.5 million procedures have been performed using the da Vinci system worldwide.¹
- Robotic surgery has been successfully adopted for patients receiving urologic, gynecologic, abdominal, otolaryngologic, cardiac, thoracic, neurosurgery, and orthopedic procedures.
- Robotic surgical systems can be classified into three types: (1) conducting robotic system; (2) robotic remote system, or (3) combined robotic system. 1. Conducting robotic system: the surgeon plans and downloads the surgical plan to the robot's software. Then, the robot performs the procedure while the surgeon oversees the procedure. 2. Robotic remote system: the surgeon controls the robot remotely and in real-time using haptic interface technology. The surgeon will receive feedback from the robotic interface and the three-dimensional views of the surgical field. 3. Combined robotic system: the robot joins the surgeon in the surgical field to enhance the accuracy of manipulating the surgical instruments.
- Robotic surgical systems allow precise and accurate geometric movements and can enhance a surgeon's ability to visualize pathologies and perform complex procedures.
- There is a great learning curve in adaptation to robotic surgical systems, which leads to increased time and the cost of robotic surgeries.
- Robotic surgery presents unique challenges for anesthesiologists, including physiologic changes associated with steep Trendelenburg positioning and difficulty accessing the patient once the robot is docked, which can lead to patient injury.
- Due to the expanding technology of robotic systems, the area of robotic surgical systems is still evolving, and many of the short-term and long-term outcomes of different robotic surgical systems still need to be clarified.

Source: Miller 10e, Chapter 67, Key Points, PDF p. 2539.

Chapter 68: Ambulatory (Outpatient) Anesthesia

- The use of ambulatory surgery continues to increase, mostly as a result of less invasive surgical techniques, improved patient selection and preparation, and an expansion of ambulatory non-operating room and office-based practice.
- Few absolute contraindications exist to ambulatory surgery. Patients should not be excluded on the basis of arbitrary limits, such as age, body mass index, or American Society of Anesthesiologists' physical status classification system. There are some restrictions on payment for some procedures in the outpatient and ambulatory surgery center setting in the United States for patients on Medicare.
- Effective preoperative assessment is required to evaluate and prepare patients and is essential for the delivery of safe, high-quality, and efficient ambulatory surgical care.
- Many anesthetics and techniques can be used for ambulatory surgery. Of prime importance, provider experience and careful attention to detail are required to deliver high-quality rapid recovery with minimal side effects.
- Spinal anesthesia may extend the range of patients and procedures suitable for ambulatory surgery, and chlorprocaine, prilocaine, and mepivacaine may offer advantages over bupivacaine.
- Sedative techniques can facilitate a wide variety of procedures performed in the hospital, office, or remote settings. However, sedation is no safer than general anesthesia and requires the same standards of personnel, monitoring, and perioperative care as for patients undergoing general or regional anesthesia.
- Multimodal analgesia, using combinations of local or regional anesthesia, acetaminophen, and nonsteroidal anti-inflammatory drugs, provides effective relief of pain and should be used whenever possible. The reduced need for opioids decreases the incidence and intensity of adverse effects including the need for admission after surgery.
- Postoperative nausea and vomiting (PONV) risk should be assessed for each patient, and antiemetic prophylaxis should be given to all patients with risk factors. Combination therapies and multimodal regimens are required for patients and procedures known to be associated with increased risk for PONV.
- Patients should be discharged with written instructions concerning aftercare, return to normal activities, follow-up evaluation, and a contact telephone number. This advice must include early warning signs and the appropriate action to take.
- Ambulatory surgery remains very popular with patients with infrequent rates of adverse events and complications.

Source: Miller 10e, Chapter 68, Key Points, PDF p. 2564.

Chapter 69: Non–Operating Room Anesthesia

- The non–operating room space represents an expanding area of anesthetic practice. Technological advancements in medicine have given way to the evolution of complex percutaneous procedures. As a result, the volume of non–operating room anesthesia (NORA) cases continues to increase. Patient safety and satisfaction have continued to be a priority. This, in turn, has increased the demand for anesthesia services to deliver a wide variety of care, ranging from sedation to general anesthesia with invasive monitoring.
- Operational constraints create additional challenges. Significant differences in practice result from the fact that NORA cases occur in locations remote from the operating room (OR) and are performed frequently by medical proceduralists who are less acquainted with the scope of anesthesiology practice. In addition, anesthesiologists may be less familiar with the demands of providing an anesthetic in a suboptimal environment in which equipment restricts mobility and access to patients, setups distinct from the OR are utilized, and where radiation hazards are significant.
- This chapter serves as a general guide to the cadence and focus of procedures performed outside of the OR, and highlights some of the adaptations, both cultural and practical, that are needed to provide a safe and optimal anesthetic.

Source: Miller 10e, Chapter 69, Key Points, PDF p. 2612.

Chapter 70: Clinical Care in Extreme Environments: Physiology at High Altitude and in Space

- High altitude or space environments present several extreme physiologic challenges that must be overcome to survive.
- Given sufficient time, humans can adapt to both hypobaric hypoxia and microgravity.
- Lack of adaptation can lead to environment-specific illnesses, such as acute mountain sickness, high-altitude pulmonary edema, decompression illness, or the acute worsening of comorbid conditions.
- These conditions can rapidly become fatal if not treated appropriately (e.g., with either descent to lower altitudes or returning to the Earth's surface).
- Providing critical care or anesthesia in such environments is further complicated by their extreme levels of remoteness.
- Exploratory missions to such environments depend on the development and vetting of robust and simple health care protocols.

Source: Miller 10e, Chapter 70, Key Points, PDF p. 2641.

Chapter 71: Clinical Care in Extreme Environments: High Pressure, Immersion, Drowning, Hypo-, and Hyperthermia

- Hyperbaric oxygen exposure (breathing oxygen at increased ambient pressure, typically 2–3 atmospheres absolute [ATA]) causes an increase in arterial and tissue partial pressure of oxygen (PO₂) and no significant change in arterial pH or partial pressure of carbon dioxide in arterial blood (PCO₂).
- During hyperbaric oxygen therapy, cardiac output and pulmonary vascular resistance are decreased; systemic vascular resistance is increased.
- Acute illnesses for which hyperbaric oxygen is indicated include carbon monoxide (CO) poisoning (based on randomized, controlled studies), gas bubble injury (gas embolism and decompression sickness [DCS]), and soft tissue necrotizing infections (the latter two based upon clinical experience and meta-analyses).
- The decision to use hyperbaric oxygen to treat a patient with arterial gas embolism (AGE) or DCS should be based on clinical criteria, including the presence of symptoms, abnormal physical examination, or a history of AGE within a few hours, even in the absence of symptoms. Neither neurophysiologic testing nor radiographic imaging is useful except, rarely, to exclude other pathologies.
- The decision to use hyperbaric oxygen to treat a patient with CO poisoning should be based on clinical criteria, including a history of impaired consciousness or other neurologic manifestation, pregnancy, or severe exposure (e.g., peak carboxyhemoglobin [HbCO] greater than 25%). The HbCO level correlates poorly with the severity of the illness and is generally useful only to make the diagnosis and a rough estimate of the degree of exposure.
- Oxygen-induced seizures are rare, self-limited and have no sequelae. Appropriate management includes discontinuation of inhaled oxygen. Chamber pressure should not be altered during the seizure, as decompression while a seizure is occurring can result in pulmonary barotrauma (pneumothorax or pneumomediastinum) and AGE.
- Recent animal and human data support that pretreatment of patients with hyperbaric oxygen may ameliorate some adverse effects of cardiac surgery and invasive cardiac procedures.
- As ambient pressure is altered, anesthetic vaporizers (except for desflurane) deliver a variable concentration but fixed partial pressure of vapor. Therefore, there is no need to adjust vaporizer settings when delivering anesthesia in a hyperbaric chamber or at altitude. Desflurane vaporizers deliver a fixed concentration, requiring upward adjustment at altitude.
- Immersion in water causes an acute redistribution of blood from the extremities and splanchnic vessels to the heart and pulmonary vessels. This can precipitate pulmonary edema (immersion pulmonary edema [IPE], swimming-induced pulmonary edema [SIPE]) in some individuals, especially during heavy exercise or in the presence of myocardial dysfunction. SIPE usually responds to removal from the water and oxygen therapy.
- Prolonged immersion induces diuresis and plasma volume loss that can predispose to severe postural hypotension after extraction from the water.
- Care of a drowning victim includes resuscitation when necessary, oxygen, and supportive care. All drowning victims who require resuscitation should be transported to the hospital for evaluation and monitoring, even if they appear alert and demonstrate effective cardiorespiratory function at the scene.

- In victims of either hypothermia or hyperthermia, optimal temperature measurement should be core (e.g., rectal or other invasive method).
- Extracorporeal life support (ECLS) for circulatory support and controlled rewarming is usually recommended for core temperatures less than 30°C.
- Suspected victims of heat exhaustion or heat stroke should be assessed with rectal temperature measurement. Victims of heat exhaustion (temperature normal or slightly above normal) can usually be treated with minimal external cooling. Heat stroke (core temperature 40°C–47°C) should be treated aggressively, ideally with immersion in ice water until a core (rectal) of 39°C or less is reached.

Source: Miller 10e, Chapter 71, Key Points, PDF p. 2671.

Chapter 72: Pediatric Anesthesia

- Age-related differences in the anatomy and physiological responses to surgery and anesthesia and the surgical lesions underlie the relevant differences between pediatric patients and their adult counterparts.
- Neonates are at highest risk for perioperative morbidity and mortality, with respiratory and cardiac-related events accounting for most of these complications.
- At birth, blood flow converts from a parallel to serial circulation. This transition places some newborns at risk of sudden hypoxemic respiratory failure due to persistence of fetal circulation. This condition may be triggered by hypoxia, hypercapnia, acidosis, and infection.
- The neonatal airway differs from the adult airway in several ways: the larynx is located higher in the neck, the glottis is shaped differently and angled over the laryngeal inlet, and the vocal cords are angled.
- Neonates have relatively larger volumes of distribution and lower clearances for most drugs. Allometric scaling (e.g., body mass) is a more accurate way to determine the dose requirement for most drugs in children rather than relying on a mg/kg calculation.
- The minimum alveolar concentration (MAC) of volatile anesthetic agents is higher in children and, for most agents, is lower in neonates when compared to adults. Neonates and infants achieve more rapid equilibration of inspired-to-brain concentrations of volatile agents.
- Former preterm infants are at risk for postoperative apnea. The use of regional anesthesia in these children may reduce the incidence of immediate postanesthesia apnea, but ongoing monitoring of the preterm infant is critically important.
- Neonates and infants are vulnerable to intraoperative hypothermia due to their large body surface-to-weight ratio and require rigorous attention to shielding from cold environment and active warming.
- Preoperative anxiety is common in children. Behavioral techniques, premedication with hypnotic drugs, and parental presence at induction have all been shown to reduce anxiety.
- Preschool children are at risk of postoperative delirium and/or agitation. Agitation may be due to many factors including pain, fear, and hunger.
- Anesthetic-induced neurotoxicity and behavioral deficits are associated with exposure to both inhaled and intravenous anesthetic drugs in laboratory settings. Some human studies have found an association between exposure to anesthesia and surgery in early childhood and subsequent neurodevelopmental issues. This may be explained by confounding factors.

Source: Miller 10e, Chapter 72, Key Points, PDF p. 2709.

Chapter 73: Anesthesia for Pediatric Cardiac Surgery

- From birth through adolescence, organ system maturation affects physiologic function and, therefore, anesthetic and surgical management and outcome.
- The understanding of congenital heart disease (CHD) and consequent anesthetic management is based on the pathophysiologic determinants of four categories of defects: shunts, mixing lesions, stenotic lesions, and regurgitant lesions.
- The chronic sequelae of CHD (repaired, palliated, or unrepaired) affect anesthetic management: ventricular failure, residual hemodynamic effects (e.g., valve stenosis), arrhythmias, and pulmonary blood flow changes (e.g., pulmonary artery hypertension).
- Preoperative assessment of cardiac status and planning are the keys to a successful anesthesia outcome.
- Intraoperative transesophageal echocardiography and central nervous system (CNS) monitoring enhance the surgical outcome and reduce morbidity.
- The induction technique needs to consider the degree of cardiac dysfunction, cardiac defects, level of sedation achieved with premedication, and the presence of an indwelling venous catheter. Other factors to consider include pulmonary hypertension, electrical rhythm disorders, and other comorbidities.
- The maintenance of anesthesia depends on the age and condition of the patient, the nature of the surgical procedure, the duration of cardiopulmonary bypass (CPB), and the need for postoperative ventilation. It is usually desirable to return the patient to an arousable, sedated state with spontaneous ventilation at the end of the procedure.
- The physiologic effects of CPB on neonates, infants, and children are significantly different from those on adults. During CPB, pediatric patients are exposed to biologic extremes not seen in adults, including deep hypothermia (18°C), increased hemodilution, low perfusion pressures, and wide variation in pump flow rates.
- After the repair of complex congenital heart defects, separating patients from CPB may be difficult. The causes may be an inadequate surgical result, pulmonary artery hypertension, low systemic vascular resistance, low hemoglobin, and right or left ventricular dysfunction.
- Modified ultrafiltration (MUF) reverses the deleterious effects of hemodilution and the inflammatory response associated with CPB in children. Furthermore, with MUF, perioperative blood loss and blood use are significantly reduced, left ventricular function and systolic blood pressure are improved, oxygen delivery is increased, and pulmonary compliance and brain function are also better.
- Neonates, infants, and children undergoing cardiac surgery with CPB have a more frequent postoperative bleeding rate than older patients. This is due to an inflammatory response that a disproportionate exposure to the nonendothelialized extracorporeal circuit produces, the type of surgery performed in neonates and infants that involves the more extensive reconstruction and suture lines, the frequent utilization of deep hypothermia or circulatory arrest, a more immature coagulation system in neonates, and an increased bleeding tendency in patients with cyanotic heart disease.
- The management of the postoperative patient warrants an understanding of both normal and abnormal convalescence after anesthesia and cardiac surgery, as it is characterized by continuous physiologic change.

- The care of adults with CHD is a new and expanding field of medicine that requires the skilled management of an experienced multidisciplinary team.
- Additional anesthetic considerations exist in patients with CHD who undergo transplantation, closed heart operations without CPB, cardiac interventional procedures, and noncardiac surgery.

Source: Miller 10e, Chapter 73, Key Points, PDF p. 2750.

Chapter 74: Regional Anesthesia in Children

- Regional anesthesia has become routine and an integral part of everyday pediatric anesthesia practice as a viable option for treating children's intraoperative and postoperative pain control.
- The blocks can be performed with greater certainty and ease if the clinician is familiar with the anatomic relationship between the nerves, surrounding bone, and blood vessels.
- Growing evidence supports the routine use of peripheral catheter techniques in pediatric patients. Data from large pediatric databases have bolstered the safety of pediatric regional anesthesia. Improved pain management and facilitated home care are just some of the benefits of these regional anesthesia techniques, which also have the potential to allow for earlier ambulation and enhanced recovery in children.
- Ultrasound guidance gradually replaces landmark techniques and nerve stimulation localization approaches in pediatric regional anesthesia. The advantages of the ultrasound-guided regional technique include visualization of targeted nerves and spaces, the spread of injected local anesthetic, and improved block safety.
- With the advent of ultrasound technology, interfascial plane blocks have entered a new era that was previously impossible.
- Using ultrasound guidance for peripheral nerve blocks allows for a reduction in local anesthetic doses, a reduction in the number of punctures, and a shortened onset time and prolonged duration of the sensitive block.
- To avoid local or systemic toxicity, the dose of local anesthetic should be calculated in volume and concentration. The use of lipid rescue (i.e., intralipid) has reduced the number of serious complications associated with regional anesthesia.
- Throughout this chapter, we have employed the most up-to-date, standardized nomenclature of regional blocks as advocated by the American Society of Regional Anesthesia and Pain Medicine (ASRA) and the European Society of Regional Anesthesia and Pain Therapy (ESRA).

Source: Miller 10e, Chapter 74, Key Points, PDF p. 2813.

Chapter 75: Pediatric and Neonatal Critical Care

- Congenital heart disease causes significant alterations in oxygenation, perfusion, and myocardial function after birth, and it can be categorized into hypoxic and normoxic lesions.
- The overall goal of therapy in shock is to treat the underlying cause, return adequate oxygen delivery to the tissues, and remove metabolic products that developed during anaerobic metabolism. The faster the body returns to adequate perfusion, the better the overall outcome.
- One of the specifics for neonatal resuscitation is the recommendation of positive pressure ventilation (PPV) with room air, unless chest compressions or medications are needed during the resuscitation, then the recommendation is still for PPV with 100% oxygen.
- Pediatric cardiac arrest is not a rare event. At least 16,000 American children (8–20 per 100,000 children per year) suffer a cardiopulmonary arrest each year.
- The four distinct phases of cardiac arrest and cardiopulmonary resuscitation (CPR) interventions are (1) prearrest, (2) no-flow (untreated cardiac arrest), (3) low-flow (CPR), and (4) postresuscitation and arrest.
- Pediatric acute respiratory distress syndrome (PARDS) diagnostic criteria have been revised and are based upon pediatric-specific consensus. Notable changes include using oxygenation index (OI) and Oxygen Saturation Index (OSI) instead of P/F ratio, and including children currently supported with noninvasive positive pressure.
- Traumatic brain injury (TBI) comprises two components: an initial primary injury due to direct mechanical deformation of brain parenchyma and a subsequent secondary injury that can develop over hours to days. Secondary injury may result from multiple mechanisms including ischemia, excitotoxicity, metabolic failure and eventual apoptosis, cerebral swelling, axonal injury, and inflammation and regeneration.
- In sickle cell disease, a vascular occlusive crisis in the lungs leads to acute chest syndrome (ACS). Acute chest syndrome is the leading cause of death and the second most common complication of sickle cell disease.
- Tumor lysis syndrome is a metabolic crisis precipitated by acute lysis of many tumor cells. Serum uric acid, potassium, and phosphate concentrations are elevated. The elevated phosphate concentrations cause hypocalcemia.
- The role of family in the pediatric intensive care unit (PICU) has evolved, and the inclusion of family in the care of their child is now recognized as an essential part of critical care.
- Accidents and trauma are the leading causes of death in children 1 to 14 years of age.

Source: Miller 10e, Chapter 75, Key Points, PDF p. 2878.

Chapter 76: The Postanesthesia Care Unit

- Emergence from general anesthesia and recovery from surgery may be accompanied by a number of physiologic disturbances that include postoperative nausea and vomiting (PONV), hypoxia, hypothermia and shivering, and cardiovascular instability.
- The ASA Practice Guidelines for Postanesthetic Care recommends assessment and monitoring of the respiratory system (respiratory rate, airway patency, oxygen saturation), cardiovascular system (pulse rate and blood pressure), neuromuscular system, mental status, pain, and nausea and vomiting for every patient during recovery.
- The most frequent cause of airway obstruction in the immediate postoperative period is decreased pharyngeal muscle tone in a sedated or obtunded patient due to the persistent effects of inhaled and intravenous anesthetics, neuromuscular blocking drugs, sedative-hypnotic medications, and opioids. These effects can be compounded in patients with obstructive sleep apnea (OSA).
- Postoperative residual neuromuscular blockade is a common occurrence in the postanesthesia care unit (PACU). Pharyngeal function is not normalized until an adductor pollicis train-of-four (TOF) ratio is greater than 0.90. Residual neuromuscular blockade should be immediately treated with neostigmine or sugammadex.
- Respiratory failure in the immediate postoperative period is often due to transient and rapidly reversible conditions such as splinting from pain, diaphragmatic dysfunction, muscular weakness, and pharmacologically depressed respiratory drive.
- Perioperative renal dysfunction from inadequate hydration, intravenous contrast agents, and/or rhabdomyolysis are common in the PACU. Patients at risk for these complications should have their renal function assessed and early intervention should be initiated to limit the development of acute kidney injury.
- Postoperative delirium is associated with worse surgical outcomes, increased hospital length of stay, functional decline, higher rates of institutionalization, higher mortality, and higher cost and resource utilization.
- The ASA Practice Guidelines for Postanesthetic Care recommend the following Guidelines for Discharge: the patient's mental status should be at their baseline, vital signs should be stable, discharge should occur after patients have met specific criteria ideally through use of a scoring system, patients should be discharged to a responsible adult, and patients should be provided with written instructions regarding postprocedure care.
- The PACU may serve as an ideal location for the care of intensive care unit (ICU) overflow patients when there are hospital capacity constraints, as exemplified during the COVID-19 pandemic. In general, it is important that PACU staffing meet the nursing staffing ratios and competencies required in the critical care units as outlined by the Joint Position Statement on ICU Overflow Patients developed by American Society of PeriAnesthesia Nurses (ASPAN), American Association of Critical Care Nurses (AACN), and American Society of Anesthesiologists (ASA). If anesthesia machines are used instead of traditional ICU ventilators in the event of a shortage, it is important to be familiar with the advantages and challenges associated with anesthesia machines.

Source: Miller 10e, Chapter 76, Key Points, PDF p. 2963.

Chapter 77: Acute Postoperative Pain

- The process of nociception is a dynamic process (i.e., neuroplasticity) with multiple points of modulation. Persistent noxious input may result in relatively rapid neuronal sensitization and possibly persistent pain.
- Postoperative pain, especially when poorly controlled, results in harmful acute effects (i.e., adverse physiologic responses) and chronic effects (i.e., delayed long-term recovery and chronic pain).
- By preventing central sensitization, preventive analgesia may reduce acute and chronic pain. Although studies overwhelmingly support the concept of preemptive analgesia, the evidence from clinical trials is equivocal mostly because of methodological issues.
- Multimodal analgesia entails use of multiple classes of analgesic drugs (acetaminophen, gabapentinoids, nonsteroidal anti-inflammatory drugs [NSAIDs], ketamine) to act on different receptors along the pain pathway. Different drugs act synergistically to enhance analgesia and reduce side effects resulting from use of an individual class of drugs. Use of multimodal analgesia is recommended whenever feasible.
- By allowing individual titration of analgesic drugs, use of patient-controlled analgesia (oral, subcutaneous, iontophoretic, intravenous, paravertebral, or epidural) may provide several advantages over traditional provider-administered analgesia (e.g., intramuscular or intermittent intravenous injections) in the management of postoperative pain.
- The incidence of respiratory depression from opioids is not significantly different among the various routes of administration (i.e., oral, intravenous versus intramuscular versus subcutaneous versus neuraxial). Appropriate monitoring of patients receiving opioid analgesics is essential to detect those with opioid-related side effects, such as respiratory depression.
- Compared with systemic opioids, perioperative epidural analgesia may confer several advantages, including a facilitated return of gastrointestinal function and decrease in the incidence of pulmonary complications, coagulation-related adverse events, and cardiovascular events, especially in higher-risk patients or procedures. However, the risks and benefits of epidural analgesia should be evaluated for each patient, and appropriate monitoring protocols should be used during postoperative epidural analgesia.
- Epidural analgesia is not a generic entity because different catheter locations (catheter-incision congruent versus catheter-incision incongruent), durations of postoperative analgesia, and analgesic regimens (local anesthetics versus opioids) may differentially affect perioperative morbidity.
- Postoperative pain management should be tailored to the needs of special populations (e.g., opioid-tolerant, pediatric, and obese patients, as well as those with obstructive sleep apnea) who may have different anatomic, physiologic, pharmacologic, or psychosocial issues.

Source: Miller 10e, Chapter 77, Key Points, PDF p. 2996.

Chapter 78: Perioperative Neurocognitive Disorders

- Concerns about postoperative cognitive function have been voiced by older patients, their families, and their caregivers for more than a century and, more recently, have been documented with objective testing.
- Perioperative neurocognitive disorders (PNDs) are the most common perioperative complications in older adults, and the strongest risk factors for PNDs are advanced age and preexisting cognitive impairment, which is usually undiagnosed.
- The new nomenclature for PNDs requires the presence of subjective cognitive complaints in addition to objective neuropsychological testing and assessment of daily function.
- The new PND nomenclature aligns with the diagnosis of cognitive disorders in the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) and thus has broader clinical relevance.
- Objective testing can take many forms and can be variously analyzed to establish whether cognitive decline has occurred.
- Further evidence is required to determine if targeting light anesthesia using electroencephalography may reduce/prevent postoperative delirium and subsequent long-term decline.
- Many mechanisms for PND have been proposed and tested in preclinical models. Neuroinflammation induced by surgery and disease in the setting of a vulnerable brain is the mechanistic hypothesis that has the most support at this time.
- Blood and EEG biomarkers may offer a unique method to identify those at risk and may prove to be useful targets for prevention of delirium and other PNDs.

Source: Miller 10e, Chapter 78, Key Points, PDF p. 3028.

Chapter 79: Critical Care Anesthesiology

- Anesthesiologists are key contributors to the field of critical care, including advancing clinical practice, developing technology relevant to the field, and caring for the sickest patients with multisystem organ failure.
- Intensive care unit structure, staffing, and utilization varies considerably across states and countries. Studies have generally shown differences in staffing and structure affect patient outcomes.
- Delivery of mechanical ventilation affects patient outcomes.
- Low tidal volume lung-protective ventilation increases survival for patients with acute respiratory distress syndrome (ARDS) but may be important in all patients receiving mechanical ventilation.
- Early identification and appropriate treatment of sepsis affects patient outcomes.
- The Surviving Sepsis Campaign's guidelines and bundles promote early recognition, administration of antibiotics, and resuscitation with fluids and vasopressors.
- Hemodynamic resuscitation in shock requires fluid responsiveness evaluation and assessment of end-organ perfusion. Evidence suggests that excess fluid administration also harms critically ill patients.
- Cancer immunotherapy continues to evolve with significant toxicities, including cytokine release syndrome, presenting new management challenges for intensivists.
- Anesthesia intensivists played a vital role in caring for patients with COVID-19.
- Point-of-care ultrasound is utilized to diagnose and manage shock in critically ill patients, but more research is needed to evaluate the impact on clinical outcomes.
- Cardiac critical care medicine has emerged as a high-volume and increasingly complex subspecialty within critical care, with significant involvement of anesthesia intensivists.
- The use of extracorporeal membrane oxygenation (ECMO), extracorporeal CO₂ removal (ECCO₂R), and mechanical circulatory support (MCS) are increasingly utilized to maintain oxygen delivery in critically ill patients.

Source: Miller 10e, Chapter 79, Key Points, PDF p. 3045.

Chapter 80: Neurocritical Care

- Critical care of the nervous system is based on control of cerebral and spinal cord physiology and the prevention of secondary insults. This goal, in turn, depends on the comprehensive maintenance and adequacy of physiologic parameters and organ function.
- Cerebral function is critically dependent on oxygen delivery matching metabolism.
- Increased intracranial volume beyond the capacity of compensatory mechanisms increases intracranial pressure (ICP) and may diminish perfusion adversely. The resulting cellular energy failure both initiates and propagates edema and inflammation.
- The resolution of cerebral edema depends on hydrostatic and osmolar forces applied to the blood–brain barrier. Excess perfusion pressure or intravascular hypotonicity worsens edema and must be avoided.
- Blood–brain barrier disruption varies over time and by pathologic process, and it affects the ability of hypertonic agents to exert a beneficial osmotic effect.
- Fever is common and challenging to manage in the neurocritical care unit. It is associated with worse patient outcomes across a range of pathologic processes.
- Neurologic monitoring comprises placement of appropriate monitoring devices as well as prompt response and institution of therapy to the changes detected. The goal is to optimize the physiologic environment. Clinical examination of neurologic function remains a crucial part of monitoring and care.
- The main principle in the treatment of traumatic brain injury involves the control of physiologic parameters such as ICP to achieve an adequate cerebral perfusion pressure.
- Although the pathophysiologic mechanisms leading to secondary brain damage are well characterized, there are no pharmacological interventions to minimize it.
- After the initial hemorrhage and early brain injury, mortality and morbidity from subarachnoid hemorrhage (SAH) arise from rebleeding and delayed cerebral ischemia.
- Early treatments of ruptured aneurysms, including medical and endovascular therapies, to improve cerebral perfusion, maintain blood volume, and optimize oxygen delivery, have improved outcomes. SAH may be accompanied by significant pulmonary, cardiovascular, or endocrine effects that must be minimized.
- Successful therapy for ischemic stroke is contingent on a time window of viability. Urgent assessment and rapid treatment are crucial to a good outcome. Endovascular therapy, in conjunction with advances in imaging, has led to significant outcome benefits.
- Injury to the spinal cord necessitates careful evaluation of respiratory mechanics to assure adequate ventilation.
- Early recognition of neuromuscular disorders and initiation of ventilatory support, treatment interventions, and supportive care in the neurocritical care unit are beneficial for patient outcomes.
- Infections of the central nervous system demand an aggressive approach to volume resuscitation, cerebrospinal fluid sampling, and early empiric antibiotic therapy, similar to treatment for sepsis.

Source: Miller 10e, Chapter 80, Key Points, PDF p. 3069.

Chapter 81: Extracorporeal Membrane Oxygenation and Cardiac Devices

- Cardiopulmonary bypass is the progenitor of extracorporeal membrane oxygenation (ECMO) and both provide circulatory support and gas exchange for patients with severe but potentially reversible respiratory and/or cardiac failure. Although the term extracorporeal life support (ECLS) might describe the system more accurately, ECMO is most often used and universally accepted for describing all its applications.
- Different configurations of ECMO (e.g., venovenous [VV], venoarterial [VA], venous to pulmonary artery [VPA]) along with hybrid circuits (VV-A, V-AV) can be utilized depending on the severity of the organ failure (e.g., respiratory failure, cardiogenic shock, cardiogenic shock associated with respiratory failure, respiratory failure in association with right ventricular failure). The most rapid way to initiate ECMO urgently (excluding patients undergoing cardiac surgery) is peripheral VA cannulation.
- ECMO must be viewed as a “bridging” therapy, where the anticipated outcome is either recovery or replacement of the failing heart, lung, or both. Assessment of the likelihood of survival using recently published scoring systems, and of the patient’s potential candidacy for transplant or cardiac device placement should be made prior to initiation. The decision to proceed with ECMO should be made by a team rather than an individual.
- Overall survival from VV ECMO is now approximately 60% in adults with acute respiratory distress syndrome (ARDS) from viral or bacterial infection, whereas survival from VA ECMO in adults with severe cardiac failure is approximately 40%. Where ECMO is urgently initiated in the setting of cardiopulmonary resuscitation, survival in adults is 29%.
- Vascular access is a critical aspect of ECMO as different sites are associated with different characteristics of flow and gas exchange in combination with the patient’s native heart and lung functions. Vascular complications are common, with arterial complications being more frequent and severe than venous complications.
- Anticoagulation management is institution specific, varying by the nature of the circuit (how much of the ECMO circuit is heparin bonded), flow (greater anticoagulation needed for lower flows), and which coagulation tests are readily available. Bleeding and clotting complications are common.
- The cardiac anesthesiologist plays an important role in the cannulation and decannulation of patients, managing sedation and cardiac support medications and providing echocardiographic assessment of cannula placement and cardiac function. The anesthesiologist-intensivist is an integral part of the management team in the intensive care unit (ICU), providing a continuum of care through ECMO management and potential transition to advanced therapies.

Source: Miller 10e, Chapter 81, Key Points, PDF p. 3103.

Chapter 82: Cardiopulmonary Resuscitation and Advanced Cardiac Life Support

- Cardiac arrest is a major public health issue worldwide. Despite significant advances in resuscitation science, survival rates remain considerably low. Improvement of patient survival and neurologic outcome relies on the development and implementation of vigorous and evidence-based resuscitation guidelines involving basic life support (BLS), advanced cardiovascular life support, and post-cardiac arrest care.
- In cardiac arrests without hypoxic causes, oxygen content in the lungs at the time of cardiac arrest is usually sufficient for maintaining acceptable arterial oxygen content during the first several minutes of cardiopulmonary resuscitation (CPR). Blood flow rather than arterial oxygen content is the limiting factor for oxygen delivery to coronary, cerebral, and systemic circulation during CPR. Thus, rescue breaths are less important than initiating effective chest compressions as soon as possible after sudden cardiac arrest (SCA).
- The mechanism through which chest compressions generate blood flow can be explained by the thoracic or cardiac pump theories. The provision of uninterrupted, high-quality chest compressions after SCA is associated with better survival and neurologic outcomes than delaying chest compressions for airway intervention in both adult and pediatric patients. "Circulation, airway, breathing" has replaced "airway, breathing, circulation."
- A single resuscitative shock should be delivered at the earliest possible opportunity after the recognition of cardiac arrest with a shockable rhythm (ventricular fibrillation or pulseless ventricular tachycardia), followed immediately by the resumption of chest compressions without post-shock cardiac rhythm analysis. Outcome studies have failed to demonstrate the benefit of a period of chest compressions before shock or for a series of stacked shocks.
- Vasopressor medications during resuscitation have been de-emphasized in deference to providing uninterrupted, high-quality chest compressions. Standard-dose epinephrine (1 mg every 3–5 minutes) is recommended for patients in cardiac arrest. Vasopressin offers no advantage as a substitute for epinephrine in cardiac arrest and has been removed from the adult cardiac arrest algorithm.
- Continuous flow left ventricular assist devices result in an unconventional, unique physiologic state of hemodynamically stable pulseless electric activity. Assessment of adequate tissue perfusion is the most important factor in determining the need for circulatory assistance such as chest compressions. Total artificial hearts (TAHs) are refractory to chest compressions, antiarrhythmic drugs, and electric therapy. Vasopressor medications are contraindicated because they increase afterload, result in complete hemodynamic collapse with pulmonary edema, and worsen TAH function.
- In consideration of opioid overdose epidemiology, patients with known or suspected opioid ingestion who are in cardiac or respiratory arrest should receive intravenous, intramuscular, or intranasal naloxone in addition to standard BLS care.
- For non-shockable rhythms, the essential step will be early detection and correction of potentially reversible underlying causes. Ultrasound technology is used to assess the etiology and the management of these patients, as well as to predict the possibility of return of spontaneous circulation (ROSC) and to justify the termination of resuscitative efforts. However, utilization of this technique should not interfere with other resuscitation efforts such as chest compressions.
- Asphyxiation is a much more common cause of cardiac arrest in infants and children than primary

cardiac events, and airway management and ventilation are therefore more important during the resuscitation of children. However, in order to facilitate training, retention, and implementation of resuscitation guidelines, the pediatric resuscitation guidelines follow similar principles as adult guidelines.

- Targeted temperature management (TTM) applied to comatose survivors of out-of-hospital cardiac arrests has significantly improved the neurologic recovery in those surviving to hospital discharge. A target temperature between 32°C and 36°C is recommended for at least 24 hours, and normothermia (to treat fever) should be maintained beyond this window. Prognostication should not occur until 72 hours after ROSC or, if TTM is provided, 72 hours after completion of TTM.
- Most deaths after SCA in both adults and children typically occur within the first 24 hours. Coordinated postresuscitation care involving access to coronary catheterization capabilities and intensive care management that includes TTM represents the best chance survivors of SCA have for optimal neurologic and cardiac recovery.
- New technologies such as individualized CPR, extracorporeal CPR, controlled automated reperfusion of the whole body (CARL), and emergency preservation for delayed resuscitation may offer opportunities for patients suffering from cardiac arrest.

Source: Miller 10e, Chapter 82, Key Points, PDF pp. 3125–3126.

Chapter 83: Burn Management

- Major burn injury results in pathophysiologic changes affecting virtually all organs from the onset of injury until wounds are healed. Anesthesiologists are often called on to care for burn injury patients throughout their hospitalization including acute airway management, resuscitation, intraoperative anesthetic care, intensive care, and management of postoperative pain.
- Burn shock is a paradigm of ischemia/reperfusion injury. The initial ischemic (ebb) phase of burn shock is a hypodynamic, hypovolemic state, with rapid loss of fluid from the intravascular space and decreased cardiac output, which typically lasts for the first 24 to 48 hours following injury. Approximately 48 hours after successful resuscitation, a hyperdynamic, hypermetabolic (flow) phase ensues consisting of tachycardia, increased cardiac output, hyperthermia, hyperglycemia, and increased protein catabolism.
- The goal of fluid resuscitation is to maintain organ perfusion by replenishing the massive loss of fluid from intravascular to extravascular compartments. Multiple fluid resuscitation formulae exist for estimating fluid needs and differ somewhat in their recommendations for the amount of crystalloid and colloid replacement. No matter which formula is used, it should serve only as a guideline, and fluid resuscitation should be titrated to physiologic end points.
- Inhalation injury is a major prognostic factor for morbidity and mortality after burn injury. Management of inhalation includes observation and monitoring. Endotracheal intubation or tracheostomy is indicated if airway patency is threatened.
- Patients with severe burn injury often suffer from nonthermal traumatic injuries. Failure to diagnose these associated injuries during initial evaluation can lead to serious morbidity and mortality. All burn patients should be approached initially as multiple-trauma patients.
- The magnitude of burns is classified according to percentage of total body surface area (TBSA) involved, depth of the burn, and the presence or absence of inhalational injury. Accurate estimation of burn magnitude is needed to guide the initial resuscitation strategy, make the referral to a burn center, ascertain the need for surgery, and estimate prognosis. Three of the most commonly used methods to estimate %TBSA are the “rule of nines,” palmar surface area, and the Lund–Browder diagram.
- “Fluid creep” refers to the trend of over-resuscitation in burn injury patients. Overly aggressive fluid administration may result in pulmonary edema, compartment syndromes, multiorgan failure, nosocomial infection, and increased mortality as well as the extension of the burn injury because of excessive local edema. Factors that contribute to “fluid creep” include overestimation of the burn size, use of supra-physiologic hemodynamic targets, increased opioid utilization, and failure to reduce the rate of fluid administration in patients with evidence of adequate tissue perfusion.
- Electrical burns can have acute and chronic effects not occurring with other types of burn injury, and with morbidity far higher than expected based on burn size estimation alone. High-voltage injuries are typically associated with loss of consciousness, arrhythmias, myoglobinuria, and extensive deep tissue damage that can result in compartment syndromes. Patients suffering from electrical injury should be evaluated for associated traumatic injury, rhabdomyolysis, and compartment syndromes. For treatment of these complications, patients may come to the operating room within 24 hours of injury.
- Airway management in the burn injury patient may be challenging and warrants particular consideration. Key features of airway assessment include preexisting airway abnormality, current airway injury (i.e., inhalation injury), and signs of glottic obstruction. The type of airway abnormalities may vary depending on the stage of the injury. In the acute burn setting, mandibular

mobility and mouth opening may be limited because of edema or, in later care, may have significant scarring and contractures in the face, mouth, nares, neck, and chest which can make airway management difficult.

- Burn-injured patients develop tolerance to most narcotics and sedatives, thereby requiring substantially higher doses than patients without thermal injury. Sedatives and narcotics should be titrated to effect while the patient is carefully monitored. Adverse effects of opioids, such as respiratory depression, acute opioid tolerance, and hyperalgesia, particularly with the need for rapidly escalating doses, have generated increasing attention to multimodal strategies.
- Muscle relaxant pharmacology is significantly and consistently altered after burn injury. Exposure to succinylcholine can result in an exaggerated hyperkalemic response, which can induce cardiac arrest. The current recommendation is to avoid succinylcholine administration in patients beginning 48 to 72 hours after burn injury. The duration of this dangerous response to succinylcholine after burn injury is unknown. Conversely, resistance develops to nondepolarizing muscle relaxants, requiring larger doses to achieve similar clinical end points in patients with burn injury.
- Surgical excision of burn wounds is often associated with substantial bleeding. It is not uncommon for the surgical team to remove eschar so rapidly that the patient becomes hypovolemic and hypotensive. Clinical judgment remains a vital component for intraoperative resuscitation, using markers of tissue and organ perfusion, erythrocyte mass, coagulation, pulse, and arterial waveform as key assessment tools. Good communication between the surgical and anesthesia teams as well as limiting the operative duration and extent of excision are also essential.
- Patients with major burn injury have an impaired ability for thermoregulation and therefore require close monitoring of body temperature. Multiple strategies are used to maintain body temperature in the operating room, including raised operating room temperature (80–100°F), use of forced-air warming blankets, thermal water mattresses, blood/fluid warmers, minimizing skin surface exposure, and wrapping the head and extremities with plastic or thermal insulation.
- Postoperatively, burn-injured patients are likely to be less stable physiologically compared with the preoperative period. Continued bleeding may be concealed by dressings, the patient may be more prone to hypothermia, emergence may be associated with delirium, and analgesic requirements will be greater. During this period of exaggerated physiologic fragility, it is important to be especially vigilant during transfer of the monitors, respiratory, and hemodynamic support equipment from the operating room to the intensive care unit.
- Burn injury leads to increased susceptibility to infection due to decreased immunity through multiple mechanisms including loss of the physical barrier of intact skin, damage to the epithelium of the respiratory tract from inhalation injury, and altered gut permeability and function. Preventive measures against infection are critical for the burn injury patient and include early excision of burn eschar to improve local perfusion and prevent microbial colonization, prudent use of invasive devices, application of antimicrobial burn dressings, and diligent compliance with aseptic techniques.
- Nearly all aspects of burn care (e.g., dressing changes, excision and grafting procedures, physical therapy, and line insertion) are associated with pain. There can be ongoing background pain, periodic breakthrough pain, procedure-related intraoperative and postprocedural pain, and eventually, chronic pain can develop. Standardized pain and anxiety guidelines are used to provide appropriate, consistent patient comfort.

Chapter 84: Occupational Safety, Infection Control, and Substance Use Disorders

- The operating environment is a complex one that has potential health risks to those working there. Exposure to risks, including waste anesthetic gas, in clinical practice is unavoidable. In the United States, the limits of exposure to waste gases are set by the National Institute for Occupational Safety and Health (NIOSH), which recommends a time-weighted average of 25 ppm for nitrous oxide and a ceiling of 2 ppm for volatile anesthetics.
- Although researchers disagree whether exposure to anesthetic gases at concentrations less than NIOSH limits affects health or performance, these limits are often exceeded. If it can be smelled, the exposure is many times greater than the safe limit.
- Occupational exposure to radiation comes primarily from X-rays scattered by the patient and surrounding equipment. A distance of 3 feet from the patient is recommended to minimize physiologic damage from occupational exposure, and a distance of 6 feet from the patient provides the same protection as 2.5 mm of lead.
- Surgical smoke is increasingly recognized as a source of potentially infectious and carcinogenic material; evacuation devices should be used where the smoke plume is created.
- Diseases can be transmitted via direct contact, droplets, or airborne particles. Some diseases are infectious only if there is direct exposure to blood or body fluid from the host. Appropriate personal protective equipment based on the suspected type of infection should always be used to prevent occupational disease transmission.
- To attenuate occupational exposure to pathogens, standard precautions at a minimum should be used at all times. The appropriate barrier precautions for intubation include eye protection, a surgical mask, and gloves.
- To prevent exposure to bloodborne pathogens, sharps safety, including the use of safety retractable needles and needleless systems, should be used.
- Occupational exposure to human immunodeficiency virus (HIV) and hepatitis B and C viruses (HBV and HCV) is most often the result of a percutaneous injury. The risk of disease transmission is generally very low but is greatest from hollow-bore needles, needles contaminated with visible blood, and exposure to a source patient with high viral titer.
- Postexposure prophylaxis (PEP) is recommended after occupational exposure to HIV or HBV. The recommended guidelines for PEP are available on the Centers for Disease Control and Prevention website. The Clinical Consultation Center PEpline service is a free expert resource for guidance on PEP (1-888-448-4911).
- The preference for and access to potent opioids contribute to the prevalence of substance use disorder among anesthesiologists. The rate of drug-related deaths is more than twice as high in anesthesiologists as internists.
- Although many recovered anesthesiologists return to the practice of anesthesia, there is a significant relapse rate. The chance of relapse is highest in physicians who become addicted to potent narcotics early in their career. Successful recovery requires a lifelong commitment to treatment. In some cases, a change in specialty is the only solution.
- Sleep deprivation has an adverse effect on physician mood, cognitive function, reaction time, and vigilance. Although sleep deprivation and fatigue adversely affect clinical performance, the full

impact on patient outcome has been difficult to determine.

Source: Miller 10e, Chapter 84, Key Points, PDF p. 3199.

Chapter 85: Emergency Preparedness in Health Care

- Anesthesiologists have a history of leadership for emergency preparedness.
- The frequency and impact of emergency events are rapidly expanding.
- This expansion includes cybersecurity events and the action of targeting by malevolent forces.
- Emphasis must move from emergency response to preventive practice, as was done in quality improvement.
- Formal training and support should be made available for anesthesia teams to support these moves.

Source: Miller 10e, Chapter 85, Key Points, PDF p. 3226.

Chapter 86: Clinical Research

- Clinical research describes the characteristics and mechanisms of disease and injury in humans and investigates drugs, devices, diagnostic products, and bundles of care with the aim of providing high-quality evidence to guide clinical practice that improves patients' lives.
- Investigators are constrained by the need to draw conclusions based on a sample of patients, providers, or systems rather than the whole of the population of interest, and this introduces random, systematic, and design error that threatens internal and external validity.
- Observational (or nonexperimental) studies involve allowing nature or clinical care to take its course, without any major modification due to study-related procedures. However, nonexperimental designs are prone to systematic error, such as selection and information bias.
- In experimental studies the investigators do not let nature (or clinical care) take its course, but actively intervene to test a new intervention. Randomization and blinding reduce the risk of random and systematic errors in experimental studies, but generalizability can be limited.
- Systematic reviews synthesize the medical literature using a transparent search strategy that every reader can replicate and update, and accompanying meta-analyses produce aggregate estimates of treatment effects.
- A carefully designed protocol is the foundation of all clinical research, facilitating the review, conduct, and eventual publication of the project. Prior planning of sample size and statistical analyses is essential.
- Appropriate ethics, registration, and regulatory approvals; financial, data, and human resource management; plans for monitoring patient safety and data integrity; and plans for publishing the results and sharing the data are vital to the proper conduct of clinical research.
- A quality improvement cycle of reflection, feedback, and forward planning is beneficial. Giving consumers an opportunity to participate in research governance gives researchers fresh insight and opens new avenues of research.

Source: Miller 10e, Chapter 86, Key Points, PDF p. 3233.

Chapter 87: Interpreting the Medical Literature

- The medical literature can be accessed in many ways, ranging from the primary literature as indexed in sources like PubMed, to sources catering to medical professionals, to the lay press and social media.
- The quality of medical literature varies widely, as does the interpretation or distillation into secondary sources of information.
- The traditional “evidence pyramid” no longer effectively reflects the diversity and impact on clinical care of the rapidly evolving body of medical knowledge.
- Understanding the design of a research study is crucial to understanding the strength and implications of its conclusions.
- Different research study designs have assessment tools, maintained by the EQUATOR Network, to assist in understanding the quality of the study.
- Statistical analysis is commonplace in original research manuscripts, but the interpretation of statistical results, particularly the probability or “P-value,” is widely misunderstood and misused.
- Although the hallmark of the primary medical literature is a rigorous peer review process, there are serious pitfalls that may be difficult to detect, including predatory journals and reviewer misconduct.
- Identifying and correcting scientific misconduct is crucial to maintaining public trust in science.

Source: Miller 10e, Chapter 87, Key Points, PDF p. 3256.

Chapter 1: The History of Anesthesia

- Anesthesiology is a young specialty historically, especially when compared with surgery or internal medicine.
- Discoveries in anesthesiology have taken decades to build upon the observations and experiments of many people, and in some instances, we are still searching. For example, the ideal volatile anesthetic has yet to be discovered.
- Much of our current anesthesia equipment is the direct result of anesthesiologists being unhappy with existing tools and needing better ones to properly anesthetize patients.
- Many safety standards have been established through the work of anesthesiologists who were frustrated by the status quo.
- Regional anesthesia is the direct outgrowth of a chance observation by an intern who would go on to become a successful ophthalmologist.
- Pain medicine began as an outgrowth of regional anesthesia.
- Organizations of anesthesia professionals have been critical in establishing high standards in education and proficiency, which in turn has defined the specialty.
- Respiratory critical care medicine started as the need by anesthesiologists to use positive-pressure ventilation to help polio victims.
- Surgical anesthesia and physician specialization in its administration have allowed for increasingly complex operations to be performed on increasingly ill patients.

Source: Barash 9e, Chapter 1, Key Points, PDF p. 76.

Chapter 2: The Practice of Anesthesiology

- Organization, self-regulation, and setting standards mark the transition from artisanal to professional practice in any trade. Anesthesiology's journey to full recognition as a medical specialty ended in 1941 when the American Board of Medical Specialties approved the American Board of Anesthesiology as an independent body.
- The increasing number and variety of subspecialty training programs reflect the growing scope of practice of anesthesiology.
- The American Society of Anesthesiologists introduced the perioperative surgical home (PSH) concept as a means to achieve the triple aim and value care goals.
- Well implemented enhanced recovery after surgery (ERAS) protocols are associated with shortened hospital stays, decreased costs and fewer complications. ERAS incorporated into PSH models may yield the best results.
- Anesthesiologists are well positioned to serve in senior leadership roles even beyond the operating room.
- Certification by the American Board of Anesthesiology (ABA) demonstrates expertise in our specialty. Graduates are eligible for initial certification upon completion of an accredited residency program and for recertification every 5 years through the ABA MOCA program. The ABA also offers certifications in several subspecialties of anesthesiology.
- Reimbursement for any patient encounter requires a diagnosis code, a procedure code, with matching anesthesia code, and, for operating room anesthesia, an anesthesia start time, and anesthesia end time.
- Because of longstanding insufficient reimbursement for anesthesia services by government payers (Medicare and Medicaid) in addition to recent restrictions on balance billing of privately insured patients, many anesthesia groups now rely on hospital subsidies to balance their budgets.
- Anesthesiology was the first medical specialty to develop a systematic approach to patient safety resulting in a dramatic decrease in anesthesia-related mortality. Consequently, anesthesiologists increasingly focus on decreasing patient morbidity, such as decreasing postoperative wound infection, postoperative cognitive decline, and opioid-related harm.
- While it is necessary that hospital and department leaders promote psychological safety in their departments, individual clinicians remain key stakeholders in establishing and maintaining a culture of safety.
- Regulatory agency surveyors closely scrutinize anesthesia departments. In-depth knowledge of, and careful compliance with, federal and state rules and regulations are essential to avoid citations and penalties.
- A well-organized anesthesia department needs the support of a variety of staff and services to allow clinicians to deliver patient care safely and efficiently.
- Although difficult to predict with certainty, current trends suggest continued growth in both the supply of and demand for anesthesiologists.
- Clinicians continue to report high levels of burnout. A multipronged approach is necessary to mitigate burnout and promote physician wellness.
- A basic understanding of personal finances is essential for all physicians, including debt

management, saving for retirement and protecting assets. While a financial advisor may be helpful, it is important to understand their compensation structure as it could influence the advice they provide.

Source: Barash 9e, Chapter 2, Key Points, PDF pp. 152–154.

Chapter 3: Occupational Health and Well-Being

- The strain imposed by the COVID-19 pandemic on health care workers and the health care system has highlighted the critical role of both occupational health and well-being in medicine and the specialty of anesthesiology.
- Vigilance is one of the most critical tasks performed by anesthesiologists and may be adversely affected by several occupationally related issues including poor equipment engineering and design, excessive noise, interpersonal conflict, production pressures, and fatigue.
- Exposure to waste anesthetic gases can be reduced to levels below those recommended by the National Institute for Occupational Safety and Health by using waste gas scavenging equipment, routinely maintaining anesthesia machines, and following appropriate work practices.
- Sleep deprivation and fatigue are common among anesthesiologists and can adversely affect patient safety due to deleterious effects on the physician's cognition, mood, and health.
- Substance use disorder, or chemical dependency, is a significant, often lethal, occupational hazard among anesthesiologists. An incidence of controlled substance abuse of 1% to 2% has been repeatedly reported within anesthesia training programs. Rates have not changed despite educational and systematic interventions.
- The risk of exposure to infectious pathogens can be reduced by the routine use of standard precautions, transmission-based precautions for infected patients, and safety devices designed to prevent needlestick injuries.
- Hepatitis B vaccine is recommended for all anesthesia personnel because of the increased risk for occupational transmission of this blood-borne pathogen.
- Anesthesiologists have a disproportionately high rate of suicide. Factors include pre-existing personality traits, substance use disorder, and stress.
- Physician well-being is integral to occupational health and patient safety. There is an evolving professional commitment to optimizing physician work style and lifestyle, as well as physical health (e.g., through better nutrition, fitness, and rest).
- Creating a culture of well-being and support for physicians and other medical personnel through organizational, systemic, and policy-level interventions has become an imperative of many governmental bodies and professional organizations.

Source: Barash 9e, Chapter 3, Key Points, PDF pp. 232–233.

Chapter 4: Anesthetic Risk, Quality Improvement, and Liability

- Anesthetic mortality has decreased, but potentially preventable deaths and disabling complications still occur.
- Risk management programs are broadly oriented toward reducing the liability exposure of the organization. Risk management programs complement quality improvement programs in minimizing liability exposure while maximizing quality of patient care.
- Continuous quality improvement is a systems approach to identifying and improving quality of care. Quality improvement programs focus on improving the structure, process, and outcome of care.
- Quality improvement programs are generally guided by the requirements of the Joint Commission that accredits health care organizations and the reporting and performance requirements of the Centers for Medicare and Medicaid Services.
- Medical malpractice refers to the legal concept of professional negligence. The patient-plaintiff must prove that the anesthesiologist owed the patient a duty and failed to fulfill this duty, that the anesthesiologist's actions caused an injury, and that the injury resulted from a breach in the standard of anesthesia care.
- The most common lawsuits against anesthesiologists (excluding dental injuries) are for death, brain damage, nerve damage, and airway injury. Chronic pain management is the source of an increasing number of malpractice claims against anesthesiologists.

Source: Barash 9e, Chapter 4, Key Points, PDF pp. 276–277.

Chapter 5: Electrical Safety and Fires in the Operating Room

- To receive a shock, the electrical circuit must contact the person at two points and there must be a voltage source to cause the current to flow.
- A microshock hazard is conductive wiring or an electrolyte solution–filled catheter that starts outside the patient’s body and terminates near the patient’s heart.
- Cardiac fibrillation can be induced by a current as low as 100 μ A directly across the heart.
- The sensory perception threshold for an electrical current at 60 Hz is 1 mA.
- A typical operating room circuit has a capacity for only 15 or 20 A and is therefore easily overwhelmed by multiple high-current draw devices (e.g., fluid warmer, forced air warmer, rapid infusion system) on a single circuit.
- Line isolation monitors notify operating room personnel about potentially unsafe equipment so that it can be serviced at a time that is safe for patient care.
- Some degree of fire hazard is present in virtually every operating room. The fire triad is necessary for a fire to ignite and consists of an oxidizing agent, a fuel, and an ignition source.
- A nasal cannula delivering 2 L/min O₂ is delivering 100% oxygen out of the nasal cannula.
- An air:oxygen ratio of at least 7:1 is required to keep the ambient oxygen concentration around the open oxygen source outlet no higher than 30%.
- The most common type of fire is an on-patient fire and happens during upper-body monitored anesthesia care (MAC) cases with nasal cannula or face mask oxygen supplementation.

Source: Barash 9e, Chapter 5, Key Points, PDF pp. 319–320.

Chapter 6: Genomic Basis of Perioperative Precision Medicine

- Phenotype-driven classification fails to accurately predict perioperative outcomes or treatment response. An increasing number of genetic association studies suggest that susceptibility to a range of common adverse perioperative events and ensuing critical illness is genetically and epigenetically determined. Ongoing emphasis is placed on prioritizing genetic variants that warrant clinical action, assessing their combined predictive ability through polygenic risk scores, and integration with the electronic health record data for clinical decision support.
- Emerging approaches utilizing “omics,” physiomarkers, and imaging allow precise perioperative diagnosis and therapy. Potential applications of biomarkers in perioperative medicine and critical illness include prognosis, diagnosis, and monitoring of adverse events, as well as informing therapeutic decisions along the surgical patient journey. Very few have been rigorously evaluated to demonstrate incremental discriminatory accuracy when added to existing risk stratification models (clinical validity), or change therapy (clinical utility). Among the most promising remain natriuretic peptides and C-reactive protein (CRP) for cardiovascular risk prediction, postoperative troponin surveillance to diagnose myocardial injury, and procalcitonin to assess infection in the critically ill.
- Interindividual variability in response to anesthetic agents is as high as 24% and has underlying genetic mechanisms.
- Individual variability in analgesic responsiveness is attributed to genetic control of peripheral nociceptive pathways and descending central pain modulatory pathways.
- Pharmacogenomic variation in genes modulating drug actions explains part of the variability in drug response and has shown promising clinical utility for several classes of drugs used perioperatively.
- To facilitate translation to medical practice, systematic evaluation of existing genomic evidence for clinical decisions in the perioperative continuum, updating the practice guidelines, as well as identifying the revenue sources to reimburse the generation and use of genomic information are still required. Deliverables of this approach include classification of patients to enhance clinical trial design both in the acute perioperative period and during long-term recovery; biomarkers to select candidates for therapeutic intervention and track the efficacy of the intervention; and predictive models to increase discrimination of recovery probabilities in surgical patients.

Source: Barash 9e, Chapter 6, Key Points, PDF pp. 410–412.

Chapter 7: Experimental Design and Statistics

- Statistics and mathematics are the language of scientific medicine.
- Good research planning includes a clear biologic hypothesis, the specification of outcome variables, the choice of anticipated statistical methods, and sample size planning.
- The crucial elements of good research design include concurrent control groups; random allocation of subjects to treatment groups; concealment of random allocation; blinding of treatment assignment to patients, caregivers, and outcome assessors; and full reporting of outcomes for all study patients. These techniques minimize the risk of bias in clinical research.
- The use of graphs, descriptive statistics (e.g., mean, standard deviation), and inferential statistics (e.g., t-test, confidence interval) are important methods for the presentation of research results.
- Bayesian statistical methods report research results as a function of both observed data and historical (prior) knowledge; the more common frequentist statistical methods report research results only as a function of observed data.
- Multivariable logistic regression and propensity score matching are statistical techniques for identifying associations between risk factors and outcomes in nonrandomized studies.
- Systematic review and meta-analysis summarize the results of individual studies and permit more powerful inferences for the comparison of interventions.
- Anesthesia data does not yet have the Volume, Velocity, and Variety characteristics of Big Data.
- Resources and guidance for experimental design and statistical methods include policy statements, textbooks, journal articles, and public domain software.

Source: Barash 9e, Chapter 7, Key Points, PDF p. 494.

Chapter 8: Inflammation, Wound Healing, and Infection

- The most crucial component of infection prevention is frequent and effective hand hygiene. Wearing gloves does not reduce the need for hand hygiene.
- The ideal hand hygiene agent kills a broad spectrum of microbes, has antimicrobial activity that persists for at least 6 hours after application, is simple to use, and has few side effects.
- There is no compelling evidence that single-use disposables reduce infections, except where adequate disinfection is not (currently) possible (e.g., vascular catheters, duodenoscopes). While the cost of acquisition is lower for disposables, the per-use cost is almost always orders of magnitude greater. In addition, life-cycle analyses have shown that disposables cause demonstrable public health harm (in the range of that caused by medical errors) by increasing waste, pollution, and greenhouse gas emissions, offsetting any theoretical protection from infection.
- Antibiotic prophylaxis has become standard for surgeries in which there is more than a minimum risk of infection. The most commonly used antibiotic for surgical prophylaxis is cefazolin, a first-generation cephalosporin, as the potential pathogens for most surgeries are gram-positive cocci from the skin.
- The exact timing for the administration of preoperative antibiotic prophylaxis depends on the pharmacology and half-life of the drug, but should generally be 0 to 60 minutes before incision. Anesthesiologists should work in consultation with surgeons to take initiative for administering prophylactic antibiotics because they have access to the patient during the 60 minutes prior to incision and can optimize timing of administration.
- The standard teaching that oxygen delivery depends more on hemoglobin-bound oxygen (oxygen content) than on arterial PO₂ may be true for working muscle, but it is not true for wound healing. Although oxygen consumption is relatively low in wounds, it is consumed by processes that require oxygen at a high concentration. Delivery of oxygen to wounds depends on arterial PO₂ rather than hemoglobin-bound oxygen (oxygen content).
- Peripheral vasoconstriction, which results from central sympathetic control of subcutaneous vascular tone, is probably the most frequent and clinically the most important impediment to wound oxygenation and wound healing. All vasoconstrictive stimuli must be corrected simultaneously to allow optimal healing. Modifiable risks for wound infections include smoking, malnutrition, obesity, hyperglycemia, hypercholesterolemia, and hypertension. These should be assessed and corrected (when possible) prior to surgery.
- Prevention or correction of hypothermia has been shown to decrease wound infections and increase collagen deposition in patients undergoing major abdominal surgery. Maintenance of a high room temperature or active warming before, during, and after the operation is significantly more effective than other methods of warming, such as application of warmed blankets, circulating water blankets placed on the surface of the operating table, and humidification of the breathing circuit.
- Optimizing the volume of perioperative fluid administration to minimize morbidity and mortality remains a significant and controversial challenge. Current best recommendations for volume management include replacing fluid losses based on standard recommendations for the type of surgery, replacement of blood loss, and replacement of other ongoing fluid losses (e.g., high urine output due to diuretic or dye administration, hyperglycemia, or thermoregulatory vasoconstriction).

- Administration of supplemental oxygen via face mask or nasal cannula increases safety in patients receiving systemic opioids. As a side benefit, it may also improve wound healing. Pain control also appears important since it favorably influences both pulmonary function and vascular tone.
- In patients with moderate to high risk of surgical site infection, anesthesiologists have the opportunity to enhance wound healing and reduce the incidence of wound infections by simple, inexpensive, and readily available means.

Source: Barash 9e, Chapter 8, Key Points, PDF pp. 547–550.

Chapter 9: The Allergic Response

- Anesthesiologists routinely administer multiple parenteral agents in the perioperative period, including drugs (antibiotics, anesthetic agents, neuromuscular blocking agents [NMBAs]), polypeptides (i.e., protamine), and blood products, or patients are exposed to environmental antigens (i.e., latex) all of which can produce an allergic reaction.
- Cytokines are inflammatory cell activators synthesized to act as secondary messengers and activate endothelial and white cells.
- Antibodies are specific proteins called immunoglobulins that can recognize and bind to specific antigens, and usually, IgE or IgG is implicated in allergic reactions.
- Immune competence during surgery can be affected by direct and hormonal effects of anesthetic drugs, immunologic effects of other drugs used, type of surgery, coincident infection, and transfused blood products.
- Most of the allergic reactions evoked by intravenous drugs occur within 5 minutes of administration. In the anesthetized patient, the most common life-threatening manifestation of an allergic reaction is circulatory collapse, reflecting vasodilation, and decreasing venous return.
- Some drugs administered perioperatively, such as vancomycin, can release histamine in a dose-dependent, nonimmunologic fashion.
- A plan for treating anaphylactic reactions should be considered by clinicians that includes airway maintenance, 100% oxygen administration, intravascular volume expansion, and epinephrine, which are essential to treat the hypotension and hypoxia resulting from vasodilation, increased capillary permeability, and bronchospasm. Norepinephrine, vasopressin, and additional diagnostic monitoring should be considered for refractory shock.
- After an anaphylactic reaction, it is important to attempt to identify the causative agent to prevent readministration.
- Health care workers, children following multiple procedures, and individuals with specific food allergies have been recognized to be at increased risk for latex anaphylaxis. However, latex has been removed as an environmental antigen over the years.
- NMBAs have several unique molecular features that make them potential antigens that can cause anaphylaxis.

Source: Barash 9e, Chapter 9, Key Points, PDF pp. 604–606.

Chapter 10: Mechanisms of Anesthesia and Consciousness

- The components of the anesthetic state include unconsciousness, amnesia, analgesia, immobility, and attenuation of autonomic responses to noxious stimulation.
- Minimum alveolar concentration (MAC) remains the most robust measurement and the standard for determining the potency of volatile anesthetics (VAs).
- Anesthetic interactions with membrane lipids may also contribute to anesthetic action. Emerging evidence indicates that anesthetic effects on lipid nanodomains may modify functionally important lipid–protein interactions.
- Direct interactions of anesthetic molecules with specific binding pockets on membrane proteins can account for most of the observed cellular and behavioral effects of anesthetics. Protein interactions satisfy the Meyer–Overton rule and provide the simplest explanation for compounds that deviate from this rule.
- While current data still support the prevailing view that neuronal excitability is only slightly affected by general anesthetics, this small effect may nevertheless contribute significantly to the clinical actions of VAs.
- The synapse is generally thought to be the most likely relevant site of anesthetic action. Existing evidence indicates that even at this one site, anesthetics produce various effects, including presynaptic inhibition of neurotransmitter release, inhibition of excitatory neurotransmission, and enhancement of inhibitory neurotransmission. Furthermore, the effects of anesthetics on synaptic function differ among various anesthetic agents, neurotransmitters, and neuronal preparations.
- The essential criteria for demonstrating a protein as an anesthetic target are (i) the function of the protein is affected by the anesthetic at relevant concentrations; (ii) the functional alteration of the protein could plausibly produce a component of anesthesia; (iii) genetic or pharmacologic manipulation of the protein alters sensitivity to the anesthetic; and (iv) the protein binds to the anesthetic.
- GABAA receptors meet all the criteria as a target for certain anesthetics.
- Hyperpolarization-activated cyclic nucleotide-gated channels are inhibited by volatile and some intravenous anesthetics and may contribute to their anesthetic effect.
- Certain background K⁺ channels have electrophysiologic, genetic, and biochemical evidence implicating them as relevant VA targets.
- Some sodium and calcium channel subtypes are inhibited by VAs and these effects may be responsible in part for inhibition of neurotransmitter release and neuronal excitability.
- The shaker potassium channel subtype (Kv1.2) is activated by VAs and has both genetic and biochemical evidence implicating it as a relevant target.
- NMDA glutamate receptors are inhibited by nitrous oxide and ketamine but mammalian genetic data is currently insufficient to link this effect to anesthesia.
- VA inhibition of mitochondria and synaptic vesicle release machinery may also contribute to anesthesia.
- While anesthetic action to produce immobility occurs largely at the spinal cord, specific molecular targets for amnesia lie in the hippocampus.

- Anesthetic-induced unconsciousness can be viewed as impairment of both arousal and awareness. These actions are mediated by targets distributed across the brain stem, hypothalamus, thalamus, and cerebral cortex.
- Anesthetic ablation of arousal relies on disruption of redundant subcortical systems that regulate sleep and patterns of cortical activity.
- Anesthetics alter the interaction of cortical networks responsible for cognitive functions and may thereby alter awareness by limiting the capacity to both represent and integrate information.

Source: Barash 9e, Chapter 10, Key Points, PDF pp. 654–657.

Chapter 11: Basic Principles of Clinical Pharmacology

- Most drugs must pass through cell membranes to reach their sites of action. Consequently, drugs tend to be relatively lipophilic, rather than hydrophilic.
- The highly lipophilic anesthetic drugs have a rapid onset of action because they rapidly diffuse into well-perfused brain tissue. They have a very short duration of action because of redistribution of drug from the central nervous system to the blood.
- The cytochrome P450 (CYP) superfamily is the most important group of enzymes involved in drug metabolism. It and other drug-metabolizing enzymes exhibit genetic polymorphism.
- The kidneys eliminate hydrophilic drugs and relatively hydrophilic metabolites of lipophilic drugs. Renal elimination of lipophilic compounds is negligible.
- The liver is the most important organ for metabolism of drugs. Hepatic drug clearance depends on three factors: the intrinsic ability of the liver to metabolize a drug, hepatic blood flow, and the extent of binding of the drug to blood components.
- The volume of distribution quantifies the extent of drug distribution. The greater the affinity of tissues for a drug relative to blood, the greater its volume of distribution (i.e., lipophilic drugs have greater volumes of distribution).
- Elimination clearance is the parameter that characterizes the ability of drug-eliminating tissues to irreversibly remove drugs from the body. The efficiency of the body in clearing drugs is proportional to the elimination clearance.
- All else being equal, an increase in the volume of distribution of a drug will increase its elimination half-life; an increase in elimination clearance will decrease elimination half-life.
- Most drugs bring about a pharmacologic effect by binding to a specific receptor that results in a change in cellular function.
- Although most pharmacologic effects can be characterized by both dose–response curves and concentration–response curves, dose–response curves cannot be used to determine whether variations in pharmacologic response are caused by differences in pharmacokinetics, pharmacodynamics, or both.
- Integrated pharmacokinetic–pharmacodynamic models allow temporal characterization of the relationship between dose, plasma concentration, and pharmacologic effect.
- In vitro drug–drug interactions due to pharmaceutical (physiochemical) properties of drugs can significantly alter drug bioavailability and produce unintended toxic byproducts.
- Novel approaches to antagonizing neuromuscular blockade have been developed that take advantage of in vivo physiochemical drug–drug interactions, thereby avoiding some of the systemic side effects associated with inhibition of plasma acetylcholinesterase.
- Distribution clearance is influenced by changes in cardiac output and regional blood flow.
- Inhibition of CYP isozyme activity can make it difficult to achieve adequate analgesia when using opioids such as codeine and tramadol that require CYP 2D6 activity for conversion to the biologically active opioid.
- Serotonin syndrome can be precipitated by a wide array of drugs with serotonergic activity, including selective serotonin reuptake inhibitors (SSRIs), serotonin–norepinephrine reuptake inhibitors (SNRIs), and phenylpiperidine opioids.

- Simulations of multicompartmental pharmacokinetic models that describe intravenous anesthetics demonstrate that for most anesthetic dosing regimens, the distribution of drug from the plasma to the inert peripheral tissues has a greater influence on the plasma concentration profile of the drug compared to the elimination of drug from the body.
- Target-controlled infusions are achieved with computer-controlled infusion pumps worldwide (not yet approved by the Food and Drug Administration in the United States), and permit clinicians to make use of the drug concentration–effect relationship, optimally accounting for pharmacokinetics and predicting the offset of drug effect.
- Classic pharmacokinetic models inaccurately describe the initial several minutes of drug distribution and, therefore, overestimate interindividual pharmacodynamic variability.
- By understanding the interactions between the opioids and the sedative-hypnotics (e.g., response surface models), it is possible to select target concentration pairs of the two drugs that produce the desired clinical effect while minimizing unwanted side effects associated with high concentrations of either drug.

Source: Barash 9e, Chapter 11, Key Points, PDF pp. 720–723.

Chapter 12: Cardiac Anatomy and Physiology

- The cartilaginous skeleton, myocardial fiber orientation, valves, blood supply, and conduction system of the heart determine its functional capabilities and limitations.
- The major determinants of coronary blood flow are Poiseuille's law, extravascular compression, metabolic regulation, pressure-flow autoregulation, and the autonomic nervous system.
- Coronary blood flow reserve is substantial, but it may be reduced by flow-limiting stenoses, pressure-overload hypertrophy, or microvascular dysfunction.
- Oxygen extraction by the left ventricle (LV) is nearly maximal under baseline conditions. As a result, increases in myocardial oxygen demand are dependent on proportional increases in coronary blood flow mediated by locally produced vasodilators.
- Metabolic regulation is a "feed-forward" mechanism based on the action of carbon dioxide and reactive oxygen species produced by cardiac myocytes and involves endothelium-derived nitric oxide (NO) and adenosine triphosphate-sensitive potassium (KATP) channels in vascular smooth muscle.
- The cardiac myocyte is engineered for contraction and relaxation, not protein synthesis.
- Laplace's law describes the mechanism whereby alterations in muscle tension and length that occur during contraction and relaxation in the sarcomere are translated into phasic changes in pressure and volume that occur in the intact heart.
- The cardiac cycle is a coordinated, temporally related series of electrical, mechanical, and valvular events.
- A time-dependent, two-dimensional projection of continuous pressure and volume during the cardiac cycle provides a useful framework for the analysis of systolic and diastolic function.
- Preload is the quantity of blood that a cardiac chamber contains immediately before contraction begins, whereas afterload is the external resistance to emptying to which the chamber is confronted after the onset of contraction.
- Myocardial contractility is quantified using indices derived from pressure-volume relations, isovolumic contraction, or the ejection phase; these indices have limitations because contractile state and loading conditions are interrelated.
- Diastolic function is defined as the ability of a cardiac chamber to effectively collect blood at a normal filling pressure.
- The pericardium exerts important restraining forces on cardiac chamber filling and is a major determinant of ventricular interdependence.
- The atria serve three mechanical functions: conduit, reservoir, and contractile chamber.
- Heart failure occurs when the LV is unable to eject blood at a sufficient rate to meet metabolic demands or it cannot fill completely without elevated pressure; heart failure is classified based on LV ejection fraction (EF) less than or greater than 50%.
- LV systolic dysfunction characterizes heart failure with reduced ejection fraction (HFrEF), whereas LV diastolic dysfunction, ventricular and arterial stiffening, left atrial (LA) dysfunction, pulmonary hypertension, and right ventricular (RV) dysfunction are common in heart failure with preserved ejection fraction (HFpEF).

Source: Barash 9e, Chapter 12, Key Points, PDF pp. 839–841.

Chapter 13: Autonomic and Cardiovascular Pharmacology

- Drugs that mimic the effects of acetylcholine are most commonly used in ophthalmology.
- Anticholinesterases prolong the actions of acetylcholine by preventing its metabolism; these drugs are used for reversal of neuromuscular blockade and treatment of central anticholinergic syndrome.
- Muscarinic antagonists inhibit the effects of acetylcholine mediated through the parasympathetic nervous system.
- The cardiovascular effects of endogenous and synthetic catecholamines are dependent on their specificity for alpha- and beta-adrenoceptor subtypes.
- Catecholamines have well-documented utility in the treatment of acute left ventricular dysfunction, but may cause arrhythmias, hypertension, and myocardial ischemia.
- Alpha1-adrenoceptor antagonists reduce arterial pressure but also produce reflex tachycardia and orthostatic hypotension, especially in patients with hypovolemia.
- The alpha2-adrenoceptor agonists clonidine and dexmedetomidine are used extensively for sedation, anxiolysis, and analgesia.
- Milrinone and levosimendan are important medications for the management of acute left ventricular dysfunction and cause synergistic positive inotropic effects when administered with catecholamines by enhancing the intracellular cAMP-mediated signaling.
- Vasopressin is the most potent arterial vasoconstrictor currently available and is useful for treatment of vasoplegia associated with sepsis or cardiac surgery.
- Beta-blockers play major roles in the treatment of hypertension, coronary artery disease, myocardial infarction, and heart failure with reduced ejection fraction (HFrEF).
- Calcium channel blockers reduce arterial pressure and dilate coronary arteries, but some of these drugs also affect sinoatrial node automaticity and atrioventricular node conduction, actions which are beneficial in the presence of supraventricular tachyarrhythmias.
- Angiotensin converting enzyme inhibitors and angiotensin receptor blockers are useful for the treatment of hypertension and HFrEF.

Source: Barash 9e, Chapter 13, Key Points, PDF pp. 936–938.

Chapter 14: Autonomic Nervous System Anatomy and Physiology

- The autonomic nervous system (ANS) includes that part of the central nervous system (CNS) and peripheral nervous system concerned with the involuntary regulation of cardiac muscle, smooth muscle, and glandular and visceral functions.
- The sympathetic and parasympathetic nervous systems (SNS, PNS) affect cardiac pump function in three ways: (1) by changing the rate (chronotropism), (2) by changing the strength of contraction (inotropism), and (3) by modulating coronary blood flow.
- SNS nerves are by far the most important regulators of peripheral circulation.
- The ANS can be pharmacologically subdivided by the neurotransmitter secreted at the effector cell: acetylcholine (ACh) released by the PNS, and the catecholamines: epinephrine (EPI) and norepinephrine (NE), which are the mediators of peripheral SNS activity.
- An agonist is a substance that interacts with a receptor to evoke a biologic response. An antagonist is a substance that at the receptor site stops the response produced by an agonist (counteracts an agonist).
- The SNS receptors (adrenoreceptors or adrenergic receptors) are G-protein–coupled receptors named adrenergic or noradrenergic, depending on their responsiveness to EPI and NE.
- The numbers and sensitivity of adrenoreceptors can be influenced by normal, genetic, and developmental factors.
- The ANS reflex comprises (1) sensors, (2) afferent pathways, (3) CNS integration, and (4) efferent pathways to the receptors and efferent organs.
- The clinical application of ANS pharmacology is based on knowledge of ANS anatomy, physiology, and molecular pharmacology.
- Clinically, anticholinesterase drugs may be divided into two types: reversible and nonreversible cholinesterase inhibitors.

Source: Barash 9e, Chapter 14, Key Points, PDF pp. 1020–1021.

Chapter 15: Respiratory Function in Anesthesia

- Elastic and restrictive forces are the major impedances to the delivery of air into the lungs.
- Functional residual capacity has a key role in determining oxygen reserves, propensity to atelectasis, and biologic effects of mechanical ventilation.
- Driving pressure is a surrogate for global lung strain. Higher values are associated with increased risk of postoperative pulmonary complications.
- Anesthesia results in smaller and stiffer lungs, primarily due to atelectasis. Trendelenburg position, obesity, and pneumoperitoneum further reduce respiratory system compliance.
- Ventilation and perfusion mismatch is common during anesthesia. Shunt physiology, due to atelectasis, is the primary etiology of hypoxemia and can be addressed with restoration of end-expiratory lung volume.
- As the amount of shunt increases, oxygen content in pulmonary venous blood has a greater contribution to arterial oxygenation and delivery.
- Dead space volume increases during anesthesia. Instrumental dead space from breathing devices can be significant when alveolar ventilation is already maximized.
- Anesthetics disrupt different aspects of ventilator control in a dose-dependent manner and cause alveolar hypoventilation.
- There is no compelling evidence that defines rules for ordering preoperative pulmonary function tests. Rather, they should only be obtained to ascertain the presence of reversible pulmonary dysfunction (bronchospasm) or to define the severity of advanced pulmonary disease.
- Individuals with morbid obesity are at high risk of postoperative pulmonary complications. Use of continuous positive airway pressure after tracheal extubation can improve their end-expiratory lung volumes.
- Multidisciplinary perioperative pulmonary rehabilitation improves pulmonary outcomes in individuals at high risk of complications.

Source: Barash 9e, Chapter 15, Key Points, PDF pp. 1102–1104.

Chapter 16: Acid–Base, Fluids, and Electrolytes

- Nearly half of all hospitalized patients will develop an acid–base disorder during their hospitalization.
- Accurate diagnosis of acid–base disturbances requires a systematic, stepwise approach.
- While control of blood glucose in the critically ill is associated with improved clinical outcomes, it is clear that tight, conservative control targeting glucose ranges of 81 to 108 mg/dL is associated with higher mortality than compared with targeting blood glucoses below 180 mg/dL.
- In healthy ASA I–II patients undergoing minor procedures, liberal fluid administration is associated with decreased risks of postoperative nausea and vomiting (PONV). However, in higher-risk patients, or those undergoing major operations, restrictive, goal-directed therapy (GDT) is associated decreased risks of postoperative ileus, pneumonia, and increased length of stay.
- While the use of albumin has failed to show clinical differences in primary outcomes in septic shock, there may be benefit to its use in the acute burns resuscitation.
- In both non-critically and critically ill patients, the use of balanced crystalloids in comparison to 0.9% saline is associated with fewer metabolic derangements, notably hyperchloremic metabolic acidosis, as well as acute kidney injury.
- Dynamic assessments for determining volume responsiveness are more sensitive and specific in determining if patients will respond well to ongoing fluid challenges.
- New oral potassium-binding agents are effective at quickly lower $[K^+]$ by increasing total body K^+ elimination.
- Ca^{2+} is essential to achieve proper hemostasis given its role in platelet activation and propagation as well as a critical co-factor in the activation of the coagulation cascade.
- The diamond of death—Hypocalcemia is now recognized as fourth component to the formerly known lethal triad of hypothermia, acidosis, and coagulopathy.
- Magnesium has demonstrated analgesic properties as a result of N-methyl-D-aspartate (NMDA) receptor antagonism.

Source: Barash 9e, Chapter 16, Key Points, PDF pp. 1145–1147.

Chapter 17: Hemostasis and Transfusion Medicine

- Modern transfusion medicine focuses on patient-centered blood component therapy.
- Blood must not only be maintained as a fluid in normal circulation, but also be capable of forming a solid clot to stanch leaks in the vascular wall, yet can be subsequently dismantled once the vascular endothelium is repaired.
- Clotting factors in the plasma are activated at sites of endothelial injury and assemble in enzymatic complexes to activate thrombin.
- All of the clotting factors are primarily produced in the liver, except for VIII, which is also released by endothelial cells and is well maintained in liver disease.
- Fibrin clots must be broken down after their job is done, and fibrinolysis is a complex process with checks and balances.
- The first screening test for hemostatic problems should always be the patient's medical history.
- A general oversight of plasma clotting factor activity is obtained by the prothrombin time (PT) for the extrinsic (tissue) pathway and the activated partial thromboplastin time (aPTT) for the intrinsic (contact) pathway.
- Disseminated intravascular coagulation (DIC) describes unchecked coagulation initiated by pathologic systemic activation of the intrinsic clotting pathway.
- The risk for venous thromboembolism is increased by intercurrent factors such as physical inactivity or immobilization, malignancy, hormone therapy, trauma, and pregnancy.
- Most anticoagulant therapies need ongoing or selective testing for assessment of therapeutic effect.
- Leukoreduction (LR) to remove WBCs from red blood cells (RBCs) and platelets reduces the risk of HLA alloimmunization, febrile nonhemolytic transfusion reactions (FNHTRs), and CMV transmission in patients who require these precautions.
- Plasma derivatives are proteins processed from plasma for therapeutic infusions.
- Many years of effort have gone into the search for an oxygen-carrying substitute for RBCs.
- Routine RBC compatibility testing includes ABO and RhD typing, an antibody screen for IgG non-ABO RBC antibodies, and an RBC cross-match.
- Over the past decade, transfusion practices for medical and surgical patients shifted from a liberal strategy to more restrictive management with lower thresholds and careful consideration of the balance between transfusion risks and the physiologic consequences of anemia.
- Oxygen delivery to the tissues (DO_2) is dependent on cardiac output (CO), regional blood flow, and oxygen-carrying capacity, also known as the oxygen content (CaO_2) of blood.
- Numerous recommendations provide guidance for the transfusion management of thrombocytopenia and acquired or inherited platelet disorders.
- Cryoprecipitate is created by a controlled thaw of frozen plasma, which allows for precipitation of large molecules, most notably fibrinogen and vWF.
- Over the past few decades, the risk–benefit ratio of blood product transfusion has been the subject of several studies and review articles.

- Given the extensive use of more sensitive methods for screening and controlling the infectious risks of blood product transfusion, noninfectious complications have emerged as the major source of transfusion-related morbidity and mortality.
- Transfusion-related acute lung injury (TRALI) is a clinical diagnosis that can be clouded by confounding comorbidities or patient acuity; therefore, TRALI tends to be underreported in the literature and is extremely difficult, if not impossible, to study with randomized prospective clinical trials.
- Preoperative autologous donation programs are most effective when double units are donated with ample time for erythropoiesis prior to the date of surgery.
- Over the past decade, RBC salvage techniques have improved drastically and now offer an efficient, cost-effective, and safe method for perioperative blood conservation.
- Disorders of hemostasis can be classified as those that cause a propensity for hemorrhage and those that facilitate inappropriate thrombosis.
- Symptomatically, disorders of primary hemostasis often present with superficial signs of bleeding on the skin or mucosa.
- Von Willebrand disease (vWD) is the most common hereditary bleeding disorder, with a prevalence of approximately 1% in the general population.
- Hemophilia is a genetic disease that results from deficiencies or dysfunction of specific clotting factors.
- Antiplatelet therapy is indicated for patients at risk for cerebral vascular accident, myocardial infarction, or other vascular thrombosis complications.
- Heparin-induced thrombocytopenia (HIT) is a clinical disorder that develops after extended use of heparin therapy. It occurs in approximately 1% to 5% of patients receiving heparin and is associated with morbidity from thromboembolic complications.
- Recombinant activated factor VII (rFVIIa) is only indicated for the treatment of hemophiliacs with inhibitors/antibodies and factor VII deficiency.
- Prothrombin complex concentrates are now the drug of choice for emergent reversal of oral anticoagulants in place of fresh frozen plasma (FFP).
- Antifibrinolytic agents have been used to prevent and treat surgical blood loss for several decades.

Source: Barash 9e, Chapter 17, Key Points, PDF pp. 1258–1261.

Chapter 18: Inhaled Anesthetics

- At equilibrium, the partial pressure of inhaled anesthetics in the brain equals the arterial partial pressure, which in turn equals the alveolar partial pressure if cardiopulmonary function is normal.
- The inspired concentration and the blood:gas solubility of an inhaled anesthetic are the major determinants of the speed of induction. Solubility alone determines the rate of elimination, provided there is normal cardiopulmonary function.
- Isoflurane is the most potent of the volatile anesthetics in clinical use, desflurane is the least soluble in blood, and sevoflurane is the least irritating to the airways.
- Nitrous oxide (N₂O) can expand a pneumothorax to double or triple its size in 10 to 30 minutes. Abrupt discontinuation of N₂O inhalation can transiently lower alveolar concentrations of oxygen and carbon dioxide, a phenomenon called diffusion hypoxia.
- Minimum alveolar concentration (MAC) is the alveolar concentration of an inhaled anesthetic at one atmosphere that prevents movement in response to a surgical stimulus in 50% of patients. Concentrations of inhaled anesthetics that provide loss of awareness and recall are about 0.4 to 0.5 MAC.
- MAC decreases approximately 6% per decade of patient age.
- Volatile anesthetics depress cerebral metabolic rate in a dose-dependent manner and at higher concentrations will increase cerebral blood flow. The latter effect may increase intracranial pressure in patients with a mass-occupying lesion of the brain.
- Hypocapnia may blunt or abolish volatile anesthetic-induced increases in cerebral blood flow depending on when the hypocapnia is induced and the nature of the cerebral disease process.
- Volatile anesthetics produce dose-dependent depression of the electroencephalogram, sensory-evoked potentials, and motor-evoked potentials.
- Volatile anesthetics in current use decrease arterial blood pressure, systemic vascular resistance, and myocardial function comparably and in a dose-dependent fashion.
- Volatile anesthetics decrease tidal volume, decrease ventilatory response to hypercarbia and hypoxia, increase respiratory rate, and relax airway smooth muscle in a dose-dependent fashion.
- Unlike halothane, volatile anesthetics in current use have minimal adverse effects on the liver and might afford some protection for hepatocytes from ischemic and/or hypoxic injury.
- Volatile anesthetics are potent triggers for malignant hyperthermia in genetically susceptible patients.
- CO₂ absorbents degrade sevoflurane, desflurane, and isoflurane to carbon monoxide when the normal water content of the absorbent (13% to 15%) is markedly decreased (<5%).
- Volatile anesthetics have a global warming potential far greater than CO₂ and should be delivered with a lower fresh gas flow to reduce their impact.

Source: Barash 9e, Chapter 18, Key Points, PDF pp. 1374–1376.

Chapter 19: Intravenous Anesthetics

- The ideal intravenous anesthetic would cause hypnosis and amnesia with a rapid onset, minimal cardiovascular and respiratory effects, and rapid metabolism.
- Context-sensitive half-time (i.e., half-life) is defined as the time to achieve a 50% reduction in concentration after stopping a continuous infusion. Context-sensitive half-time demonstrates the influence of the distributive process in governing drug disposition.
- The mechanism by which the unconscious state is attained by propofol is complicated, but primarily occurs via enhancement of gamma-aminobutyric acid (GABA) inhibitory pathways.
- The rapid and smooth induction and emergence from anesthesia helped transform propofol into an intravenous sedative–hypnotic that is a viable alternative to standard inhalational agents and other intravenous drugs.
- Etomidate is a hemodynamically stable induction medication with a relatively large safety margin. Adrenocortical suppression is a recognized adverse effect of etomidate.
- Ketamine causes “dissociative anesthesia” acting through the N-methyl-D-aspartate (NMDA) receptor, and is associated with nystagmus, significant analgesia, and unconsciousness.
- Dexmedetomidine, an alpha-2 agonist, has been used for intensive care unit (ICU) sedation in mechanically ventilated patients, procedural sedation, and as an adjunct to general anesthesia. Dexmedetomidine is unique as a sedative in that it has limited respiratory depressant effects.
- Benzodiazepines are frequently used to produce several clinically desirable effects: anxiolysis, anterograde amnesia, sedation, and hypnosis.
- The mechanism of action of barbiturates involves cortical and brainstem GABA inhibitory pathways, leading to loss of consciousness, as well as respiratory and cardiovascular depression.
- New drugs are being developed in an effort to achieve the ideal intravenous anesthetic with a short duration of action, a short context-sensitive half-time that allows for infusion administration, and minimal adverse effects.

Source: Barash 9e, Chapter 19, Key Points, PDF pp. 1462–1464.

Chapter 20: Opioids

- Opioids produce analgesia as well as serious side effects. All physicians who prescribe opioids for relief of acute or chronic pain need to know how to use these drugs safely. This requires an in-depth understanding of the pharmacokinetics and pharmacodynamics of opioids as well as the acquisition of sufficient clinical experience in their use.
- Opioids can be classified according to strength or potency based on the plasma concentrations at which they exert their effects (C50 or the plasma concentration causing a 50% effect). Strong opioids include fentanyl, sufentanil, and remifentanyl. Weak opioids include codeine and tramadol. An intermediate group includes morphine, methadone, oxycodone, and buprenorphine.
- Opioids act through specific opioid receptors on neuronal tissues such as peripheral nerves and neurons in the spinal cord and brain. The most important receptors include the μ -opioid receptor (MOR), δ -opioid receptor (DOR), and κ -opioid receptor (KOR). For anesthesia and pain relief, the MOR is most important. Opioids also act through nonneuronal pathways, such as those affecting the immune system, which may be of relevance in the treatment of inflammatory pain.
- Endogenous opioid pathways are activated in cases of stress-induced analgesia, placebo-induced analgesia, and conditioned pain modulation (CPM). CPM occurs when pain arising from one focus is decreased by application of a second painful stimulus (pain inhibits pain).
- Opioid-induced hyperalgesia (OIH) is a paradoxical opioid effect whereby pain sensitivity increases during or following escalating opioid treatment. It is also observed postoperatively following the use of a remifentanyl infusion during anesthesia. Greater and more frequent doses of morphine are required to treat postoperative pain. OIH may be treated and prevented by administration of a low dose of ketamine, an N-methyl-D-aspartate (NMDA) receptor antagonist.
- Tolerance to opioid-induced respiratory depression develops more slowly than tolerance to pain relief. Tolerance is a dynamic process and after a long period of abstinence from opioids, the risk of opioid toxicity increases.
- The pharmacokinetic characteristics of a drug determine its behavior in the body of a patient. One important pharmacokinetic concept for opioids is that of context-sensitive half-time (CSt $\frac{1}{2}$). This is the time needed for the opioid's plasma concentration to decrease by 50% from a steady-state concentration. This value is not a constant but is dependent on the duration of the infusion of the opioid. For example, the CSt $\frac{1}{2}$ of fentanyl increases rapidly with the infusion duration. In contrast, the CSt $\frac{1}{2}$ of remifentanyl is independent of the infusion duration due to its rapid elimination from plasma by nonspecific plasma esterases in plasma.
- Opioid metabolism is affected by drugs that interfere with metabolizing cytochrome P450 (CYP) enzymes, most importantly CYP3A4. Opioid metabolites can be active or inactive. Active metabolites need to be considered when treating patients. For example, the active metabolite of morphine, morphine-6-glucuronide, can accumulate in patients with renal impairment. Genetic variability in the CYP system can have important clinical implications, especially in case of variations in the copies of a gene coding for the metabolizing enzyme. An example is the enzyme CYP2D6, which catalyzes the conversion of codeine into morphine. Patients with multiple copies of the CYP2D6 gene and who receive codeine will have large plasma morphine concentrations with all related beneficial and adverse side effects.
- Opioid effects are variable among patients. Dosing is optimum when opioids are titrated to the effect. It is also important to consider the delay between the administration of an opioid and its effect, which is defined as the blood–effect site equilibration half-life, or t $\frac{1}{2}$ ke0. This will allow proper and timely dosing, particularly when anticipating a stressful event (laryngoscopy,

intubation, skin incision, etc.) and administering opioids to prevent the occurrence of a large hemodynamic response to these stimuli. Since the $t_{1/2ke0}$ of morphine is about 90 minutes, it is important to give an initial bolus dose of morphine at least 60 minutes before the end of surgery when using the drug for postoperative pain relief.

- Opioids reduce the requirement of inhalational anesthetics and propofol during anesthesia, which makes rapid awakening from anesthesia possible. Using known pharmacokinetic and pharmacodynamic data, it can be determined what doses and plasma concentrations will permit the shortest time to awakening. For example, termination of drug infusions at plasma concentrations of propofol 1.5 µg/mL and remifentanyl 9.0 ng/mL will lead to patient awakening within 6.5 minutes.
- In the perioperative setting, careful, slow infusion of opioids allows the gradual accumulation of arterial CO₂, which serves as a respiratory stimulant at the chemoreceptors and lowers the probability of apnea.
- Administration of opioids can potentially lead to life-threatening respiratory depression. The incidence of serious perioperative respiratory events is approximately 0.5% (1 in every 200 patients). In the acute setting, risk factors include sleep-disordered breathing, obesity, renal impairment, pulmonary disease, neurologic disorders, and CYP450 enzyme polymorphism. In the pediatric population, risk factors include morphine administration in patients with renal impairment, adenotonsillectomy for recurrent tonsillitis, and/or obstructive sleep apnea and codeine use in patients with CYP2D6 gene polymorphism associated with the ultrarapid metabolizer phenotype. In chronic pain patients, risk factors included renal failure, sensory deafferentation, and drug–drug interactions.
- The risk of opioid-induced respiratory depression increases when opioids are consumed in combination with other centrally acting respiratory depressants such as alcohol, benzodiazepines, antidepressants, and antipsychotics. This is a serious cause of enhanced mortality, as a large number of patients that die from opioids also had sedatives in their system.
- The nonspecific opioid receptor antagonist, naloxone, is currently the drug of choice to reverse opioid-induced respiratory depression. The required dose of naloxone depends on the pharmacokinetic and pharmacodynamic properties and the dose of the opioid that needs reversal. Postoperatively, when there is persistent apnea, opioid concentrations are often just above the threshold for respiratory depression. Intravenous administration of naloxone using incremental doses of 40 to 80 µg, to a cumulative dose of less than 400 µg, may be sufficient for breathing to resume.
- New drugs are being developed that reverse or prevent opioid-induced respiratory depression without affecting analgesia. Drugs that stimulate opioid-depressed breathing include ketamine, nicotinic acetylcholine receptor agonists, ampakines, and potassium channel blockers acting at the carotid bodies.
- Apart from respiratory depression, opioids cause a large number of side effects that require treatment: nausea, smooth muscle spasms, skeletal muscle rigidity, histamine release, pruritus (especially after spinal administration), miosis, sedation, and dizziness. The cardiovascular side effects of opioids include bradycardia and hypotension but are generally mild at usual clinical doses. However, when combined with anesthetics, even at usual clinical doses, or in severely ill patients, opioids may produce hemodynamic instability, which requires treatment.

Chapter 21: Neuromuscular Blocking Drugs

- It should be recognized, from the outset, that neuromuscular blocking drugs do not alter consciousness or cause amnesia. For this reason, neuromuscular blocking drugs have been implicated in contributing to accidental awareness under anesthesia.
- Depending on the specific drugs used for neuromuscular blockade and reversal, the occurrence of postoperative residual neuromuscular blocking drug effect (known as residual paralysis) has been estimated to occur in approximately 40% of patients.
- Nondepolarizing neuromuscular blocking drugs bind and inhibit presynaptic $\alpha_3\beta_2$ acetylcholine receptors, leading to a reduction in muscle contraction with subsequent nerve signals.
- Nondepolarizing neuromuscular blocking drugs bind to one or both alpha subunits of the nicotinic acetylcholine receptor without causing any activation of these ion receptor channels.
- The four most common neuromuscular blocking drugs currently used in clinical practice have an ED₉₅ of either 0.3 or 0.05 (succinylcholine and rocuronium ED₉₅ 0.3, cisatracurium and vecuronium ED₉₅ 0.05). There is a highly significant variation between patients in response to all neuromuscular blocking drugs.
- Because of its molecular similarity to acetylcholine, succinylcholine depolarizes both postsynaptic and extrajunctional receptors.
- A prolonged duration of succinylcholine blockade may occur when the concentration of normal pseudocholinesterase is very significantly reduced or when an atypical variant of pseudocholinesterase is present.
- Nondepolarizing neuromuscular blocking drugs can be classified by their chemistry and molecular structure: aminosteroid (e.g., pancuronium, vecuronium, rocuronium) or benzylisoquinolinium (e.g., atracurium, cisatracurium, mivacurium). Clinically they are classified by their expected duration of action: long-, intermediate-, or short-acting.
- Different classes of nondepolarizing neuromuscular blocking drugs should not be mixed in the same patient because this can lead to an unpredictable extended duration of muscle paralysis.
- All clinical assessments (e.g., maximum inspiratory force, vital capacity, tidal volume, eye opening, handgrip, tongue protrusion, handgrip, leg lift, and head lift) are insensitive and thus render the clinician incapable of identifying the presence of residual paralysis.
- In the absence of any tactile or visual recognition of fade with train-of-four subjective assessment, the train-of-four ratio can be as low as 0.3 to 0.4.
- Quantitative monitoring is the only way to objectively measure train-of-four ratio, supporting expert recommendations for this technology to be used in clinical practice.
- Blocking the breakdown of acetylcholine by acetylcholinesterase (with anticholinesterase drugs) results in an increase in the available pool of acetylcholine at the synaptic cleft and improves the chances of competing with a nondepolarizing neuromuscular blocking drug.
- Only when the residual neuromuscular block is minimal (e.g., train-of-four count of 4 without fade or train-of-four ratio ≥ 0.4) is neostigmine a highly effective reversal agent.
- For a train-of-four count 2 or more, a sugammadex dose of 2 mg/kg will reverse to a train-of-four ratio greater than 0.9 in 2 to 4 minutes.

Source: Barash 9e, Chapter 21, Key Points, PDF pp. 1595–1597.

Chapter 22: Local Anesthetics

- Local anesthetics provide anesthesia and analgesia by blocking the transmission of pain sensation along nerve fibers.
- The key target of local anesthetics is the voltage-gated sodium channel. The binding is intracellular and is mediated by hydrophobic interactions.
- The degree of nerve blockade depends on both drug concentration and volume.
- Most clinically relevant agents contain a lipid-soluble benzene ring connected to an amide group and are categorized as either aminoesters or aminoamides based on their chemical linkage.
- Potency is related to lipid solubility and physiochemical properties of the agent. In general, more potent agents are more lipid-soluble.
- Clinical use of local anesthetics may be increased by addition of dexamethasone, epinephrine, opioids, and α_2 -adrenergic agonists. The value of alkalization of local anesthetics and liposomal bupivacaine appear to be debatable as a clinically useful tool to improve anesthesia.
- The rate of local anesthetic system absorption depends on the site of injection, the dose, the drug's intrinsic pharmacokinetic properties, and the addition of a vasoactive agent.
- Systemic toxicity from the clinical use of local anesthetics is an uncommon occurrence. Patients with cardiovascular collapse from bupivacaine, ropivacaine, and levobupivacaine may be especially difficult to resuscitate; however, intravenous lipid infusion is an effective new therapy.

Source: Barash 9e, Chapter 22, Key Points, PDF pp. 1670–1671.

Chapter 23: Preoperative Patient Assessment and Management

- The goals of a preoperative evaluation are to reduce patient risk and morbidity associated with surgery and anesthesia, prepare the patient medically and psychologically, and also promote efficiency and reduce costs.
- The anesthesiologist is responsible for assessing the medical condition of the patient and developing the anesthesia plan of care. The American Society of Anesthesiologists (ASA) published basic standards for preoperative care, as well as an updated practice advisory for preanesthesia evaluation that details evidence-supported recommendations.
- It is important for the evaluation to be complete, accurate, and clear, not only to allow the information to be relayed to others who may care for the patient perioperatively but also for medicolegal purposes.
- The preoperative evaluation serves as a screening tool to anticipate and avoid airway difficulties or problems with anesthetic drugs. In addition to the history and physical examination, previous anesthetic records should be reviewed and contraindications to specific drugs, such as succinylcholine, nitrous oxide, or volatile agents, should be sought.
- A review of the patient's medication list, including over-the-counter and herbal preparations, should investigate potential drug interactions and potential indications for stress dose steroid coverage. The anesthesiologist should be aware of the patient's allergies and previous drug reactions, including the possibility of latex allergy.
- When evaluating the patient with hypertension, diabetes mellitus, or obesity, it is important to determine the presence of end-organ damage, such as cardiovascular disease.
- Exercise tolerance is a significant predictor of cardiac risk. Multiple specialty groups have contributed to formal guidelines for the perioperative cardiovascular evaluation and management of patients undergoing noncardiac procedures. The algorithms provide useful guides for further testing and evaluation.
- Preoperative laboratory tests should be ordered on the basis of positive findings from the history and physical examination or from anticipated physiologic disturbances during surgery, such as blood loss.
- Optimization of the patient's health status prior to surgery includes clear instruction regarding fasting times as well as which medications to continue until the time of surgery. In general, most medications for hypertension or cardiac disease should be continued, and consideration should be given to initiating β -blocker therapy before the day of surgery in appropriate patients who are at risk for cardiac adverse events. The need for subacute bacterial endocarditis prophylaxis should be anticipated. Likewise, drugs for asthma or chronic obstructive pulmonary disease should be continued or administered prophylactically. Medications taken for the treatment of esophageal reflux should be continued or initiated for those patients with untreated symptoms. For diabetic patients, oral hypoglycemic agents should often be held, but patients requiring insulin will need to continue to take adjusted doses.
- Although preoperative sedation is generally limited to drugs given immediately prior to anesthesia, the administration must be carefully planned to allow optimal effect and avoid operating room delays.

Chapter 24: Rare Coexisting Diseases

- The cytoskeleton of the muscle membrane in patients with muscular dystrophy is fragile. In these patients, succinylcholine and/or halogenated anesthetics can cause the massive release of intracellular contents resulting in hyperkalemic cardiac arrest.
- Myotonic dystrophy produces cardiac conduction delay that can manifest as third-degree atrioventricular block.
- Patients with myasthenia gravis are exquisitely sensitive to nondepolarizing muscle relaxants. Short-acting muscle relaxants and objective monitoring of neuromuscular function are indicated.
- Small-cell carcinoma of the lung and other cancers are associated with the development myasthenic syndrome; in some cases, the syndrome is diagnosed before the cancer is discovered.
- Patients with multiple sclerosis can have an exacerbation of their neurologic symptoms despite a well-managed anesthetic.
- An unexpected increase in end-tidal carbon dioxide is the most sensitive sign of malignant hyperthermia.
- Hypoglycemia and metabolic acidosis are constant risks to patients with glycogen storage diseases.
- The deposition of mucopolysaccharides in the upper airway of patients with mucopolysaccharidoses often complicates airway management during anesthesia.
- Repeated episodes of sickling in patients with sickle cell disease cause pulmonary hypertension that increases perioperative risks.
- Rheumatoid arthritis is a multisystem disease that causes subclinical cardiac and pulmonary dysfunction.
- Patients with rheumatoid arthritis can have significant degeneration of the cervical spine with few neurologic symptoms. Cervical manipulation during laryngoscopy and tracheal intubation requires special precautions.
- Esophageal dysfunction in patients with scleroderma and dermatomyositis increases the risk of aspiration pneumonitis.
- Patients with epidermolysis bullosa can have undiagnosed dilated cardiomyopathy.

Source: Barash 9e, Chapter 24, Key Points, PDF pp. 1823–1825.

Chapter 25: The Anesthesia Workstation and Delivery Systems for Inhaled Anesthetics

- The oxygen failure cutoff valves (previously known as “fail-safe” valves, “hypoxic guards,” or “proportioning systems”) minimize the likelihood of delivery of a hypoxic gas mixture, but they are not foolproof. Delivery of a hypoxic mixture may still result from (1) the wrong supply gas, either in the cylinder or in the main pipeline; (2) a defective or broken safety device; (3) leaks downstream from the safety devices; (4) inert gas administration (e.g., helium); and (5) dilution of the inspired oxygen concentration by high concentrations of inhaled anesthetics.
- In the event of a gas pipeline crossover, two actions must be taken. The backup oxygen cylinder must be turned on (since the tank valve should always be turned off when not in use), and the wall/pipeline supply sources must be disconnected.
- Because of desflurane’s low boiling point (22.8°C) and high vapor pressure (669 mmHg at 20°C), delivery of desflurane requires specially designed vaporizers, such as the GE Healthcare Tec 6, the Dräger D-Vapor, and the GE Healthcare Aladin cassette vaporizing system.
- Erroneous filling of an empty variable bypass vaporizer with desflurane could theoretically be catastrophic, resulting in delivery of a hypoxic mixture and a massive overdose of inhaled desflurane.
- Inhaled anesthetics can interact with CO₂ absorbents and produce toxic compounds. During sevoflurane-only anesthesia, compound A can be formed, particularly at low fresh gas flow (FGF) rates. Carbon monoxide may be produced when volatile anesthetics are utilized, particularly with desiccated absorbents.
- Desiccated strong base absorbents (particularly barium hydroxide lime, known as Baralyme) can react with sevoflurane, producing extremely high absorber temperatures and combustible decomposition products. These, in combination with the oxygen- or nitrous oxide-enriched environment of the circle system, have produced very high temperatures and fires within the breathing system. For this reason, Baralyme is no longer available in the United States. Lithium-based absorbents are nonreactive.
- With older design machines, use of the oxygen flush valve during the inspiratory phase of mechanical ventilation could cause barotrauma, particularly in pediatric patients. The newer workstations have fresh-gas decouplers or peak-inspiratory pressure limiters that were designed to prevent these complications. Ventilators that use fresh gas decoupling (FGD) technology virtually eliminate the possibility of barotrauma by oxygen flushing during the inspiratory phase because FGF and oxygen flush flow are diverted to the reservoir bag. However, if the reservoir bag has a large leak or is missing, patient awareness under anesthesia and delivery of a lower-than-expected oxygen concentration could occur due to entrainment of room air.
- Anesthesia ventilators with ascending bellows (bellows that ascend during the expiratory phase) were initially thought to be safer than descending bellows. This is because a breathing system disconnection would be obvious since the ascending bellows would not refill and rise during exhalation. Contemporary machines with descending bellows, however, have been carefully redesigned to address their initial limitations. Currently used descending bellows ventilators have featherlight bellows, an electric eye at the bottom of the bellows housing to detect bellows movement, and the bellows housing is subjected to positive end-expiratory pressure (PEEP), such that in case of a disconnect, the bellows would rise and stay up.
- Modern ventilators compensate for the changes in FGF, respiratory rate, and I:E (inspiratory to

expiratory) ratio so that the delivered tidal volume does not change from that set to be delivered. This compensation is achieved either by “fresh gas decoupling” (in Dräger Fabius, Tiro, and Apollo workstations) or by “fresh gas compensation” (in GE Healthcare workstations).

- Two vital but often overlooked items in the preuse checkout of the anesthesia workstation are a readily available functioning self-inflating resuscitation bag (SIRB) and a full auxiliary tank of oxygen.
- The low-pressure circuit (LPC) is the “vulnerable area” of the anesthesia workstation because it is most subject to breakage and leaks. The LPC is located downstream from all anesthesia machine safety features except the oxygen analyzer (or the ratio controller in some machines), and it is the portion of the machine where a leak is most likely to go unrecognized if an inappropriate LPC leak test is performed. Leaks in the LPC can cause delivery of a hypoxic or subanesthetic mixture, leading to hypoxic injury or awareness during anesthesia.
- Because some GE Healthcare anesthesia machines have a one-way check valve in the LPC, a negative-pressure leak test is required to detect leaks in the LPC. A positive-pressure leak test will not detect leaks in the LPC of a machine with an outlet check valve.
- Before administering an anesthetic, the circle breathing system must be checked for leaks and for correct flow. To test for leaks, the circle system is pressurized to 30-cm water pressure, and the circle system airway pressure gauge is observed (static test). To check for appropriate flow to rule out obstructions and faulty valves, the ventilator and a test lung (breathing bag) are used (dynamic test). In addition, the manual/bag circuit must be actuated by compressing the reservoir bag, in order to rule out obstructions to flow in the manual/bag mode.
- Internal vaporizer leaks can only be detected with the vaporizer turned to the “on” position. In the “off” position, the vaporizer is excluded from the LPC.
- Many new anesthesia workstation self-tests do not detect internal vaporizer leaks unless each vaporizer is individually turned on during repeated self-tests.

Source: Barash 9e, Chapter 25, Key Points, PDF pp. 1924–1928.

Chapter 26: Commonly Used Monitoring Techniques

- The purpose of monitoring equipment is to augment the situational awareness of the anesthesiologist by providing clinical data either more rapidly than can be achieved manually or more precisely than can be achieved by direct examination of the patient. The value of any particular monitoring technique lies in its ability to inform the anesthesiologist so that the patient's physiologic condition can be maintained within satisfactory parameters. No monitoring technique, however sophisticated, can substitute for the judgment of the anesthesiologist.
- It is mandatory to measure inspired oxygen concentration for patients receiving general anesthesia to prevent the inadvertent administration of a hypoxic gas mixture. However, inspired oxygen concentration monitoring cannot reliably detect disconnection of the circuit nor does it guarantee adequate arterial oxygenation.
- Pulse oximetry provides a noninvasive means to detect the onset of hypoxemia rapidly. Modern pulse oximeters are noninvasive, continuous, auto-calibrating, have quick response times, and are suitable for use during transport. However, pulse oximetry is a poor indicator of adequate ventilation; desaturation is a late sign of apnea or respiratory insufficiency.
- Monitoring of expired carbon dioxide is now recommended for procedures involving moderate to deep sedation in addition to procedures performed under general anesthesia. Advances in infrared absorption spectroscopy have led to the production of conveniently portable devices for expired gas analysis.
- During direct invasive arterial pressure monitoring, the fidelity of the system is optimized when the catheter and tubing are stiff, the mass of the fluid is small, and the length of the connecting tubing is not excessive.
- Automated noninvasive blood pressure monitors use the oscillometric method to estimate arterial blood pressure. Pulsatile flow generates oscillations in the internal pressure of an inflated blood pressure cuff; these oscillations are greatest when the cuff is inflated to mean arterial pressure. During prolonged surgical cases, it may be prudent to relocate the blood pressure cuff every few hours to reduce the risk of neurapraxia or cutaneous injury.
- On the basis of available evidence, it is difficult to draw meaningful conclusions regarding the effectiveness of pulmonary artery catheter (PAC) monitoring in reducing morbidity and mortality in critically ill patients. Expert opinion suggests that perioperative complications may be reduced if PACs are used in the appropriate patients and settings and if clinicians interpret and apply the data provided by the PAC correctly.
- New noninvasive devices have been developed to generate similar cardiac output (CO) parameters to the PAC as well as potentially to be able to predict fluid responsiveness. However, certain common comorbidities can impair the accuracy of these devices; the PAC thermodilution technique remains the clinical gold standard for measuring CO.
- Clinical studies have demonstrated that patients in whom intraoperative hypothermia develops are at a higher risk for development of postoperative myocardial ischemia and wound infection compared with patients who are normothermic in the perioperative period. Although liquid crystal skin temperature strips are convenient to apply, they do not correlate well with core temperature measurements.
- Although the algorithms used by processed electroencephalogram (EEG) monitors are proprietary, the general features of the EEG that they use are well described. Processed EEG monitors have not been demonstrated to be superior to end-tidal agent concentration monitoring in the

prevention of awareness under anesthesia, although neither technique is sufficient to avoid awareness with complete reliability.

- The advent of “smarter” and more technically sophisticated monitoring devices does not relieve anesthesiologists of their obligation to employ their clinical judgment wisely. On the contrary, it requires anesthesiologists to understand the operation of these devices intimately so that they can be used safely, accurately, and appropriately.

Source: Barash 9e, Chapter 26, Key Points, PDF pp. 2090–2093.

Chapter 27: Echocardiography and Point-of-Care Ultrasound

- Understanding the basic principles, utility, and pitfalls of ultrasound is required to optimize image quality and patient benefit, and to minimize patient harm.
- Comprehensive standard-of-care echocardiographic examination involves a combination of two-dimensional (2D) and three-dimensional (3D) imaging, and Doppler techniques.
- While global left and right ventricular systolic function can be visually estimated or quantitatively measured, several quantitative measurements are required for the diagnosis and grading of diastolic function.
- Regional wall abnormalities are not always straightforward to evaluate, even among expert echocardiographers.
- Speckle tracking and artificial intelligence are newer modalities that are evolving and will be more widely applied in the future.
- The assessment of valve pathology should include consideration of the mechanism, etiology, severity, and the associated sequelae.
- Peak velocity, mean gradient, and aortic valve area are commonly used to grade the severity of aortic stenosis.
- Echocardiographic features differentiating cardiac tamponade from pericardial effusion include atrial collapse during ventricular systole and right ventricular collapse during diastole.
- Point-of-care ultrasound (POCUS) for the anesthesiologist goes beyond assessment of cardiac function and hemodynamic status and includes assessment of lung pathology and gastric contents.
- Certification and training for anesthesiologists in echocardiography and POCUS are available and evolving worldwide.

Source: Barash 9e, Chapter 27, Key Points, PDF pp. 2169–2170.

Chapter 28: Airway Management

- Management of the difficult airway, especially when unanticipated, requires flexibility and is paramount to safe perioperative care. Preoperative airway evaluation and planning favorably affects outcomes.
- From infancy to childhood, the airway undergoes significant changes in its size, shape, and relation to the cervical spine.
- The advent of the laryngeal mask airway, as well as other supraglottic airways, has revolutionized both routine and emergency airway management.
- Airway management always begins with a thorough airway-relevant history and physical examination.
- Preoxygenation (also termed denitrogenation) should be practiced in all cases when time allows.
- Oxygenation should be optimized throughout the process of airway management.
- The gold standard for verification of tracheal intubation is sustained detection of exhaled carbon dioxide.
- The technique of rapid-sequence induction is performed to gain control of the airway in the shortest period of time after the ablation of protective airway reflexes with the induction of anesthesia.
- In terms of airway management, the period of extubation may be far more treacherous than that of induction of anesthesia and tracheal intubation.
- In most instances, awake intubation can be accomplished successfully if approached with care and patience. Awake airway management remains a mainstay of the American Society of Anesthesiologists' difficult airway guidelines.
- Airway access via the extrathoracic trachea may be warranted when mask ventilation, supraglottic airway, and intubation all fail.

Source: Barash 9e, Chapter 28, Key Points, PDF pp. 2291–2292.

Chapter 29: Patient Positioning and Potential Injuries

- The etiologies of postoperative peripheral neuropathies are often not clear. Although there are potential anatomic and neurophysiologic reasons, perioperative inflammation, resulting in microvasculitis, may be an important etiologic factor.
- Stretching nerves 5% or more beyond their resting lengths may kink feeding arterioles and result in ischemia.
- Padding provided by any number of different materials (e.g., gel or foam pads, blankets) should be used to widely disperse point pressure on body parts and soft tissues.
- When possible, patients to be sedated or anesthetized should be placed in anticipated positions while they are awake to assure they are comfortable.
- Permanent loss of vision can occur after nonocular surgical procedures, especially spine surgery performed in a prone position.

Source: Barash 9e, Chapter 29, Key Points, PDF pp. 2426–2427.

Chapter 30: Monitored Anesthesia Care

- Monitored anesthesia care (MAC) is a specific anesthesia service provided by a qualified anesthesiologist, thus can safely encompass the complete spectrum of sedation to general anesthesia if required.
- The standards for preoperative evaluation, intraoperative monitoring, and the continuous presence of a member of the anesthesia care team are no different from those for general or regional anesthesia.
- Indications for monitored anesthesia care include, but are not limited to, the nature of the procedure, the patient's clinical condition, and/or the need for deeper levels of analgesia and sedation than can be provided by moderate sedation (including potential conversion to a general or regional anesthetic).
- The qualified anesthesiologist provider of monitored anesthesia care must be prepared to convert to general anesthesia and respond to the pathophysiology (airway and hemodynamic changes) of procedure and position in the management in induction of general anesthesia when necessary.
- If the patient loses consciousness and the ability to respond purposefully, the anesthesia care is a general anesthetic, irrespective of whether airway instrumentation is required.
- The important mechanisms whereby respiratory function may be compromised during monitored anesthesia care include the effects of sedatives and opioids on respiratory drive, upper airway patency, and protective airway reflexes.
- The leading cause of death and severe nervous system injury during MAC is hypoxia due to suppression of spontaneous respiration by sedative–hypnotic drugs.
- Burns, particularly around the head and neck, are an important cause of patient injury during MAC. The combination of electrocautery, supplemental oxygen, alcohol prep, and flammable drapes is particularly dangerous.
- As a general principle, to avoid excessive levels of sedation, drugs should be titrated in small increments or by adjustable infusions rather than administered in larger doses according to predetermined notions of efficacy.
- At the present time, no single drug can provide all the components of monitored anesthesia care (i.e., analgesia, anxiolysis, and hypnosis) with an acceptable margin of safety or ease of titratability.
- If anesthesiologists are not willing or able to provide MAC or sedation/analgesia services, others, who are less qualified, are prepared to assume that role.

Source: Barash 9e, Chapter 30, Key Points, PDF pp. 2479–2481.

Chapter 31: Ambulatory Anesthesia

- Procedures appropriate for ambulatory surgery are those associated with low rates of postoperative complications that do not require intensive physician or nursing management.
- Society for Ambulatory Anesthesia issued a 2021 statement supporting cataract surgery in medically stable American Society of Anesthesiologists (ASA) IV patients at ambulatory surgery centers (ASCs), recognizing the procedure's low-risk profile, the enormous benefit of sight restoration, and ease of performing the procedure with topical anesthesia.
- Medically stable ASA physical status III or IV patients may be appropriate candidates for ambulatory surgical procedures.
- ASA, Society for Ambulatory Anesthesia, and American College of Chest Physicians agree that a procedure typically performed as an ambulatory procedure with local or regional anesthesia can also be performed as an ambulatory procedure in patients with obstructive sleep apnea (OSA).
- The most recent ASA revised practice guidelines for preoperative fasting were published in 2017. The guidelines continue to allow a light meal up to 6 hours before an elective procedure and support a fasting period for clear liquids of 2 hours for all patients.
- Regional and general anesthesia in the ambulatory setting are equally safe. However, even for experienced anesthesiologists, there is a failure rate associated with regional anesthesia.
- Nerve blocks improve postoperative patient satisfaction and decrease postoperative nausea and vomiting (PONV) and postoperative pain.
- Two primary concerns for ambulatory anesthesia are speed of wake-up and incidence of PONV.
- It is important to distinguish between wake-up time and discharge time. Patients may emerge from anesthesia with desflurane and nitrous oxide significantly faster than after propofol or sevoflurane and nitrous oxide, though the ability to sit up, stand, and tolerate fluids and the time to fitness for discharge may be no different.
- Nausea, with or without vomiting, is probably the most important factor contributing to a delay in discharge and an increase in unanticipated admissions after ambulatory surgery.
- Widespread use of sugammadex in ambulatory anesthesia has not been shown to reduce postoperative pulmonary complications, and dosing neuromuscular blockade reversal should be guided by careful monitoring of neuromuscular block depth to ensure patient safety.

Source: Barash 9e, Chapter 31, Key Points, PDF pp. 2552–2554.

Chapter 32: Office-Based Anesthesia

- An office-based anesthetic is one that is performed in an outpatient venue such as a freestanding medical or surgical office or procedure room that is not accredited as either an ambulatory surgery center (ASC) or a hospital.
- Cost containment and patient convenience are the two major advantages of office-based procedures over traditional hospital-based procedures.
- The disadvantages of office-based procedures are related to patient safety, reporting, quality improvement, and peer review. Thus, great care must be taken concerning these issues.
- Office-based morbidity and mortality is usually the result of inadequate perioperative patient monitoring, oversedation, and thromboembolic events.
- The injuries that occur in offices tend to be of greater severity than those that occur in ambulatory surgical centers. Twenty-one percent of the reported injuries sustained in offices were temporary and nondisabling in nature and 64% were permanent or led to death, whereas 62% of the injuries sustained in ambulatory surgical centers were temporary and nondisabling and only 21% were permanent or led to death.
- The American Society of Plastic Surgeons acknowledged that the ideal patient for an office-based procedure has an American Society of Anesthesiology (ASA) physical status of 1 or 2. They recommended that ASA physical status 3 patients undergo an office-based procedure only after an anesthesia consultation and only have an office-based procedure performed under local anesthesia without sedation.
- Pulmonary embolism arising from deep vein thrombosis remains an important risk of an office-based procedure.
- When formulating a quality improvement program, there should be random chart reviews as well as key sentinel events that trigger a case review.
- The anesthesiologist should function as a zealous patient advocate in assuring that an anesthetic only is performed in a safe anesthetizing location. Issues that must be sorted out and codified include equipment, monitoring in conformance with ASA standards, emergency drugs and equipment, controlled substance handling and storage, medical director and governance, plan for hospital admission if needed, fire prevention and preparedness, and accreditation.
- Nonfatal injuries occurring during monitored anesthesia care were more likely to be permanent than the nonfatal injuries occurring during general anesthesia, which were more frequently temporary. Of all injuries occurring during monitored anesthesia care, 34% were fatalities.
- In terms of smooth operation of an office-based anesthesia practice and patient satisfaction, prevention of postoperative nausea and vomiting and pain are paramount.

Source: Barash 9e, Chapter 32, Key Points, PDF pp. 2586–2588.

Chapter 33: Nonoperating Room Anesthesia

- Nonoperating room (NOR) locations are remote from a hospital's main operating room sites.
- A three-step approach that considers the patient, the procedure, and the environment is useful in considering any anesthetic in the NOR setting.
- All significant patient comorbidities must be carefully assessed, and the appropriate level of sedation, general anesthesia, and level of monitoring determined by the anesthesiologist caring for the patient.
- Procedural considerations are both general (e.g., duration, position, and level of discomfort) and specific to individual specialties.
- Patients must receive the same standard of care at a NOR site as they do in the operating room, and the American Society of Anesthesiologists (ASA) has defined guidelines to be applied to the administration of anesthesia in NOR locations.¹
- The anesthetic and monitoring equipment used for nonoperating room anesthesia (NORA) must meet the same standards as equipment provided in the operating room.
- Following NORA, the patient should be transported to an appropriate postanesthesia care unit (PACU), accompanied and monitored by anesthesia-trained personnel.
- Environmental considerations in NOR locations include hazards such as radiation, magnetic resonance imaging, and the side effects of intravenous contrast agents.

Source: Barash 9e, Chapter 33, Key Points, PDF pp. 2651–2652.

Chapter 34: Anesthesia for the Older Patient

- The aging of America presents a medical and economic challenge to the entire healthcare system as older patients present for surgery in ever-increasing numbers.
- Aging involves a gradual, cumulative process of damage and deterioration. The aging process affects connective tissue and cellular function, including the mitochondria, and inevitably leads to decreased function and, ultimately, frailty.
- Determining the preoperative frailty status of a patient is essential to understanding perioperative risk. Frailty is associated with poorer postoperative outcomes and preoperative identification of a frail patient should prompt a tailored, risk-mitigation perioperative care plan.
- Decreased organ reserve and increased sensitivity to anesthetic agents result from generalized body composition changes such as connective tissue stiffening and decreased muscle mass, plus impaired autonomic reflexes, and increased sensitivity to drugs.
- Preoperative patient-centered care and multi-system evaluation focused on goals and risk-mitigation are critical for enhancing recovery of function after surgery. Planning for geriatric surgery has unique ethical considerations and should include discussions about informed consent, living wills, and advance directives. Risk modification should target education, prehabilitation, medication scrutiny, and discharge planning.
- Intraoperative management must consider the increased sensitivity to drugs in the elderly patient, as well as an increased likelihood of hemodynamic, pulmonary, and thermoregulatory instability.
- As older patients progress through the days following surgery, vigilance for neurologic and physiologic perturbations that can derail recovery is critical.
- Analgesia is an important component of postoperative care but is made more difficult by the increased likelihood of adverse consequences from the analgesic regimen.
- Perioperative complications, most notably pulmonary, cardiac, and central nervous system complications such as delirium or cognitive decline, occur more commonly in the elderly patient because of an interaction between comorbid disease and the decreased physiologic reserve of aging.

Source: Barash 9e, Chapter 34, Key Points, PDF pp. 2705–2707.

Chapter 35: Neuraxial Anesthesia

- A growing body of evidence suggests that spinal and epidural anesthesia are associated with less morbidity and mortality than general anesthesia. This association is strongest in patients undergoing major hip or knee surgery, and data continues to evolve.
- The epidural space is not a homogeneous, continuous cylinder surrounding the subarachnoid space. Instead, discrete pockets of fat lie in the posterior and lateral epidural spaces. Epidural veins pass through the anterior and lateral space. Blood in an epidural needle or catheter suggests that you are not in the midline.
- Develop a mental model of spinal anatomy that will allow you to systematically locate the subarachnoid or epidural space.
- The hemodynamic effects of spinal and epidural anesthesia are proportional to the extent of local anesthetic block.
- Patient variables (age, height, weight, etc.) minimally affect the spread of intrathecal and epidural medications.
- Unrecognized intrathecal (subarachnoid) injection of local anesthetics is a common cause of morbidity during epidural block. Early signs of unintended intrathecal local anesthetic injection can be subtle and easily missed. Commonly used epidural doses, if injected intrathecally, can produce excessive levels of sensory and motor block.
- Intravenous local anesthetics can produce seizures and cardiac arrest. Test doses and incremental injections are key safety steps to prevent this complication.
- Patients receiving antithrombotic or thrombolytic medications are at increased risk of developing a spinal or epidural hematoma. Refer to the American Society of Regional Anesthesia and Pain Medicine Regional Guidelines for Antithrombotic and Thrombolytic Therapy for current recommendations.

Source: Barash 9e, Chapter 35, Key Points, PDF pp. 2786–2787.

Chapter 36: Peripheral Nerve Blockade

- Peripheral nerve blocks (PNBs) deliver effective anesthesia and analgesia in a site-specific manner, with the potential for long-term benefits.
- PNB safety and success require precise nerve identification and local anesthetic delivery. Improved ultrasound image resolution has recently shifted regional anesthesia from targeting particular nerves to the expanded use of interfascial plane blocks.
- Ultrasound imaging provides real-time visualization of needle movement when approaching nerve structures, minimizing the risk of inadvertent needle contact with critical anatomy and potentially reducing complications. Although ultrasound technology is highly desirable as a tool for providing regional anesthesia, it requires extensive training and a complete understanding of both the equipment and the cross-sectional anatomy relevant to regional nerve blocks.
- Peripheral nerve stimulators are excellent ultrasonography adjuncts for deep blocks, but they do not eliminate the risk of nerve injury. An awake and responsive adult patient may improve safety by reporting nerve contact or discomfort during injection.
- Pediatric nerve block techniques are broadly comparable to those performed on adults, with the notable exception that they are routinely performed with the child under general anesthesia or sedation.
- Nerve blocks that are anatomically related to bone or vascular features are more reliably and more accessibly performed using ultrasound guidance.
- Larger quantities of local anesthetic may boost PNB success, but the dosage must be restricted to minimize systemic toxicity. Increased local anesthetic concentrations produce more reliable motor blocks. Ultrasound imaging may permit successful blocks with lower volumes of local anesthetics, but this has not been completely validated.
- This chapter uses the most recent standardized nomenclature for regional anesthetic procedures, as recommended by the American Society of Regional Anesthesia and Pain Medicine (ASRA) and the European Society of Regional Anaesthesia and Pain Therapy (ESRA).

Source: Barash 9e, Chapter 36, Key Points, PDF pp. 2884–2886.

Chapter 37: Anesthesia for Neurosurgery

- The brain receives approximately 70% of its blood supply from two internal carotid arteries anteriorly and 30% from two vertebral arteries posteriorly forming the basilar artery, that subsequently converge to form the circle of Willis, an anastomotic ring at the base of the skull.
- The spinal cord receives its blood supply from one anterior spinal artery and two posterior spinal arteries. The anterior spinal artery originates from 6 to 8 major radicular arteries derived from the aorta, with the largest one being the artery of Adamkiewicz (usually occurring at T11 or T12 and generally supplying T8 to the conus medullaris terminus).
- Cerebral blood flow (CBF) is regulated by "flow-metabolism coupling," whereby increases in regional neuronal electrical activity require corresponding increases in regional blood flow. Such coupling occurs on the order of seconds, with very little variation in the amount of oxygen extraction by the brain tissue (i.e., CBF matches cerebral metabolic rate of oxygen consumption (CMRO₂) very quickly and efficiently in the healthy brain).
- Moderate changes in mean arterial pressure (MAP) (or cerebral perfusion pressure) will yield a consistent CBF of 50 mL/100 g/min, due to the normal brain's ability to autoregulate its blood flow. Cerebral autoregulation of blood flow is thought to remain intact between a MAP of approximately 60 and 160 mmHg and functions by altering cerebrovascular resistance (CVR) on the order of 5 to 60 seconds. The alteration in CVR is accomplished in both a rapid phase ("dynamic autoregulation") and a slow phase ("static autoregulation").
- CBF is linearly associated with arterial carbon dioxide tension (PaCO₂) between 20 and 80 mmHg. Hyper- and hypoventilation, both patient-determined and iatrogenic, play critical roles in decreasing or increasing CBF, respectively. A change in PaCO₂ of 1 mmHg roughly correlates to a similar change in CBF of 1 to 2 mL/100 g/min. Below the lower limit of this linear effect (i.e., with PaCO₂ below 20 mmHg), maximal cerebral vasoconstriction leads to tissue hypoxia and a reflex vasodilation.
- Intravenous drugs, such as propofol, etomidate, benzodiazepines, and barbiturates, decrease CBF by virtue of a drug-induced decrease in CMRO₂ and subsequent flow-metabolism coupling. Autoregulation and PaCO₂ responsiveness remain intact with these agents.
- Potent volatile anesthetics, such as isoflurane, sevoflurane, and desflurane, are direct cerebral vasodilators. However, this direct vasodilation is offset by a drug-induced decrease in CMRO₂, and via flow-metabolism coupling, an attenuation of the direct effect on CBF. This leads to minimal, if any, increase in CBF at lower doses. However, at high doses, where maximal suppression of CMRO₂ has occurred, are direct vasodilatory effects observed leading to a dose-dependent increase in CBF. Furthermore, autoregulation is inhibited with potent volatile anesthetic drugs in a dose-dependent fashion, although the cerebral vasculature remains responsive to changes in PaCO₂.
- The Monro-Kellie doctrine states that "an increase in the volume of one intracranial compartment will lead to a rise in intracranial pressure (ICP) unless it is matched by an equal reduction in the volume of another compartment." Since the brain parenchyma is relatively incompressible, cerebrospinal fluid (CSF) and cerebral blood volume play an integral role in accommodating increases in ICP.
- The most commonly used modalities of evoked potential monitoring are somatosensory evoked potentials, motor evoked potentials, and electromyography, with brainstem auditory evoked potentials and visual evoked potentials being less commonly used. Anesthetic drugs play a major role in facilitating the success of intraoperative evoked potential monitoring.

- Reliable pharmacologic and nonpharmacologic therapies to prevent neuronal ischemic injury are currently not readily available for use in the perioperative period. At the present time, one can only hope to attenuate injury by preventing secondary insults to surrounding neuronal tissue (e.g., ensuring adequate oxygen and substrate delivery).
- The preoperative evaluation of the neurosurgical patient is of paramount importance to ensure a safe and successful anesthetic. Specific problems must be identified so as to formulate appropriate plans for intraoperative and postoperative management. For patients with intracranial mass lesions, the most important fact to ascertain is the presence and extent of intracranial hypertension and this should be assumed until information proves otherwise. Choice of drugs for maintenance of general anesthesia depends to a great extent on the ICP and whether neuromonitoring is being employed.

Source: Barash 9e, Chapter 37, Key Points, PDF pp. 3051–3054.

Chapter 38: Anesthesia for Thoracic Surgery

- Determine prior to the commencement of anesthesia whether the patient will be able to tolerate the planned lung resection.
- Preoperative assessment of vital capacity is critical because at least three times the tidal volume is needed for an effective cough.
- Smoking increases airway irritability, decreases mucociliary transport, and increases secretions. It also decreases forced vital capacity and forced expiratory flow 25% to 75%, thereby increasing the incidence of postoperative pulmonary complications
- The absolute indications for lung separation using a double-lumen tube are for protection against spillage of blood, infectious material, or lavage fluid from one lung to the other, or for ventilation in the case of a bronchopleural fistula or bullae. Pneumonectomy or lobectomy are relative indications.
- The most important advance in checking the proper position of a double-lumen tube or a bronchial blocker is the pediatric flexible fiberoptic bronchoscope.
- During one-lung ventilation (OLV), the dependent lung should be ventilated using a tidal volume that results in a plateau airway pressure <25 cmH₂O at a rate adjusted to maintain a PaCO₂ of 35 ± 3 mmHg.
- The choice of anesthetic technique for OLV must take into consideration the effects on oxygenation and therefore on hypoxic pulmonary vasoconstriction.
- The need for OLV is much greater with video-assisted thoracoscopic surgery than with open thoracotomy because it is not possible to retract the lung during VATS, unlike during open thoracotomy.
- The potential advantages offered by high-frequency positive pressure ventilation are that lower tidal volume and inspiratory pressure result in a quiet lung field for the surgeon, with minimal movement of the airway, lung tissue, and mediastinum.
- Myasthenia gravis is a disorder of the neuromuscular junction, characterized by weakness and fatigability of voluntary muscles with improvement following rest. Surgical thymectomy is a commonly performed therapy.
- Adequate pain relief is essential to avoid atelectasis from limited inspiratory thoracic cage expansion. It also makes the patient more comfortable postoperatively.

Source: Barash 9e, Chapter 38, Key Points, PDF pp. 3132–3134.

Chapter 39: Anesthesia for Cardiac Surgery

- The hemodynamic goals for patients with coronary artery disease are to control heart rate, avoid hypo- and hypervolemia, and assure adequate perfusion pressure.
- Volatile anesthetics precondition the myocardium against reperfusion injury and help reduce myocardial infarct size.
- Manipulating preload, afterload and heart rate helps to promote forward flow in patients with regurgitant valvular disease.
- In patients with aortic dissection, preventing hypertension and tachycardia are crucial to avoiding progression of dissection.
- Mechanical Circulatory Support (MCS) reduces left ventricular stroke work and myocardial oxygen demand and provides adequate systemic and coronary perfusion in patients with cardiogenic shock.
- Contemporary anesthetic regimens for elective cardiac surgical procedures provide a balanced anesthetic that is based on use of short-acting, fast-emergence drugs to facilitate early recovery.

Source: Barash 9e, Chapter 39, Key Points, PDF pp. 3281–3282.

Chapter 40: Anesthesia for Vascular and Endovascular Surgery

- Atherosclerosis is the most common underlying pathophysiologic mechanism for the development of cardiovascular disease.
- The same risk factors that lead to coronary artery disease also lead to vascular disease in other major vascular beds, including the cerebral, aortic, and peripheral circulation.
- Improvements in the medical management of patients with atherosclerotic risk factors with antiplatelet agents, β -blockers, angiotensin-converting enzyme inhibitors, statins, and strict glucose control have revolutionized the management of vascular surgery patients. Meticulous optimization of comorbid conditions plays a critical role in the reduction of perioperative morbidity and mortality in this patient population.
- Most vascular surgery patients are at elevated perioperative risk of major adverse cardiac events, which remain the single most important cause of both short- and long-term morbidity and mortality following vascular surgery.
- The pathophysiology of aneurysmal aortic disease is a clinically distinct degenerative process, although many of the risk factors are shared between degenerative and atherosclerotic vascular disease.
- Endovascular procedures have transformed the field of vascular surgery with an increasing number of procedures being performed via a minimally invasive approach. Perioperative improvements in morbidity and mortality must be weighed against the long-term risks of decreased durability and increased need for repeat intervention.

Source: Barash 9e, Chapter 40, Key Points, PDF pp. 3380–3381.

Chapter 41: Obstetric Anesthesia

- Airway edema may be particularly severe in women with preeclampsia, in those who have pushed during the second stage of labor, or with concurrent use of tocolytic agents.
- As oxygen consumption increases during pregnancy, the maternal cardiovascular system adapts to meet the metabolic demands of a growing fetus.
- The driving force for placental drug transfer is the concentration gradient of free drug between maternal and fetal blood.
- Labor analgesia may benefit mother and fetus and should not be withheld if requested.
- Benefits of neuraxial anesthesia for cesarean delivery include avoiding airway manipulation, decreasing aspiration risk, minimizing fetal anesthetic exposure, and allowing the mother to remain awake during childbirth.
- A rapid-sequence induction of anesthesia, application of cricoid pressure, and intubation with a cuffed tracheal tube are usually performed for pregnant women receiving general anesthesia after 20 weeks of gestation.
- Pregnant women are at a higher risk for developing postdural puncture headache due to their age and sex.
- Pregnancy and parturition are considered “high risk” when accompanied by conditions unfavorable to the well-being of the mother, fetus, or both.
- Preeclampsia includes a spectrum of end-organ manifestations of the disease—severe hypertension or other end-organ disease is termed preeclampsia with severe features.
- Hemorrhage is the leading cause of maternal mortality worldwide.
- Cardiovascular conditions are now responsible for over a quarter of all pregnancy-related deaths in the United States.
- Obese parturients are more likely to have antenatal comorbidities, which may adversely affect outcomes.
- In the setting of nonobstetric surgery during pregnancy, no data suggest that any one anesthetic technique is preferred over another, provided oxygenation and blood pressure are maintained and hyperventilation is avoided.

Source: Barash 9e, Chapter 41, Key Points, PDF pp. 3471–3473.

Chapter 42: Neonatal Anesthesia

- Understanding the physiologic changes that occur during the transition from fetal to neonatal life is crucial to the anesthetic management of the neonate. The circulatory, pulmonary, hepatic, and renal systems are all affected in this process.
- Important physiologic and anatomic factors account for the rapid rate of desaturation observed in neonates. These include an increase in oxygen consumption, a higher closing volume, a high ratio of minute ventilation to functional residual capacity, and a pliable rib cage.
- Persistent pulmonary hypertension of the newborn is a pathologic condition that can be primary but often is secondary to other conditions, including meconium aspiration, sepsis, congenital diaphragmatic hernia, or pneumonia. Understanding the pathophysiologic characteristics of this condition helps guide therapy.
- Knowledge of the major anatomic differences between the infant and the adult airway helps one understand why the infant's airway is often described as "anterior" and why airway management may be challenging. These differences include a relatively large tongue, a cephalad glottis with anterior slanting vocal folds, a larger occiput, and a narrowing at the cricoid ring.
- Careful attention must be given to the choice of anesthetic agents and dosing of such agents in the neonatal population. Ongoing maturational changes in the renal and hepatobiliary systems, which occur during the first 30 days of life, will affect the metabolism and elimination of many anesthetic agents.
- Although a host of anesthetic techniques are available, including regional anesthesia, multiple factors are considered when choosing an anesthetic plan for the neonate. These include the surgical requirements, the need for postoperative ventilation, the cardiovascular stability of the neonate, and the anticipated method of postoperative pain control.
- Special considerations must be addressed when planning an anesthetic for a neonate. Some of the controversial issues include the risk of postoperative apnea, the role of oxygen concentration in the development of retinopathy of prematurity, and the neurocognitive effects of anesthetic agents on the fetal and neonatal brain.
- True surgical emergencies are uncommon in the neonatal period. Knowledge of conditions with comorbidities, such as tracheoesophageal fistula, omphalocele, and congenital diaphragmatic hernia, and a thorough preoperative evaluation and stabilization of such neonates cannot be overemphasized.

Source: Barash 9e, Chapter 42, Key Points, PDF pp. 3564–3566.

Chapter 43: Pediatric Anesthesia

- Children with certain syndromes have facial features that can make tracheal intubation and/or mask ventilation difficult. Pediatric anesthesiologists should be familiar with these syndromes.
- The process of “informed consent” entails obtaining permission from parents/legal guardians and may involve seeking assent from the child depending on their capacity to understand and make medical decisions.
- Perioperative critical events including pulmonary aspiration, respiratory adverse events, cardiovascular instability, malignant hyperthermia, and anaphylaxis can be mitigated with adequate preoperative planning, multidisciplinary care, and use of standardized protocols.
- The preoperative assessment is critical in optimizing patients and guiding the perioperative plan, including considerations for decreasing airway hyperreactivity, management of upper airway obstruction, anxiolysis, and glycemic control.
- Management of the difficult airway in children should be approached differently than adults, as awake tracheal intubation is unlikely to be tolerated. Use of video laryngoscopy should be considered for the first attempt.
- Induction of anesthesia can be achieved by inhalation of anesthetic gasses or IV medications, and care should be taken to make this experience pleasant and atraumatic for pediatric patients, with use of premedication and distraction.
- We recommend avoidance of nitrous oxide and desflurane due to environmental concerns, as they are potent greenhouse gasses, and safe alternatives exist. Total intravenous anesthesia plus regional anesthesia where appropriate, can provide a superior experience for the patient and are less harmful to the environment.
- Regional anesthesia (both neuraxial and peripheral nerve techniques) are safe in pediatrics and should be an integral component of multimodal analgesia.
- Pediatric patients are at increased risk for respiratory complications during induction, emergence, and recovery from anesthesia. Perioperative respiratory adverse events (PRAEs) include bronchospasm, laryngospasm, severe persistent coughing, airway obstruction, apnea, oxygen desaturations, and stridor. The patients at highest risk include neonates, those with a current or recent upper respiratory infection, and those with chronic lung disease.
- Emergence delirium is more common in the pediatric population. Emergence delirium typically manifests as a period of restlessness, inconsolability, lack of eye contact with caregivers, lack of purposeful activity, and lack of awareness of surroundings. A variety of prevention techniques have been trialed in the pre- and intraoperative phases of care. Dexmedetomidine is often the first-line agent chosen for treatment due to demonstrated efficacy.

Source: Barash 9e, Chapter 43, Key Points, PDF pp. 3697–3699.

Chapter 44: Anesthesia for Laparoscopic and Robotic Surgeries

- Small incisions, decreased postoperative pain, and lower surgical complication rates are some of the benefits of laparoscopy over laparotomy.
- Pneumoperitoneum and positioning cause physiologic changes that are important for the anesthesiologist to understand.
- Patients with a body mass index (BMI) >40 kg/m² and obesity-related comorbidities tend to have more perioperative complications when undergoing laparoscopic surgery compared to those without such conditions.
- Advances in robotic-assisted laparoscopy have expanded its application to multiple subspecialties.
- Access to the patient during robotic-assisted surgery may be limited, which is especially significant in the event of an intraoperative cardiopulmonary or airway emergency.
- Severe hypercarbia and acidosis from absorbed carbon dioxide (CO₂) can lead to reduced inotropy, dysrhythmias, and arterial vasodilation.
- High intra-abdominal pressures (IAP), combined with hypovolemia, can severely impair venous return and cardiac filling.
- Migration of the endotracheal tube into a bronchus is a risk during laparoscopic surgery. This is due to displacement of the diaphragm into the thorax caused by abdominal insufflation, as well as Trendelenburg positioning.
- Renal blood flow, glomerular filtration, and urine output are reduced during pneumoperitoneum.
- The assessment of neuromuscular blockade during laparoscopic surgery can be highly subjective.
- Major vascular injuries occur rarely during abdominal entry but are associated with significant morbidity and mortality.
- Severe hypotension during pneumoperitoneum should be treated with desufflation and may require conversion to an open procedure.
- Risk factors for complications related to subcutaneous emphysema include operative time more than 200 minutes, elevated BMI, high IAP, and Nissen fundoplication surgery.
- Tension capnothorax is a life-threatening condition that requires a high index of suspicion and immediate action from the operating room team.
- Preemptive multimodal pain management strategies and postoperative nausea and vomiting prophylaxis facilitate rapid patient recovery after laparoscopic surgery.

Source: Barash 9e, Chapter 44, Key Points, PDF pp. 3795–3797.

Chapter 45: Anesthesia and Obesity

- Obese patients may have significant cardiovascular disease and appear asymptomatic because they often have limited exercise tolerance.
- Elevated liver function tests (mostly elevated alanine aminotransferase) are seen in many obese patients, but no clear correlation exists between abnormalities of routine liver function tests and the capacity of the liver to metabolize drugs.
- A neck circumference of ≥ 40 cm is the biggest predictor of difficult mask ventilation, difficult laryngoscopy, and difficult endotracheal intubation.
- Evidence of obstructive sleep apnea (OSA) and obstructive hypoventilation syndrome (OHS) should be obtained preoperatively because they are frequently associated with difficult airway management and increased perioperative pulmonary complications.
- Blood pressure measurements can be falsely elevated if a cuff is too small. In the supine position, forearm measurements with a standard cuff can be used when upper arm measurements are challenging.
- The “ramped” position elevates the obese patient’s upper body above the chest and improves laryngoscopic view compared with the standard “sniffing” position.
- Larger doses of induction agents may be required by obese patients because blood volume, muscle mass, and cardiac output increase linearly with the degree of obesity. An increased dose of succinylcholine is necessary because of an increase in pseudocholinesterase activity.
- Positive end-expiratory pressure is the only ventilatory parameter that has consistently been shown to improve respiratory function in obese patients but it decreases venous return, cardiac output, and subsequent oxygen delivery.
- Prompt but safe extubation reduces the likelihood that the morbidly obese patient will become ventilator dependent, especially in patients with cardiopulmonary disease.
- Because of the risk of perioperative hypoxemia and apnea in obese patients, postoperative pain management should include opioid-sparing multimodal analgesic techniques. Regional anesthetic techniques reduce the risk of opioid-related complications.
- Obese patients who have received either neuraxial or parenteral opioids require careful postoperative monitoring. Delayed respiratory depression with centrally administered neuraxial opioids, when coupled with a potentially difficult airway in the obese patient, suggests that close monitoring is prudent.
- Morbid obesity is a major independent risk factor for deep venous thrombosis and sudden death from acute postoperative pulmonary embolism. Mobilization in the morbidly obese is often difficult but critically important in the prevention of postoperative complications.

Source: Barash 9e, Chapter 45, Key Points, PDF pp. 3839–3841.

Chapter 46: The Liver: Surgery and Anesthesia

- The liver is the largest internal organ, accounting for 2% of the total body mass of adults. It receives 25% of the cardiac output via a dual afferent blood supply. The portal vein supplies 75% of the hepatic blood flow, the hepatic artery supplies the remaining 25%. Because of the higher oxygen content in the hepatic artery, each vessel provides roughly 50% of the hepatic oxygen supply.
- The liver plays a preeminent role in the intermediary metabolism of nutrients (glucose, nitrogen, and lipids) and the detoxification of chemicals, including lipophilic medications. Liver dysfunction affects the metabolism of nutrients and xenobiotics, and negatively impacts nearly every other organ system.
- Portal hypertension, the end result of hepatic injury and fibrotic changes, results in portosystemic shunts that bypass the liver's metabolic and detoxification capabilities. When nitrogenous waste and other substances normally cleared by the liver enter the central circulation, hepatic encephalopathy ensues.
- Additional complications of portal hypertension include variceal hemorrhage, ascites, and hepatorenal syndrome. Cardiac sequelae include hyperdynamic circulation due to decreased systemic vascular resistance, which results in an increase in cardiac output.
- Perioperative complications encountered by cirrhotic patients include liver failure, postoperative bleeding, infection, and renal failure. Patients with a model for end-stage liver disease (MELD) score of less than 11 have a low postoperative mortality and represent an acceptable surgical risk. End-stage liver disease patients with a risk of postoperative liver failure should have elective abdominal surgery at institutions with a liver transplant program. In patients with a MELD score of 20 or higher (or Child Class C), high mortality risk contraindicates elective procedures until after liver transplantation.
- Medical management undertaken to optimize cirrhotic patients undergoing surgery should be directed toward treating active infection, minimizing vasoactive infusions, optimizing central blood volume and renal status, minimizing ascites, and improving encephalopathy and coagulopathy.
- The perioperative risk of patients with end-stage liver disease depends more on the operative site and the degree of liver impairment than the anesthetic technique.

Source: Barash 9e, Chapter 46, Key Points, PDF pp. 3904–3906.

Chapter 47: Endocrine Function

- The major risk of anesthesia in the poorly controlled thyrotoxic patient is thyroid storm, which must be aggressively treated with β -blockers, iodide, and antithyroid drugs.
- Asymptomatic or mild hypothyroidism does not appear to significantly increase anesthetic risk and is not a contraindication to surgery. Moderate to severe hypothyroidism should be corrected before surgery to prevent multisystem complications.
- Patients who have received corticosteroids for more than 3 weeks in the past year may have adrenal suppression; supplemental steroids in the perioperative period should be considered.
- Preoperative preparation of the pheochromocytoma patient with α -blockers decreases intraoperative hemodynamic instability.
- Pheochromocytoma manipulation is associated with severe hypertension that should be treated aggressively with nitroprusside, phentolamine, or other rapidly acting vasodilators.
- The major perioperative risks to the diabetic patient come from unrecognized or poorly managed coexisting diseases, especially coronary artery disease.
- Very tight control of perioperative blood glucose levels appears to increase the risk of hypoglycemic complications without clearly reducing the risk of hyperglycemic complications.
- Endotracheal intubation may be unpredictably difficult in patients with acromegaly.

Source: Barash 9e, Chapter 47, Key Points, PDF p. 3985.

Chapter 48: Anesthesia for Otolaryngologic Surgery

- The restricted spaces in the airway require an understanding and cooperative relationship between surgeon and anesthesiologist and the use of specially adapted equipment suitable to these cramped areas.
- Despite only mild-to-moderate tonsillar enlargement on physical examination, children with obstructive sleep apnea have upper airway obstruction while awake and apnea during sleep. The clinician should not underestimate the severity of the problem based on tonsillar size alone.
- Patients with Obstructive Sleep Apnea have increased sensitivity to opioids, and consequently, the dose administered should be reduced by as much as 50%.
- Post-tonsillectomy hemorrhage may result in unappreciated large volumes of swallowed blood originating from the tonsillar fossa. These patients must be considered to have a full stomach, and anesthetic precautions addressing this situation must be taken.
- The middle ear and sinuses are air-filled, nondistensible cavities. During procedures in which the eardrum is replaced, or perforation is patched, nitrous oxide should be discontinued or, if this is not possible, limited to a maximum of 50% during the application of the tympanic membrane graft to avoid pressure-related displacement.
- During functional endoscopic sinus surgical procedures, systemic absorption of vasoconstrictive agents may cause hypertension, bradycardia, tachycardia, and arrhythmias. Preoperative evaluation should include a thorough investigation of the patient's cardiovascular status. Rapid response by the anesthesiologist to these effects is necessary for preventing complications.
- Patients with a history of head and neck cancer may have undergone prior chemotherapy that can impact specific organ systems or radiation, leading to fibrosis and ankylosis of the temporomandibular joint, rendering direct laryngoscopy difficult.
- Facial trauma is commonly associated with other injuries such as cervical spine and head injuries that can affect patient care beyond airway management.

Source: Barash 9e, Chapter 48, Key Points, PDF pp. 4075–4077.

Chapter 49: Anesthesia for Ophthalmologic Surgery

- Cataract surgery is one of the most frequently performed surgical procedures worldwide but represents just one aspect of the various ophthalmic subspecialties, which include cornea, retina, glaucoma, uveitis, strabismus, oculoplastic, and oncology surgeries.
- Eye surgery patients are often at the extremes of age, ranging from premature babies with retinopathy of prematurity to nonagenarians with multiple coexisting diseases in which age-related anesthetic considerations are key.
- With intraocular procedures, globe akinesia, patient movement, and control of intraocular pressure (IOP) are important variables; however, with extraocular surgery, the significance of IOP fades and elicitation of the oculocardiac reflex becomes a concern.
- Inhalation anesthetics cause dose-related reductions in IOP. The exact mechanisms are unknown, but postulated causes include depression of a control center in the diencephalon, reduction of aqueous humor production, enhancement of aqueous outflow, and/or relaxation of the extraocular muscles.
- The oculocardiac reflex is triggered by pressure on the globe and by traction on the extraocular muscles as well as on the conjunctiva or on the orbital structures. This reflex, whose afferent limb is trigeminal and efferent limb is vagal, may also be elicited by performing a regional eye block, by ocular trauma, and by direct pressure on tissue remaining in the orbital apex after enucleation.
- Ophthalmic drugs may significantly alter the patient's reaction to anesthesia. Similarly, anesthetic drugs and maneuvers may dramatically influence intraocular dynamics.
- Several anesthetic options are available for the many types of ocular procedures, including general anesthesia, retrobulbar (intraconal) block, peribulbar (extraconal) anesthesia, sub-Tenon block, topical analgesia, and intracameral injection.
- The complications of ophthalmic anesthesia can be both vision- and life-threatening.

Source: Barash 9e, Chapter 49, Key Points, PDF pp. 4131–4133.

Chapter 50: The Renal System and Anesthesia for Urologic Surgery

- Kidney filtration and reabsorption are susceptible to alterations by surgical illness and anesthesia. Autoregulation of renal blood flow (RBF) is effective over a wide range of mean arterial pressures (50 to 150 mmHg). Autoregulation of urine flow does not occur, but a linear relationship between mean arterial pressure above 50 mmHg and urine output is observed.
- Kidney medullary blood flow is low (2% of total RBF) but central to the kidneys' ability to concentrate urine. During periods of reduced kidney perfusion, the metabolically active medullary thick ascending limb may be especially vulnerable to ischemic injury.
- The physiologic response to surgical stress invokes intrinsic mechanisms for sodium and water conservation. Kidney cortical vasoconstriction causes a shift in perfusion toward juxtamedullary nephrons, a decrease in glomerular filtration rate, and retention of salt and water result.
- The stress response may induce a decrease in RBF and glomerular filtration rate, causing afferent arteriolar vasoconstriction. If this situation is not reversed, ischemic damage to the kidney may result in acute renal failure (ARF).
- Anesthetic-induced reductions in RBF have been described for many agents but are usually clinically insignificant and reversible. Likewise, anesthetic agents have not been shown to interfere with the kidney response to physiologic stress.
- Isolated ARF carries a mortality of up to 60% in surgical patients, with acute tubular necrosis being the cause of ARF in most of these patients.
- Surgical patients with nondialysis-dependent chronic kidney disease are at higher risk of developing end-stage renal disease. The single most reliable predictor of new postoperative need for dialysis is preoperative kidney insufficiency.
- Overall, there are no conclusive comparative studies demonstrating superior kidney protection or improved kidney outcome with general versus regional anesthesia.
- Maintaining adequate intravascular volume and hemodynamic stability with aggressive management of kidney hypoperfusion is a basic principle of anesthetic care to prevent acute kidney injury.
- Urologic patients are often elderly, have numerous comorbidities, and require critical evaluation prior to any urologic procedure.
- Combining epidural with general anesthetic techniques for some major urologic surgeries may offer advantages for accelerated recovery, improved analgesia, and even better outcomes, but these techniques must be conducted with respect for other perioperative issues, including thromboprophylaxis for prevention of deep venous thrombosis.
- Watchful waiting, minimally invasive principles, and technologic innovation (e.g., laparoscopy, robotics) have changed the favored approach to many kidney, bladder, and prostate disorders, in some cases reducing the number of high-risk surgeries, in others creating other safer less morbid alternate treatments.
- Absorption of irrigating solution related most often to transurethral prostate or bladder tumor resections can cause "TUR syndrome," a condition that, while becoming less common, has the potential to be serious and even life threatening during the several hours following surgery. Knowledge of specific concerns relevant to the different irrigating solutions, and vigilance of the

anesthesiologist to factors that minimize absorption, recognition of signs and symptoms, and appropriate treatment are key to favorable outcomes with this condition.

Source: Barash 9e, Chapter 50, Key Points, PDF pp. 4224–4226.

Chapter 51: Anesthesia for Orthopedic Surgery

- Optimal anesthetic management for orthopedic surgery requires expertise in regional anesthesia and analgesia techniques.
- Regional anesthesia can provide physiologic benefits and facilitate recovery compared to general anesthesia.
- General anesthesia is appropriate for surgeries not amenable to and/or patients with contraindications to regional techniques.
- Orthopedic surgery patients may have limited mobility and may require special attention to avoid positioning-related injury.
- Patients presenting for spine surgery should be evaluated carefully for potential airway challenges and/or impaired respiratory function.
- Major spine surgery frequently involves significant bleeding, and a blood conservation strategy should be considered as part of the anesthetic plan.
- Intraoperative monitoring of spinal cord function should be used for surgeries where the cord is at risk of injury.
- Catastrophic venous air embolism during spine surgery may present as unexplained hypotension with high end-tidal nitrogen and low end-tidal carbon dioxide.
- Nerve injury may result from surgical trauma and/or nerve blockade in the setting of pre-existing neurologic deficits.
- In the sitting position, for every 20 cm of difference in height of the head from the heart, there is a 15-mmHg difference in mean arterial pressure.
- Ephedrine, atropine, and glycopyrrolate should be available for management of hypotensive bradycardic events occurring during surgery in the beach chair position.
- Interscalene blocks cause hemidiaphragmatic paresis and can cause respiratory compromise in patients with reduced pulmonary function.
- Venous thromboembolism is a common complication of lower extremity orthopedic surgery performed with inadequate thromboprophylaxis.
- The American Society of Regional Anesthesia has released guidelines for safe use of regional anesthesia in the setting of antithrombotic or thrombolytic therapy.

Source: Barash 9e, Chapter 51, Key Points, PDF pp. 4351–4353.

Chapter 52: Transplant Anesthesia

- Brain death is declared when the clinical picture is consistent with irreversible cessation of all brain functions.
- Anesthesiologists have a significant role to play in organ procurement and should consult with local organ procurement personnel for protocols to optimize healthy graft retrieval.
- Extended criteria organs, including donation after cardiac death (DCD), are used because of ongoing organ shortage; cold ischemia times should be minimized for extended criteria donor organs.
- Living kidney and liver donors must be healthy and without significant cardiopulmonary, neurologic, or psychiatric disease, diabetes, obesity, or hypertension.
- Renal transplant recipients are often anemic, with hyperdynamic cardiac indices.
- For renal transplantation, the major anesthetic consideration is maintenance of renal blood flow, typically achieved with volume expansion using balanced salt solutions rather than vasopressors.
- Patients with end-stage liver disease have multiorgan dysfunction with secondary cardiac, pulmonary, renal, and neurologic complications.
- Liver transplantation is traditionally described in three phases: dissection, anhepatic phase, and neohepatic phase, with graft reperfusion marking the start of the neohepatic phase.
- Intraoperative management of lung transplant patients should focus on fluid and ventilatory strategies designed to minimize acute lung injury and primary graft dysfunction.
- Mechanical circulatory support devices are increasingly common in patients presenting for heart transplantation.
- Nonischemic cardiomyopathy has replaced ischemic cardiomyopathy as the most common indication for heart transplantation.
- For all transplant recipients, antibiotic, antiviral, antifungal, immunosuppressive, and disease-specific drug regimens should be disrupted minimally in the perioperative period.

Source: Barash 9e, Chapter 52, Key Points, PDF pp. 4404–4406.

Chapter 53: Anesthesia for Trauma and Burns

- Perioperative care of acutely injured, hemorrhaging patients requires both timely operative control and well-balanced resuscitation based on understanding the mechanism of each injury, pathophysiology of trauma, and trauma-induced coagulopathy.
- Airway management in traumatized patients is generally achieved by rapid sequence induction with manual in-line stabilization to minimize movement of cervical spine, in conjunction with either direct or video laryngoscopy.
- In “cannot intubate, cannot ventilate” situations, invasive airway access by cricothyroidotomy should be promptly undertaken.
- Vasopressor use during early trauma resuscitation for correction of blood pressure has not been shown to improve outcomes and is associated with high mortality risk.
- Neurologic injury is a leading cause of death in trauma. Prevention of secondary injury caused by hypotension and hypoxemia is important for improvement of long-term outcome following traumatic brain injury.
- The cornerstone of trauma resuscitation is “hemostatic” blood products administration with emphasis on plasma:erythrocyte (RBC) ratios, as well as platelet:RBC ratios greater than 1:2, preferably closer to 1:1.
- Group O whole blood with low hemagglutinin (anti-A and anti-B) titer offers logistical advantages over blood component therapy and has been increasingly used during initial trauma resuscitation. Early data showed similar survival benefits and safety profile of this to component therapy.
- Viscoelastic hemostatic assays (VHA) are increasingly utilized for trauma resuscitation. However, VHA-guided resuscitation has not shown survival benefit over traditional coagulation tests.
- Compared to adults, perioperative considerations in pediatric trauma include a predominance of blunt and traumatic brain injury; airway anatomical differences; and varying respiratory, cardiovascular and hematologic physiology.
- Major burns can cause systemic pathophysiologic effects correlating to severity of burn injury, manifesting in two temporal phases: (1) burn shock phase (24 to 48 hours after injury) characterized by intravascular volume loss resulting in reduced circulatory volume and possible organ hypoperfusion, and (2) hypermetabolic phase (48 to 72 hours after injury) characterized by hyperdynamic circulation, increased body temperature, oxygen consumption and drug metabolism.
- Inhalational injury increases mortality of burn-injured patients. It can occur as thermal and chemical injury to the upper airway, noxious chemical injury to the lower airway (tracheobronchial-alveolar), and systemic toxicity from toxic gases. A high index of suspicion and repeated monitoring is crucial, as patients can require early tracheal intubation to prevent airway obstruction.

Source: Barash 9e, Chapter 53, Key Points, PDF pp. 4495–4497.

Chapter 54: Postanesthesia Recovery

- Postoperative planning begins when a patient is scheduled for surgery. With emerging protocols for Enhanced Recovery After Surgery (ERAS), specific evidence-based and best practice structure for patient care has the goal of providing care that is coordinated with the surgical team to provide the best outcomes and reduce unnecessary use of resources.
- The level of PACU care depends on the type/approach of surgery, type of anesthetic, intraoperative course of events, as well as patient pre-existing and evolving comorbidities. Typical recovery settings include inpatient recovery, ambulatory recovery (phase I for more intensive needs and phase II for less intensive needs), short stay (23-hour admit), and recovery from specific procedures (e.g., computed tomography, magnetic resonance imaging, invasive radiology, cardiac, pediatric, and radiation procedures).
- The transfer of care to a PACU nurse includes assuring that the patient has had appropriate monitoring applied, admission vital signs taken, a direct and thorough report received that allows for rapid evaluation should complications arise, as well as a nurse capable of handling the acuity of the patient's medical/surgical problems.
- Relative hypovolemia should be evaluated and managed in PACU based on the patient's comorbidities, preoperative status (i.e., bowel preparation, postdialysis), type and duration of surgery, blood loss, and urine output.
- Postoperative analgesia should be individualized to requirements and expectations. A multimodal approach includes the appropriate use of nonsteroidal anti-inflammatory drugs, narcotics, adjuncts, regional and local anesthetics, as well as anxiety relief and appropriate emotional support.
- Discharge criteria should be tailored to the individual patient's underlying disease, recovery course, and postdischarge level of care.
- The cardiac risks during the postoperative stay include myocardial ischemia, which may be minimized with continued use of beta-blockers, analgesia, nitrates, supplemental oxygen, adequate circulating volume, oxygen-carrying capacity, heart rate control, and an understanding of hypercoagulable states.
- The respiratory risks of a patient must take into account the preoperative respiratory disease status. Residual anesthetics, residual muscle relaxants, opioids, and sedatives all impair responsiveness to increasing CO₂ and decreasing O₂ levels. Pain itself can decrease respiration and minute ventilation, leading to CO₂ retention and hypoxemia. Supplemental O₂ application alone does not guarantee hypoxemia will not occur.
- The evaluation of a patient's ability to void may be affected by type of surgery (i.e., genitourinary surgery, hernia repairs) or type of anesthetic (i.e., regional, neuraxial, or opioids).
- Glycemic monitoring and control should persist as a continuum from intraoperative management. Good glycemic control may help with fighting infection, improve wound healing, which can result in better surgical outcomes. Hypoglycemia occurs because of preoperative fasting, intraoperative administration of insulin, as well as the patient using programmable insulin pumps.
- Hypothermia can lead to an increased length of stay in PACU, lethargy, decreased minute ventilation, decreased strength, and increased cardiac demand. It is important to assure that the patient is dry and insulated. The use of air warming blankets, warming mats, and intravenous fluid warmers all minimize hypothermia.

- Many elderly patients experience a varied degree of postoperative confusion, delirium, or cognitive dysfunction in the PACU. Many pediatric patients also experience postemergence delirium leading to increased length of stay in the PACU.
- Postoperative nausea and vomiting is a major cause of patient discomfort and dissatisfaction, as well as an aspiration risk and causes prolonged PACU stay.

Source: Barash 9e, Chapter 54, Key Points, PDF pp. 4553–4556.

Chapter 55: Acute Pain Management

- Inadequate relief of postoperative pain has adverse physiologic effects that can contribute to significant morbidity and mortality, resulting in the delay of patient recovery and return to daily activities.
- The pain pathway is not “hard wired” and nociceptive input is not passively transmitted from the periphery to the brain. Tissue injury tends to fuel neuroplastic changes within the nervous system, which results in both peripheral and central sensitization.
- In order for preventive analgesia to be successful, three critical principles must be adhered to: (1) The depth of analgesia must be adequate to block all nociceptive input during surgery, (2) the analgesic technique must be extensive enough to include the entire surgical field, and (3) the duration of analgesia must include both the surgical and postsurgical periods.
- The various opioid analgesics available today have distinct pharmacologic differences that we can credit to their intricate interaction with the classical three main opioid receptors mu, delta, and kappa (μ , δ , and κ). The opioid receptors are members of a G protein–coupled (guanosine triphosphate regulatory proteins) receptor family, which signals via a second messenger such as cyclic adenosine monophosphate or an ion channel.
- Opioid-sparing multimodal pharmacotherapy, regional anesthesia techniques, and the continuous monitoring of patient ventilation with pulse oximetry and capnography can decrease the risk of opioid-related respiratory depression, particularly in high-risk individuals.
- The therapeutic benefit of nonsteroidal anti-inflammatory drugs is believed to be mediated through the inhibition of cyclooxygenase enzymes, types 1 and 2, which convert arachidonic acid to prostaglandins.
- Short-term use of parecoxib and valdecoxib in patients following coronary artery bypass surgery is associated with an increased risk of thromboembolic events. The authors, therefore, do not recommend prescribing a cyclooxygenase-2 inhibitor for patients with a known history of coronary artery disease or cerebrovascular disease.
- The five variables associated with all modes of patient-controlled analgesia include (1) bolus dose, (2) incremental (demand) dose, (3) lockout interval, (4) background infusion rate, and (5) 1- and 4-hour limits. A typical patient-controlled analgesia regimen in an otherwise healthy adult would be an incremental dose of 1 to 2 mg of morphine with an 8- to 10-minute lockout. The authors do not recommend a background infusion of opioid in the opioid-naïve patient.
- Epidural analgesia is a critical component of multimodal perioperative pain management and improved patient outcome. Meta-analyses investigating the efficacy of epidural analgesia found epidural analgesia to be superior to systemically administered opioids.
- The opioid-dependent patient is often identified just moments prior to surgery and the anesthesia team needs to be innovative. The anesthesiologist needs to be flexible enough to tailor an individual anesthetic that incorporates a multimodal approach, combining regional anesthesia with general anesthesia and nonopioid coanalgesics with opioid analgesics. Opioids remain the mainstay of perioperative pain management, and an adequate dose of opioid needs to be maintained to avoid precipitating withdrawal symptoms.
- Perioperative management of the opioid-tolerant patient requires prudent use of both opioid and nonopioid analgesics as well as the application of site-specific regional anesthesia/analgesia.
- Establishing a successful perioperative pain management service begins with an institutional

commitment to support the service. The team must be built around a physician leader with training and experience in pain medicine. There must be other anesthesiologists available to support the service.

Source: Barash 9e, Chapter 55, Key Points, PDF pp. 4632–4635.

Chapter 56: Chronic Pain Management

- Under normal conditions, A delta and C fibers transmit nociceptive information (pain) to the spinal cord from free nerve endings in the periphery. In chronic pain conditions, the A beta fibers, which normally transmit non-noxious information, also participate in nociceptive transmission.
- Radicular symptoms of pain, paresthesia, and numbness in a dermatomal distribution in the presence of objective signs of weakness, diminished reflexes, and positive straight-leg raise are secondary to pathology or dysfunction of the spinal nerve roots.
- Most randomized controlled studies evaluating efficacy of epidural steroid injections confirm temporary relief of radicular pain. Studies on thermal rhizotomy of the medial branch nerves for relief of facet syndrome show benefit of 3- to 12-month duration. These therapies mitigate use of opioids.
- Postherpetic neuralgia is typically managed pharmacologically, although interventional techniques may be used in refractory cases.
- Antidepressants and antiepileptics are effective for treatment of neuropathic pain syndromes. Antiepileptics are first-line treatment due to a more favorable side-effect profile and speed of therapeutic effect, but efficacy is improved when combined with an antidepressant.
- Complex regional pain syndrome that does not respond to nerve blocks, physical therapy, and/or pharmacologic management may respond to spinal cord stimulation.
- Pain secondary to cancer is generally managed most effectively with a combination of antiepileptic, antidepressant, and opioid pharmacologic agents in addition to interventional procedures.
- Neurolysis of the visceral sympathetic system for pain secondary to abdominal or pelvic cancer relieves pain, decreases opioid consumption, and improves patient quality of life.
- Vertebroplasty and kyphoplasty are indicated for vertebral compression fractures, although some studies question their efficacy.
- Spinal cord stimulation is effective in patients with failed back syndrome and complex regional pain syndrome and is likely effective in treatment of angina and critical limb ischemia.
- Intrathecal drug delivery systems are valuable options in cancer pain patients in whom opioids are ineffective at high doses or cause intolerable side effects. Chronic noncancer pain patients have improved outcomes if using no or minimal opioids prior to initiation of intrathecal therapy.

Source: Barash 9e, Chapter 56, Key Points, PDF pp. 4733–4735.

Chapter 57: Critical Care Medicine

- Simple and inexpensive interventions in the intensive care unit (ICU), such as the utilization of checklists, protocols, and care bundles can result in substantial improvements in patient outcomes.
- The goal of resuscitation in brain injury is to prevent continuing cerebral insult after a primary injury has already occurred. This is accomplished by restoring cerebral blood flow, maintaining adequate cerebral perfusion pressure, reducing intracranial pressure, evacuating space occupying lesions, and avoiding fever, hyperglycemia, and hypoxia. 3. "Triple-H" therapy consisting of hypertension, hypervolemia, and hemodilution is no longer recommended for the treatment of cerebral vasospasm in subarachnoid hemorrhage. Instead, maintenance of euvolemia and a controlled stepwise trial of blood pressure augmentation in patients with suspected vasospasm are recommended.
- Administration of thrombolytic therapy (rtPA) to patients presenting within 4.5 hours of onset of acute ischemic stroke results in improved neurologic outcome.
- There are four general types of shock: hypovolemic, cardiogenic, distributive, and obstructive. The 28-day mortality for septic and cardiogenic shock, two of the most common types encountered in the ICU, are 20% to 40% and 70% to 80%, respectively.
- In patients with septic shock, early and aggressive use of fluid resuscitation, appropriate antibiotics, infectious source control, and vasopressors/inotropes as needed improves survival.
- Despite strong regional, local, and individual biases, there is little evidence to suggest that the mode of mechanical ventilation contributes significantly to any major outcome measure, and the choice of mode is largely one of clinician preference.
- Separation from mechanical ventilation in patients who are recovering from respiratory failure is accelerated by respiratory therapy–driven protocols and daily trials of spontaneous breathing.
- Ventilation with low tidal volume (6 mL/kg) in patients with acute lung injury and acute respiratory distress syndrome reduces mortality, compared to traditional tidal volumes (12 mL/kg).
- Red blood cell transfusion in the ICU should be restricted (transfusion threshold hemoglobin <7 g/dL) with the possible exception of patients with active bleeding, early septic shock, acute myocardial infarction, unstable angina, or primary neurologic or neurosurgical problems.
- Available evidence suggests that practices favoring light sedation, bolus administration versus continuous infusion of sedatives, and daily trials of awakening in ICU patients result in a variety of benefits, including a shortened duration of mechanical ventilation, decreased time in the ICU, and reduced mortality.
- Delay in treatment of nosocomial infections is associated with increased mortality. Treatment should not be delayed pending diagnostic evaluation; rather treatment should be started after culture specimens are sent, and antibiotics then "de-escalated" after 48 to 72 hours to ensure adequate initial antibiotic treatment, but avoid long-term overuse of antibiotics.
- End-of-life care is an important aspect of ICU care, and good communication between medical teams, patients, and their family members is required to ensure that delivered care is consistent with patient values and preferences. Patients at high risk of death or severely impaired functional recovery should be offered care focused on comfort and not just curative treatment. Palliative care can also be offered in conjunction with curative care.

Source: Barash 9e, Chapter 57, Key Points, PDF pp. 4804–4806.

Chapter 58: Cardiopulmonary Resuscitation

- Brain adenosine triphosphate is depleted after 4 to 6 minutes of no blood flow. It returns to nearly normal within 6 minutes of starting effective cardiopulmonary resuscitation (CPR).
- The major components of resuscitation from cardiac arrest are airway, breathing, circulation, drugs, and electrical therapy (ABCDE).
- Currently the two dominant theories for the mechanism of blood flow during closed-chest compressions are: cardiac pump and thoracic pump.
- During CPR, myocardial perfusion is 20% to 50% of normal, whereas cerebral perfusion is maintained at 50% to 90% of normal.
- Approximately 1.2% of adults admitted to U.S. hospitals suffer in-hospital cardiac arrest (IHCA). Outcomes from IHCA continue to improve with a current survival rate of 25%—significantly better than the outcomes from out-of-hospital cardiac arrest (OHCA), which carries a survival rate of 10%. The rate of survival after perioperative cardiac arrest is much higher—about 40%.
- End-tidal carbon dioxide is an excellent noninvasive and readily available monitor of the effectiveness of closed-chest compressions.
- Effective uninterrupted chest compressions and defibrillation should take precedence over medications when cardiac arrest is due to ventricular fibrillation or ventricular tachycardia. Effective chest compressions and epinephrine are the primary treatment for cardiac arrest with a nonshockable rhythm. The underlying cause of cardiac arrest should be identified concurrently and immediately reversed if possible.
- Although there is limited evidence that vasopressors result in better early restoration of spontaneous circulation, there is no strong evidence that they improve long-term survival in cardiac arrest.
- After vasopressors, the drugs most likely to be of benefit during CPR are those that help suppress ectopic ventricular rhythms.
- Ventricular fibrillation is the most common electrocardiogram pattern found during witnessed sudden cardiac arrest in adults.
- Untreated ventricular fibrillation is a time-sensitive model with three phases: electrical, circulatory, and metabolic.
- In the pediatric age group, cardiac arrest is more likely related to progressive deterioration of respiratory and circulatory function and less likely to be a sudden event.
- For optimal outcome, successful restoration of spontaneous circulation must be followed by correction of reversible causes of arrest, including immediate coronary reperfusion and aggressive supportive care.
- Excellent teamwork and communication are key for leveraging procedural skills, resuscitation expertise, and diagnostic acumen during and after a cardiac arrest.
- Patients have the right to place limitations on medical treatment, including establishing do-not-resuscitate orders, even while undergoing surgical procedures, particularly palliative procedures. Anesthesiologists must be prepared to provide informed consent that includes matching the patient's goals of care with requested limitations on resuscitative interventions.

Source: Barash 9e, Chapter 58, Key Points, PDF pp. 4891–4893.

Chapter 59: Disaster Preparedness

- There are certain principles that are common to all disaster events, independent of their etiology. As a group, anesthesiologists are well prepared to assist their communities in planning for and in caring for patients who sustain injury or harm from such events.
- Often overlooked during emergency preparedness and disaster management training is the development of a family plan and a personal preparedness plan. The former is important whether one lives alone; has a pet, family, or friends living with them, or has legal responsibility for a loved one (elderly parents, disabled person). A plan should be in place that provides care for and information to loved ones.
- Anesthesiologists' basic understanding of physiology and pharmacology, airway skills, fluid resuscitation expertise, ability to manage ventilators, and provide anesthesia both in and out of the operating rooms, is invaluable.
- If assigned to triage patients, the anesthesiologist will be expected to classify patients into four groups: Those requiring immediate care, delayed care, minimal care (first aid), and expectant (no care).
- If a patient presents with life-threatening injury, the patient is treated first and decontaminated afterward; otherwise, all other patients are decontaminated in the triage area before they are evaluated and treated.
- During a disaster, the anesthesiologist in charge of the schedule for the day should become the operating room medical director, co-locate with the operating room nurse in charge, and determine the status of ongoing cases. Surgeons in the midst of a procedure should be contacted and urged to finish as soon as possible. Elective cases should be postponed or cancelled.
- In many disasters, infrastructure degradation plays a large part in reducing surgical capabilities at a time when demand is increased. Surgeons and anesthesiologists must consider what types of procedures can safely be undertaken and must prioritize care based on urgency and practicality.
- There are three categories of biologic weapons. Category A are those weapons that are highly contagious, are associated with a high mortality rate, and have all the characteristics of a relatively ideal weapon of mass destruction.
- The principle of disaster management following the release of ionizing radiation always involves containment (avoidance of bringing patients with material emitting ionizing radiation to the hospital). Therefore, as part of the containment process, to the extent possible, patients should be decontaminated at the site. Removal of clothing is critically important. β and γ rays and neutrons will no longer be present unless there is still material emitting this radiation on a person's clothing. Rather than guess whether radiation is still present it is best to disrobe patients and wash them with warm soapy water.
- Orally administered potassium iodide can attenuate most radiation-induced thyroid effects, but must be given as quickly as possible because after 24 hours, there is little protective effect.
- Because of the possibility of blast, thermal, and crush injuries that may occur along with radiation injury, the care of the injured may require the care of patients who have multiple combined injuries.
- Patients who have sustained traumatic injury from an improvised explosive device but do not have third-degree burns should receive minimal amounts of intravenous crystalloid and should undergo "damage control resuscitation/surgery" as soon as possible.

- Infrastructure degradation worsens the consequences of mass casualty events. Preparing to deliver care under austere circumstances, developing creative responses, and practicing (conducting simulations) regularly will mitigate the effects of a disaster and increase resilience for individuals, teams, and institutions.

Source: Barash 9e, Chapter 59, Key Points, PDF pp. 4970–4973.